IELTS ACADEMIC READING

Volume 2

10 complete Academic Reading tests — three full passages and 40 questions each — with a complete answer key and passage evidence for every answer

IB

10
FULL TESTS

400
QUESTIONS

Full
ANSWER KEY

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Created by Teacher: Ibodullaev Bekhruz

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Introduction

Welcome to **IELTS Academic Reading — Volume 2**, a printable practice book of 10 complete Reading tests, compiled and prepared by Teacher Ibodullaev Bekhruz.

Every test follows the real exam format: three passages of increasing difficulty and 40 questions covering the full range of IELTS question types — True/False/Not Given, Matching Headings, Matching Information, Multiple Choice, Summary and Sentence Completion, and more. Answer circles and blank lines are left for you to write your answers, just like on test day.

Part One contains all 10 tests with questions only, so you can practise without seeing any answers. **Part Two** contains the complete Answer Key for each test, with the keywords from each question and the exact sentence in the passage that proves the correct answer.

How to use this book:

- Sit each test in one go, under timed conditions — 60 minutes for all three passages.
- Write your answers directly in the circles and blanks provided.
- Check your answers in Part Two and read the passage evidence for every question you got wrong.
- Use the band-score table on the answer key pages to estimate your result, then revisit weak question types.

Good luck with your preparation, and follow **@bekhruzposts** for more free IELTS materials.

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IELTS Academic Reading — Test 1

Volume 2 · 3 Passages · 40 Questions

Reading Passage 1

Sweet Trouble

Problems in the Australian sugar industry

A The Australian town of Mossman in the state of Queensland sits in a tropical landscape between the rainforest and the Great Barrier Reef. Eco-tourism is important there; more than 80% of Douglas Shire, of which Mossman is the administrative centre, is protected by World Heritage listing. But for most of the town's history, forest and reef have been largely irrelevant: since the sugar mill was built in 1894, the town has relied on sugarcane.

B Now Mossman is holding its breath. For two years the mill used by all the farms has been close to bankruptcy. It is at the centre of the economic shocks that have shaken Australia's sugar industry, and for lifetime farmers and a long list of cane industry workers a way of life will disappear if the mill closes.

C Mossman has roughly 160 growers, who now produce less than one million of the nation's annual cane harvest of 30-40 million tonnes. But it is a microcosm of the industry. All across Australia, the cane-growing business is being squeezed between the pincers of economics and the environment.

D The ten-year average return to sugar growers throughout the 1990s was about $350 a tonne. In early 2004, sugar prices plummeted, resulting in a 25-year-low average price for Australian sugar of about $232 a tonne. Although figures vary widely across farms and regions, that was about what it cost to grow a tonne of sugar in Australia. To forestall social and economic disaster, the Government offered more than $400 million to encourage growers to leave the industry. By the end of the year, 21 farmers had taken up the offer to leave, but another 1,000 are thought to be seriously considering it, allowing those remaining to buy the vacated land and improve their economies of scale.

E Fourth-generation Mossman grower Bill Phillips-Turner is one who plans to fight on. 'The consequences of losing the mill would be catastrophic,' Bill says. 'Sugar has a big economic multiplier effect: for every dollar generated from sugar, an additional $7 is generated in the wider community. Because of limited options around here, most people now employed by the industry would have to leave the area to find work.'

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F The farmer-shareholders have so far saved the mill by accepting substantial cuts to cane payments, but this has come at a big cost to everyone. As chairman of the board of the mill, Bill has presided over some tough and unpopular decisions; the hardest was slashing the mill workforce. Assets were sold and maintenance costs cut. The board has also worked hard to find new ways of doing business. Ethanol production, where sugar is used to produce fuel, has potential, and co-generation, using cane waste to generate electric power, is another possibility. However, the most radical but preferred alternative is to create a future for the mill as a food factory, turning out quality sugar-based foods.

G In addition to the economic struggle, there is the environmental one. The sugar industry has the reputation of being environmentally damaging, but it has some surprising supporters. Douglas Shire mayor, Mike Berwick, is a well-known environmentalist, and might be expected to be anti-cane. 'There's no question of the past damage it's done to the reef through chemical and nutrient run-off,' he says. 'But there's a formula for sustainable cane production and Mossman has nearly reached it.'

H Another surprise endorsement for cane comes from the Queensland Environmental Protection Agency's sugar liaison officer, Karen Benn. 'I'm less worried about the effect of cane on the environment than I am about other agriculture,' she says. 'There are good growers everywhere, but at Mossman they seem to have taken up the challenges faster.' For example, sediment run-off, previously one of the main environmental problems caused by cane growing, is now nothing like it once was, according to Dr Brian Roberts, co-ordinator of the Douglas Shire Water Quality Improvement Program. 'North Queensland used to hold the record for soil loss,' he says. 'Now..... cane country is accumulating soil.'

I However, these improvements have been achieved at a great cost to growers. Now in his 60s, Tom Watters has spent a lifetime on the same farm. Fourteen years ago, he was alerted to the effect his cane could be having on Mackay Creek, the narrow waterway that receives all his run-off, and so Tom planted a 5,000-tree buffer along the edge of the creek, built up the edges of the creek with rocks to prevent erosion, and began exploring farming methods that cause minimal soil disturbance. However, none of these costly initiatives has helped him get better cane prices.

J Another cane grower, Doug Crees, comments that 'Economically, it [cane growing] doesn't make much sense. But there's more to life than money.' It's this addiction to the way of life that keeps many cane farmers growing an under-performing crop. 'It's a good lifestyle,' Doug says. 'I spend eight months working on the farm and four months looking after our kids while my wife works in town. I've been looking at alternative crops, like forestry and cocoa, and it turns out that working away from the farm is the best diversification we could do. However, I still don't want to do that.'

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K It is difficult to see how anyone can deal satisfactorily with the passing of a way of life. Cane farmers have been part of eastern Queensland for more than a century, but despite the efforts they have put into fighting the good environmental fight, there is no guarantee that the new way of life evolving there will include cane.

Questions 1–4

Look at the following statements and the list of people below. Match each statement with the correct person, **A–F**.

LIST OF PEOPLE

A Bill Phillips-Turner
B Mike Berwick
C Karen Benn
D Brian Roberts
E Tom Watters
F Doug Crees

Q	Statement	Answer
1	Mossman cane farming practices are close to an environmentally friendly model.	
2	Financial return is not the only important factor for cane growers.	
3	Cane sugar may not harm the environment as much as other crops do.	
4	The local population would decline if the sugar-processing plant closed.	

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Questions 5–8

Choose the correct letter, **A, B, C** *or* **D**.

5 In the first paragraph, the writer says that the town is 'holding its breath' because

A it has environmental problems.
B its tourism business is threatened.
C most of its people have left the town.
D a key processing plant may shut down.

6 According to the writer, cane growers who refuse the government offer are expected to

A expand their farms.
B sell their land at a low price.
C find jobs in other industries.
D seek financial help from banks.

7 Which of the following did Bill Phillips-Turner find most difficult to do?

A sell mill property
B reduce spending on upkeep
C lower mill workers' wages
D cut the number of mill staff

8 Cane grower Doug Crees says that he

A would prefer to grow cocoa.
B wants to remain on his farm.
C wants his family to live together.
D will look for part-time work in town.

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Questions 9–13

Do the following statements agree with the information given in Reading Passage 1?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

9 In 2004, the cost of producing sugar in Australia was similar to the selling price.

10 Farmers who accepted the Government offer have mostly moved to the city.

11 Fuel production is regarded as the most desirable alternative business for the sugar mill.

12 Tom Watters reduced his use of pesticides.

13 Environmentally friendly farming practices have been profitable for Tom Watters.

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Reading Passage 2

Global Warming in Europe

It was that summer, scientists now realise, when global warming at last made itself unmistakably felt. People in the Northern Hemisphere knew that summer 2003 was remarkable. Britain had record high temperatures; Europe had out-of-control forest fires, great rivers drying to a trickle and thousands of heat-related deaths. But how remarkable that summer was is only now becoming clear.

June, July and August were the warmest three months recorded in western and central Europe. And they were the warmest by a very long way. Like Britain, Portugal, Germany and Switzerland had record national highs. Over a great rectangular block stretching from west of Paris to northern Italy, taking in Switzerland and southern Germany, the average temperature for the summer months was 3.78°C higher than the long-term norm, says the University of East Anglia's Tyndall Centre for Climate Change Research, one of the world's leading institutions for monitoring and analysing temperature records.

That might not seem a lot until you are aware of the usual pattern. But then you realise it is enormous. There is nothing like this in previous data. It is considered so exceptional that Professor Phil Jones, the unit's director, says openly - in a way that few scientists have done - that 2003's extreme temperatures may be directly attributed to global warming caused by human actions, rather than natural climate variations.

Meteorologists have hitherto contented themselves with the formula that recent high temperatures are 'consistent with predictions' of climate change. For the great block of the map in question, the unit has reliable temperature records dating back to 1781. Using as a baseline the average summer temperature between 1961 and 1990, departures from the temperature norm, 'anomalies', can easily be plotted. Over the past 200 years, there have been at least half a dozen excess temperature anomalies approaching, or even exceeding, 2°C. But there has been nothing remotely like that year, when the anomaly was nearly 4°C.

'That is quite remarkable,' Professor Jones says. 'It's very unusual in a statistical sense. If this series had a normal statistical distribution you wouldn't get this number. The return period (i.e. how often it could be expected to recur) would be something like one in 1,000 years. If we look at an excess above the average of nearly 4°C then perhaps nearly 3°C of that is natural variability, because we've seen that in past summers. But the final degree of it is likely to be due to global warming, caused by human action.'

That year's summer had in a sense been one that climate scientists had long been anticipating. Until then, the warming had been manifesting itself mainly in winters that were less cold rather than in summers that were much hotter. Last week, the UN predicted

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that winters were warming so quickly that some of Europe's lower-level ski resorts will die out. But sooner or later the unprecedented hot summer was bound to come - and that year it did.

Over a large swathe of the western part of the European continent, records were broken in all three months. It wasn't only monthly averages, but daily extremes and the lengths of spells above thresholds. National records were set in at least four countries.

One of the most dramatic features of the summer was the hot nights, especially in the first half of August. The high night-time temperatures were related to the 15,000 extra deaths in France during August, compared with previous years. They gradually increased during the first 12 days of the month, peaking at about 2,000 a day on August 12 and 13 and severely overloading the medical services. Then they dropped dramatically after August 14 when minimum temperatures fell by about 5°C. The elderly were most affected - their death rate rose 70 per cent.

For Britain, the year as a whole is likely to be the warmest recorded. But despite the temperature record on August 10, the summer itself - defined as the June, July and August period - comes behind 1976 and 1995, when there were longer periods of intense heat. At the moment, the year is likely to be the third-hottest in the global temperature record (which goes back to 1856), behind 1998 and 2002. But when the records for October, November and December are collated, it might move into second place. The ten hottest years in the record have occurred since 1990.

Professor Jones is in no doubt about the astonishing nature of that year's European summer. 'The temperatures recorded that year were out of all proportion to the previous record,' he says. 'It was the warmest summer in the last 500 years and probably way beyond that. It was enormously exceptional.'

His colleagues at the Tyndall Centre are planning a study of it. 'It was a summer that had not been experienced before, either in terms of the temperature extremes that were reached, or the range and diversity of the effects of the extreme heat,' says the centre's executive director, Professor Mike Hulme. 'It will certainly have left its mark on a number of countries as to how they think and plan for climate change, much as the 2000 floods revolutionised the way the Government is thinking about flooding in the UK. The 2003 heatwave will have similar repercussions across Europe.'

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Questions 14–19

Do the following statements agree with the information given in Reading Passage 2?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

14 In 2003, the average temperature for June to August in parts of Europe was almost 4°C higher than usual.

15 According to Phil Jones, the impact people have on the planet was one reason for 2003's record summer.

16 Temperatures are recorded twice daily in major cities.

17 Professor Phil Jones believes that the exceptional summer temperatures are within normal variation.

18 Before 2003, global warming generally caused milder winter months, not warmer summers.

19 New ski resorts are being built at higher altitudes in some countries.

Questions 20 and 21

Answer the questions below. Choose **NO MORE THAN THREE WORDS AND/OR NUMBERS** *from the passage.*

20 In Britain, which TWO years had hotter summers than the one under discussion?

21 Apart from the extreme summer of 2003, what other natural event has affected the way Britain is planning for the future?

Questions 22–25

Complete the summary below. Choose **ONE WORD AND/OR A NUMBER** *from the passage for each answer.*

The three warmest years since global temperature records began in 22 _____ have been 1998, 2002 and 2003. It is significant that the warmest ten years have all been after 23 _____. Recorded temperatures in Europe in summer 2003 were the highest for at least 24 _____ years and had serious consequences. For example, in August alone, thousands of people died in 25 _____.

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Question 26

26 Which of the following is the most suitable title for Reading Passage 2?

A The Causes of Global Warming

B The Effects of Extreme Heat

C The History of Temperature Recording

D The Year of European Heat

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Reading Passage 3

Improving Patient Safety

How improved drug packaging could provide some answers to patient safety issues

One of the most prominent design issues in pharmacy is that of drug packages and the patient information leaflets (PILs) included in them. Many pharmacists are concerned that current designs are 'accidents waiting to happen'. The UK government shares this concern and, in 2003, the National Patient Safety Agency created a new role, appointing Colum Lowe, who has 14 years' experience as a designer in the private sector, as head of design and human factors.

Packaging design in the pharmaceutical industry is handled by either in-house teams or external design agencies. For packaging design of over-the-counter medicines, which do not have to be dispensed by a pharmacist but can be bought directly from a sales assistant, characteristics such as attractiveness and distinguishability are important and so these are usually commissioned from an external design team. The marketing team prepares the initial brief and the designers come up with six or seven designs. Two or three of these are then tested on a consumer group. In contrast, most designs for prescription-only products are created in-house. In some cases, this may simply involve the company's design team applying the house design and then handing it over to design engineers rather than testing the design on a consumer group. Clearly this process cannot adequately address the needs of the wide variety of patients using medication.

Design considerations

In her book Information Design for Patient Safety, Thea Swayne highlighted a multitude of design problems. For example, drugs that look or sound alike can lead to confusion, small type sizes and even the glare on silver foil packaging can lead to names or instructions being misread. One such example is a drug that was accidentally injected into a patient through the spine (intrathecally) rather than through the veins (intravenously). Investigations following this tragedy attributed some blame to the poor choice of typescript used on the drug container. Furthermore, according to Swayne, real situations in which medicines are used include a parent giving a cough medicine to a child in the middle of the night; packaging should be designed for moments such as these rather than for the ideal world of a hospital.

Safety and compliance

Child protection is another area that gives designers opportunities to improve safety. According to the Child Accident Prevention Trust, 70% of children admitted to hospital with suspected poisoning have swallowed medicines, and although child-resistant lids have helped, they are not yet fully effective. There is scope for improving what is currently

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available, according to Richard Mawle, a freelance product designer who feels it is not just children who are blocked by child-proof closures. 'Many child-resistant packs are based on strength but older people may have the same level of strength as a child,' he explained, and suggested that better designs could rely on cognitive skills (e.g. removing the lid using a three-step process).

Mawle also worked on a project which involved applying his skills to packaging and PILs. Commenting on the information presented, he said: 'There can be an awful lot of junk at the beginning of PILs. For example, why are company details towards the beginning of a leaflet when what might be more vital for the patient is that the medicine should not be taken with alcohol?'

Design principles and guidelines

Most designers work according to basic principles, for example, certain print styles are known to be more difficult to read than others. Look-alike boxes present the potential for errors and an obvious solution would be to use colours to highlight a larger dosage of a drug. However, according to Thea Swayne, designating a colour to a particular dosage is not recommended because this could lead to the user not reading the text on a box.

Design features can provide the basis for lengthy debates. One argument is that if all packaging was white with black lettering, people would have no choice but to read every box carefully. The problem is that trials of drug packaging are few; common signage studies concern road traffic signs and visual display units. Although some designers take results from such studies into account, proving that a particular feature is beneficial can be difficult. For example, current UK legislation requires packaging to include the name of the medicine in Braille, but, according to Karel van der Waarde, a design consultant to the pharmaceutical industry, it is not known how much visually impaired patients will benefit nor how much the reading of visually able patients will be impaired.

Van der Waarde is sceptical about current legislation and says that many regulatory authorities do not have the resources to handle packaging information properly. 'They do not look at the use of packaging in a practical context they only see one box at a time and not several together as pharmacists would do,' he said.

Innovation

On a positive note, a recent innovation exhibition revealed several new designs. 'The popper' aims to help arthritis sufferers remove tablets from blister packs, and 'Pluspoint' is an adrenaline auto-injector (a device that allows diabetics to inject themselves) aimed at overcoming the fact that many patients do not carry their medication due to its prohibitive size. The aim of good design is to try to make things more user-friendly as well as safer.

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The guidelines in Information Design for Patient Safety are not intended to be legally binding. Rather, the book's purpose is to create a basic design standard and to stimulate innovation. The challenge for the pharmaceutical industry is to adopt such a standard.

Questions 27–32

Look at the following statements and the list of people or groups below. Match each statement with the correct person or group, **A–D**.

LIST OF PEOPLE OR GROUPS

A Thea Swayne

B The Child Accident Prevention Trust

C Richard Mawle

D Karel van der Waarde

Q	Statement	Answer
27	The elderly would benefit from drug containers that do not require force to open them.	
28	Adapting packaging for the blind may disadvantage people who can see.	
29	Specially designed containers have not been able to eliminate drugs being swallowed accidentally.	
30	Designers have to consider how drugs are used in the home.	
31	Governing bodies need to compare different drug containers rather than studying individual ones.	
32	Information provided with medicine is not listed in the right order.	

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Questions 33–37

Complete the summary using the list of words, **A–G**, *below.*

Packaging design in the pharmaceutical industry

A consumers **B** design engineers **C** external design team **D** in-house design team
E marketing team **F** pharmaceutical industry **G** pharmacists

Over-the-counter drugs: First, a proposal is written by the (33) _____. Then several designs are produced by the (34) _____. Finally, selected designs are shown to (35) _____.

Prescription-only drugs: The (36) _____ create the design. The design is then passed to (37) _____.

Questions 38–40

Choose the correct letter, **A, B, C** *or* **D**.

(38) In the accident mentioned in the passage, what was the 'design consideration' that caused a drug to be given incorrectly?

A a printing error
B the style of print
C an incorrect label
D the shape of the bottle

(39) What do some people say about the use of only black and white as a design feature?

A Consumers would dislike this option.
B Drug containers would all look too similar.
C People would pay more attention to label information.
D Partially sighted people would find these colours more helpful.

(40) Why does the writer refer to 'the popper' and 'Pluspoint'?

A to show that progress is being made in pharmaceutical packaging design
B to give an example of pharmaceutical design problems that can cause accidents
C to prove that a lot of work still needs to be done to improve pharmaceutical packaging design
D to point out that patients need to be more informed about pharmaceutical products

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IELTS Academic Reading — Test 2

Volume 2 · 3 Passages · 40 Questions

Reading Passage 1

Good Nuggets of Australia

Australia was, and is, home to some of the most amazing discoveries of gold the world has seen. This is mainly due to the fact that found on the surface. Some finds changed the nation entirely, such as those in Ballarat in Victoria and Bathurst in New South Wales. Gold nuggets are naturally occurring lumps of gold. These are recovered either by placer mining (mining the gold in deposit where the veins (thin layers of gold contained in rock) are weathered.

A single, small-to-medium sized gold nugget could turn a poor miner into a rich man and nugget discoveries could turn under-worked fields or empty farmland into thriving tent villages with thousands of miners looking to locate the next "big one". Some of these places where gold was found over 150 years ago are towns today. Many of the largest gold nuggets ever formed on Earth were found in Australia. In fact, eight out of the top ten were found there.

The Welcome nugget was by far the largest nugget ever found when it was discovered. It was found in June 1856 by a team of approximately 20 miners at Bakery Hill, Ballarat. It was dug out of a tunnel at a depth of 55 metres. The Welcome nugget weighed 69.98 kilograms, and it was melted down a little over a year later, in 1859. Despite its size, the Welcome's fame was short lived.

An even bigger nugget was found ten years later, when on 5th February 1869 the Welcome Stranger nugget was discovered in a small town named Moliagul. A miner named Deason was digging at Bulldog Gully when he first thought he had struck a large stone just under the surface. However after digging only three centimetres, he was delighted to see the twinkle of gold. With the help of his companion, Oates, Deason was able to free the monster from the roots of a native Australian tree. This discovery was the world's largest ever recorded gold nugget. There are some differing reports on the exact weight of the Welcome Stranger nugget, but the most well-known rating was 2284 troy ounces (71 kilograms). It measured 60 by 30 centimetres and could not be weighed in the district as they did not have scales large enough. At today's gold exchange rate, the Welcome Stranger would be worth over $2,000,000. A stone monument marks where it was found.

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In 1855 at the Mount Alexander goldfield in Victoria, a group of inexperienced miners were sent by other miners to a site in Golden Gully that was believed to be empty of gold. Such spots were called duffer claims because they were thought to be useless. However, in this case the plan backfired. On their second day of digging the miners unearthed a 1008 ounce (28.5 kilograms) gold nugget which would change their lives forever. The nugget was named after a gold commissioner called Mr Heron. The Heron nugget was the seventh largest nugget in the world when discovered.

More recently, the Hand of Faith was found behind a primary school in the state of Victoria by an amateur metal-detector named Hillier. The nugget was in the vertical position only about 15 centimetres below the surface. It measures 47 by 20 centimetres, and is still on display at a casino in Las Vegas, named The Golden Nugget, which bought it for US $1,000,000. Interestingly, the school where it was found is only approximately 30 miles from where the Welcome Stranger nugget was found in 1869. The nugget Hillier found in 1980 was, and is, the largest known nugget found anywhere in the world by a metal detector. It weighs 27.21 kilograms, making it also the largest still in existence. As noted above, much larger nuggets were discovered in the 1850s, but these were melted down to decrease the chances of theft and to change the gold into a form that would be easily sold.

While the largest nugget of gold ever found is the Welcome Stranger, it is not the largest mass of gold ever found. The Holtermann 'nugget' is not technically a nugget: in 1872 it was cut from an incredibly rich vein at Hill End, New South Wales, by a prospecting company in which Holtermann was a partner. The specimen also had quartz in it and weighed a whopping 286 kilograms. It was estimated that 3,000 ounces (85 kilograms) of gold were extracted when the giant was crushed. Only the top of the specimen remains, cut off by Holtermann as a souvenir.

Amazingly, this was not the only gigantic piece of gold cut from Hill End's mine; an even larger specimen was drilled from the same vein some months later. This second monster was broken up below ground as there seemed no point struggling with it up to the surface only to have it crushed. Its estimated weight was around 318 kilograms and it yielded approximately 5,000 ounces (142 kilograms) of gold.

Gold has been a major part of Australia's development and continues to have huge effects on the nation. It is possible that even larger nuggets and gold deposits still exist, although as years and technology progress, it is not very likely.

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Questions 1–7

Do the following statements agree with the information given in Reading Passage 1?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

1 Placer mining involves searching for gold near a present or past source of water.

2 The locations of former tent villages still attract gold diggers today.

3 A small minority of the ten largest gold nuggets in the world were found in Australia.

4 The Welcome Stranger nugget was smaller than the Welcome nugget.

5 The people who found the Heron nugget were sent to the site by people who were trying to trick them.

6 Holtermann kept a piece of the gold his company had found.

7 A more significant specimen of gold that was found at the same site as the Holtermann 'nugget' was cut into pieces as soon as it was brought above ground.

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Questions 8–13

Complete the table below. Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** *from the passage for each answer.*

Name of nugget	When found	Where found	Notes
Welcome nugget	1856	55 m beneath ground in a tunnel	found by team of 20 miners weighed 69.98 kg
Welcome Stranger nugget	1869	just (8) _____ beneath ground, among roots of tree	area unequipped to weigh the nugget point of discovery now indicated by a (9) _____
Heron nugget	1855	Mount Alexander	nugget given the name of a (10) _____ at that time
Hand of Faith nugget	1980	at the back of a (11) _____, 15 cm beneath ground	discovered by using a metal detector can now be seen at a (12) _____
Holtermann 'nugget'	1872	Hill End, New South Wales	gold was found in a mass that also contained (13) _____ yielded 85 kg of gold

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Reading Passage 2

Nature's Useful Designs

A What has fins like a whale, skin like a lizard, and eyes like a moth? The future of engineering. Almost all living organisms are uniquely adapted to the environment in which they live, some so well that scientists study them in the hope of replicating their natural designs in technology. This process, called biomimetics, is the crossroads where nature and engineering meet.

B Perhaps the best example of biomimetics is Velcro. In 1948 a Swiss scientist, George de Mestral, had trouble removing a plant's prickle which was stuck to his dog's fur, so he studied it under a microscope. Impressed by the stickiness of the prickle's hooks, he copied the design, engineering a fastener made of two pieces. One piece has stiff hooks like the prickle, while the other has soft loops that allow the hooks to stick. De Mestral named his invention Velcro - a combination of the words "velour" and "crochet".

C Andrew Parker, a research fellow at the Natural History Museum in London and at the University of Sydney, is a leading proponent of biomimetics - applying designs from nature to problems in engineering, materials science, medicine, and other fields. He has investigated iridescence in butterflies and beetles and antireflective coatings in moth eyes - studies that have led to brighter screens for cellular phones and an anticounterfeiting technique so secret he cannot say which company is behind it. He is working to make cosmetics that mimic the natural sheen of diatoms (a type of algae) and, with the British Ministry of Defense, to emulate the water-repellent properties of these same creatures. He even draws inspiration from nature's past: on the eye of a 45-million-year-old fly trapped in amber that he studies in a museum in Poland, Parker noticed microscopic corrugations that reduced light reflection. This feature is now being built into solar panels.

D To Parker, every species, even those that have become extinct, is a success story, optimized by millions of years of natural selection. He asks: why not learn from this? Parker explained how the metallic sheen and dazzling colors of certain birds derive not from pigments but from neatly spaced microstructures that reflect specific wavelengths of light. Such structural color, fade-proof and more brilliant than pigment, is of great interest to people who manufacture paint and holograms on credit cards. Glowworms produce a cool light with almost zero energy loss (a normal light bulb wastes 98 percent of its energy as heat) and bombardier beetles have a highly effective combustion chamber in their posterior that heats chemicals and fires them at would-be predators.

E For all nature's sophistication, many of its clever devices are made from simple substances like keratin, calcium carbonate, and silica, which are manipulated into structures of fantastic complexity and toughness. The abalone, for example, makes its shell

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out of calcium carbonate, the same stuff as soft chalk. Yet by coaxing this substance into walls of staggered, nanoscale bricks through a subtle play of proteins, it creates an armor 3000 times harder than chalk. Understanding the microscale and nanoscale structures responsible for a living material's exceptional properties is critical to re-creating it synthetically.

F Though impressed by biological structures, Robert Cohen, an engineer at MIT in the United States, considers biology merely a starting point for innovation. "You don't have to reproduce a lizard skin to make a water-collection device or a moth eye to make an antireflective coating," Cohen says. "The biological structure provides a clue to what is useful. But maybe you can do it better." Ultimately, he considers a biomimetics project a success only if it has the potential to make a useful tool for people. "Looking at pretty structures in nature is not sufficient," says Cohen. "What I want to know is, can we actually transform these structures into something with true utility in the real world?"

G This, of course, is the tricky bit. Potentially one of the most useful embodiments of natural design is the bio-inspired robot, which could be deployed in places where people would be too conspicuous, bored to tears, or killed. But such robots are notoriously hard to build. Ronald Fearing, a professor of electrical engineering at the University of California, Berkeley, has taken on one of the biggest challenges of all: to create a miniature robotic fly that is swift, small, and maneuverable enough for use in surveillance or search-and-rescue operations.

H The key to making his micromechanical flying insect (MFI) work, Fearing says, is not to attempt to copy the fly, but to isolate the structures crucial to its feats of flying. The fly's wing is driven by 20 muscles, some of which only fire every fifth wing beat, and all you can do is wonder, "what on earth just happened there?" says Fearing. Some things are just too mysterious and complex to be able to replicate.

I For all the power of the biomimetics paradigm, and the brilliant people who practice it, bio-inspiration has led to surprisingly few mass-produced products, and arguably only one household word - Velcro. Some biomimetists blame industry, whose short-term expectations about how soon a project should be completed and become profitable clash with the time-consuming nature of biomimetics research. Others lament the difficulty in coordinating joint work among diverse academic and industrial disciplines, which is required to understand natural structures and mimic what they do. But the main reason biomimetics has not yet come of age is that, from an engineering standpoint, it is fiendishly difficult to reproduce such intricate nanopuzzles. Nonetheless, the gap with nature is gradually closing.

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Questions 14–19

Reading Passage 2 has nine paragraphs, **A–I**. *Which paragraph contains the following information?*

Q	Statement	Answer
14	how an organism turns a basic material into something incredibly strong	
15	a claim that biomimetics has had limited commercial impact so far	
16	a difficulty that led to a researcher's accidental discovery	
17	an example of nature being far more efficient than a common household object	
18	an ancient specimen that inspired a modern innovation	
19	situations where it is preferable to replace a human with a machine	

Questions 20–22

Look at the following statements and the list of people below. Match each statement with the correct person, **A–D**.

LIST OF PEOPLE

A George de Mestral

B Andrew Parker

C Robert Cohen

D Ronald Fearing

Q	Statement	Answer
20	All living things provide something useful for humans to discover.	
21	Natural designs are sometimes impossible for people to copy.	
22	Biomimetics achieves nothing unless it has a practical application.	

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Questions 23–26

Complete the sentences below. Choose **ONE WORD ONLY** *from the passage for each answer.*

Andrew Parker is studying the shiny surface of diatoms in order to develop (23) _____.

Paint companies are interested in the way some (24) _____ get their colour.

Bombardier beetles protect themselves by shooting hot (25) _____ at their enemies.

One scientist is studying the (26) _____ to build a tiny robot that can help people in danger.

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Reading Passage 3

The Art of Deception

Do tiny changes of facial expression show whether someone is telling lies?

Forty years ago, the research psychologist Dr Paul Ekman was addressing a group of young psychiatrists in training when he was asked a question, the answer to which has kept him busy ever since. Suppose, the group wanted to know, a particular patient swears they are telling the truth. They look and sound sincere. So here is the question: is there any way you can be sure they are telling the truth? Ekman did not know the answer then, but wanted to find out.

As part of his research, he had already filmed a series of 12-minute interviews with psychiatric patients. In a subsequent conversation, one of the patients told him that she had lied to him. So, Ekman sat and looked at the film but saw nothing noteworthy. Then he slowed it down and looked again. Then he slowed it even further. And suddenly, there, across just two frames of the film, he saw it: an intense expression of extreme anguish. It lasted less than a 15th of a second, but once he had spotted the first expression, he soon found three more examples in that same interview. He termed his discovery "micro-expressions": very rapid, intense demonstrations of emotion that the subject intended to be concealed.

Over the course of the next four decades, Ekman successfully demonstrated a proposition first suggested by Charles Darwin: that the ways in which we express rage, disgust, contempt, fear, surprise, happiness and sadness are universal. The facial muscles triggered by those seven basic emotions are, he has shown, essentially standard, regardless of language and culture, from the US to Japan and Brazil to Papua New Guinea. What is more, expressions of emotion are impossible to suppress and, particularly when we are lying, micro-expressions of powerfully felt emotions will inevitably flit across our face before we get the chance to stop them.

Fortunately for liars, most people will fail to spot these fleeting signals of inner torment. Of the 15,000 Ekman has tested, only 50 people, whom he calls "naturals", have been able to do it. But given a little more training, Ekman says, almost anyone can develop the skill. He should know, since these tests were completed in the mid-1980s and the first publication of his research, he has been called in by the FBI and CIA (among countless more law-enforcement and other agencies around the world), not just to solve cases, but to teach them how to use his technique for themselves. He has held workshops for defense and prosecution lawyers, health professionals, even jealous spouses, all of them wanting to know exactly when someone is not being 100 per cent candid.

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Most recently, Ekman's research has resulted in a new television series about the exploits of the fictional Dr Cal Lightman, a scientist who studies involuntary body language to discover not only if you are lying, but why you might have been motivated to do so. According to the publicity blurb, Lightman is a human lie detector, even more accurate than a polygraph test. Ekman concedes he was skeptical when the producer first approached him with the idea of turning his life's work into a TV series, and initially would have stopped the project if he could. In particular, he was fearful that the show would exaggerate the effectiveness of his techniques and create the quite inaccurate impression among audiences that criminals could no longer hope to get away with lying. In the worst-case scenario, he was concerned about unfair convictions, that one day someone not properly trained in his techniques might be sitting on a jury and wrongly find someone guilty of a crime simply on the basis of a television programme.

In the end though, he was won over because the series is unusual in several respects. It is the first time, as far as Ekman is aware, that a commercial TV drama has been based on the work of just one scientist. That scientist is also deeply involved in the project, talking through plot ideas and checking five successive drafts of each script to ensure details are correct. He was also impressed with the producer's manifestly serious and well-intentioned reasons for making the programme. Now that the first series has been completed, he believes probably 80-90 per cent of the show is based on fact and that's good enough for what is, after all, a drama, not a documentary.

Ekman, incidentally, professes to have been a terrible liar ever since he was a small boy and observes that the ability to detect a lie and the ability to lie successfully are completely unrelated. He has been asked by people running for high office if he could teach them to become "more credible" with the public but has always refused to use his skills in that way on ethical grounds. He also insists that there are various kinds of lies. A "true" lie can be identified by having two essential characteristics: there must be a deliberate intent to mislead and there must be no notification that this is what is occurring. This means that an actor or a poker player isn't a true liar: they are supposed to be deceiving you, it's part of the game, and the same is true of flattery. He prefers to focus on the kinds of lies where the liar would be in grave trouble if they were found out, and where the target would feel properly aggrieved if they knew.

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Questions 27–31

Choose the correct letter, **A, B, C** *or* **D**.

27 According to the writer, Ekman became interested in lying after a question from his

A peers
B patients
C students
D teachers

28 The writer refers to the 12-minute interviews in order to

A illustrate how frequently patients lie
B describe the origins of Ekman's theories
C compare Ekman's research to previous studies
D show how patients' behaviour is affected by filming

29 What is the writer's point in the third paragraph?

A micro-expressions are common to all people
B recent research has refuted an old idea
C with practice we can learn to control our micro-expressions
D human society is too complex to allow for generalisations.

30 What are we told about Ekman's conclusions from his tests?

A it's natural for people to lie
B few untrained people can detect lying
C most liars suffer from periods of depression.
D all of his subjects were trained to identify micro-expressions.

31 What point does the writer make about Ekman's techniques in the fourth paragraph?

A they take decades to teach
B they have been in great demand
C they have aroused the suspicions of some agencies
D they can be used by a limited range of occupations.

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Questions 32–36

Complete the summary using the list of words, **A–I**, *below.*

The television series based on Ekman's work

A consequences **B** crimes **C** false beliefs **D** motives **E** justice **F** accuracy
G acting **H** research **I** ratings

A new TV series based on Ekman's work features a hero named Lightman, who detects lies. Initially, Ekman was unenthusiastic about the TV project because he feared the possibility of encouraging viewers (32) _____. For example, he was worried that one day the programme could result in (33) _____ not being carried out. Ultimately though, he has given the show his blessing because he is not aware of any other comparable programme based on a single person's (34) _____. The (35) _____ of the show's producer have been another pleasant surprise and, considering the genre of the programme, Ekman is happy with the show's overall (36) _____.

Questions 37–40

Do the following statements agree with the claims of the writer in Reading Passage 3?

TRUE *if the statement agrees* • **FALSE** *if the statement contradicts* • **NOT GIVEN** *if there is no information*

(37) Ekman regrets the lies he told as a child.

(38) People who are good at lying tend to be good at detecting lies.

(39) Ekman has worked with poker players to help them lie more convincingly.

(40) Ekman is more interested in the types of lies with serious consequences.

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IELTS Academic Reading — Test 3

Volume 2 · 3 Passages · 40 Questions

Reading Passage 1

Footprints in the Muds of Time

The dinosaurs may have risen to power in as little as 10,000 years ...

Everybody knows that the dinosaurs became extinct as a result of a large asteroid; something big hit the Earth 65 million years ago and, when the dust had fallen, so had the great reptiles. There is thus a nice, if ironic, symmetry in the idea that a similar impact brought about the dinosaurs' rise. That is the thesis proposed by Paul Olsen of Columbia University.

Dinosaurs first appear in the fossil record 230 million years ago, during the Triassic period. But they were mostly small, and they shared the Earth with lots of other sorts of reptile. It was in the subsequent Jurassic period, which began 202 million years ago, that they overran the planet and turned into the monsters realistically depicted in modern books and movies. Dr Olsen and his colleagues are not the first to suggest that the dinosaurs inherited the Earth as the result of an asteroid strike. But they are the first to show that the takeover did, indeed, happen in a geological eye blink.

Dinosaur skeletons are rare. Dinosaur footprints are, however, surprisingly abundant. And the size of the prints is as good an indication of the size of the beasts as are the skeletons themselves. Dr Olsen and his colleagues therefore concentrated on prints, not bones. The prints in question were made in eastern North America, a part of the world then full of rift valleys similar to those in East Africa today. Like the modern African rift valleys, the Triassic/Jurassic American ones contained lakes, and these lakes grew and shrank at regular intervals because of climatic changes. Rocks from this place and period can be dated to within a few thousand years. As a bonus, squishy lake-edge sediments are just the things for recording the tracks of passing animals.

By dividing the labour between them, the research team were able to study such tracks at 80 sites and look at 18 so-called 'ichnotaxa'. These are recognisable types of footprint that cannot be matched precisely within the species of animal that left them. But they can be matched with a general sort of animal, and thus act as an indicator of the fate of that group, even when there are no bones to tell the story.

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Their findings show that five of the ichnotaxa disappear before the end of the Triassic, and four march confidently across the boundary into the Jurassic. Six, however, vanish at the boundary, or only just splutter across it; and three appear from nowhere, almost as soon as the Jurassic begins.

That boundary itself is suggestive. The first geological indication of the impact that killed the dinosaurs was an unusually high level of iridium in rocks at the end of the Cretaceous period, when the beasts disappear from the fossil record. Iridium is normally rare at the Earth's surface, but it is more abundant in meteorites. When people began to believe the impact theory, they started looking for other Cretaceous-end anomalies. One that turned up was a surprising abundance of fern spores in rocks just above the boundary layer - a phenomenon known as 'fern spike'.

That matched the theory nicely. Many modern ferns are opportunists. They cannot compete against plants with leaves, but if a piece of land is cleared by, say, a volcanic eruption, they are often the first things to set up shop there. An asteroid strike would have scoured much of the Earth of its vegetable cover, and provided a paradise for ferns. A fern spike in the rocks is thus a good indication that something terrible has happened.

The surprises are how rapidly the new ichnotaxa appeared and how quickly they increased in size. Dr Olsen and his colleagues suggest that the explanation for this may be a phenomenon called ecological release. This is seen today when reptiles (which in modern times tend to be small creatures) reach islands where they face no competitors. The most spectacular example is on the Indonesian island of Komodo, where local lizards have grown so large that they are often referred to as dragons. The dinosaurs, in other words, could flourish only when the competition had been knocked out.

That leaves the question of where the impact happened. No large hole in the Earth's crust seems to be 202 million years old. It may, of course, have been overlooked. Old craters are eroded and buried, and not always easy to find. Alternatively, it may have vanished. Although continental crust is more or less permanent, the ocean floor is constantly recycled by the tectonic processes that bring about continental drift. There is no ocean floor left that is more than 200 million years old, so a crater that formed in the ocean would have been swallowed up by now.

There is a third possibility, however. This is that the crater is known, but has been misdated. The Manicouagan 'structure', a crater in Quebec, is thought to be 214 million years old. It is huge - some 100 kilometres across - and seems to be the largest of between three and five craters that formed within a few hours of each other as the lumps of disintegrated comet hit the Earth one by one. Such an impact would surely have had a

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perceptible effect on the world, but the rocks from 214 million years ago do not record one. It is possible, therefore, that Manicouagan has been misdated. That will be the next thing to check.

Questions 1–6

Do the following statements agree with the claims of the writer in Reading Passage 1?

YES *if the statement agrees with the claims of the writer* • **NO** *if the statement contradicts the claims of the writer* • **NOT GIVEN** *if it is impossible to say what the writer thinks about this*

1 There is still doubt about the theory that an asteroid strike killed the dinosaurs.

2 Books and the cinema have exaggerated the size of dinosaurs.

3 Other scientists have rejected Olsen's idea of a sudden dinosaur occupation of the Earth.

4 Dinosaur footprints are found more frequently than dinosaur skeletons.

5 Ichnotaxa offer an exact identification of a dinosaur species.

6 There is evidence that some groups of dinosaur survived from the Triassic period into the Jurassic period.

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Questions 7–13

Complete the summary below. Choose **NO MORE THAN THREE WORDS** *from the passage for each answer.*

Dr Olsen's group believe that the sudden increase in the size of dinosaurs may have been due to something known as (7) _____. A current example of this can be found on Komodo Island in Indonesia, where some of the lizards are commonly called (8) _____ because of their size. Apparently, they have grown this big because they do not have any (9) _____. The asteroid strike that may have cleared the way for dinosaurs to become the dominant group probably occurred 202 million years ago.

According to the writer, there are three possible reasons why we have not found a large hole in the Earth's crust dating back 202 million years. First, it may have been (10) _____ by scientists, because craters are easily covered up. Or, it could have (11) _____ – for example, if the hole had been in the ocean, it would no longer exist because of the (12) _____ that produce continental drift. Thirdly, the hole could still exist but have been (13) _____.

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Reading Passage 2

The Constant Evolution of the Humble Tomato

Heirloom tomatoes – varieties passed down through generations of a family for their good flavour – are really no more 'natural' than grocery-store varieties. New studies promise to restore their lost, healthy genes.

A Famous for their taste, color and organic appearance, heirloom tomatoes are favorites of gardeners and advocates of locally grown foods. The tomato enthusiast might conclude that, given the immense varieties, heirlooms must have a more diverse and superior set of genes than the tomatoes available in grocery stores, those ordinary hybrid varieties such as cherry and plum. However, their seeming diversity is only skin-deep: heirlooms are actually feeble and inbred - the defective product of breeding experiments that began hundreds of years ago, and exploded thanks to enthusiastic backyard gardeners. 'The irony of all this,' says Steven Tanksley, a geneticist at Cornell University, 'is all that diversity of heirlooms can be accounted for by a handful of genes. There're probably no more than 10 mutant genes that create the diversity of heirlooms you see.' But rather than simply proving that the myth about the heirloom's diversity is wrong, Tanksley's deconstruction of the tomato genome, along with work by others, is showing how a small berry-like fruit from the Andes became one of the world's top crops.

B The cultivated tomato is a member of the nightshade family that includes New World crops such as the potato, which spread around the globe after Christopher Columbus brought them back to Spain in the 15th century. But whereas scientists have uncovered a wealth of archaeological evidence on early farming practices in the New World, the record is blank when it comes to the tomato. The modern tomato seems to have its origins in the Andes in South America, and may have been domesticated in Vera Cruz, Mexico. Primitive varieties still grow throughout the Americas. All told, botanists call as many as 13 species 'tomatoes', and consider an additional four to be closely related.

C One might assume that one of these known wild species became today's cultivated crop, but that's not the case: the Mother Tomato has never been found. The closest relative is the currant tomato which, based on genetic comparisons, split from today's tomato some 1.4 million years ago. So researchers like Tanksley have to work backward, crossing tomato varieties and species in order to understand how various genes influence shape and size. Once isolated, Tanksley later inserts those genes into other tomato varieties to make his case with a dramatic transformation.

D Tanksley concludes from his analyses that in their effort to make bigger, tastier and faster-growing fruit, our ancestors ultimately exploited just 30 mutations out of the tomato's 35,000 genes. Most of these genes have only small effects on tomato size and

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shape, but recently Tanksley and his colleagues reported that they found a gene that increases fruit size by 50 percent. It was probably the most important event in domestication. The first written record of tomatoes - from Spain in the 1500s - confirms that this mutation, which enlarges tomatoes by producing compartments known as locules, existed back in the same yellow tomatoes that gave Italians the word pomodoro, or golden apple. Besides size, tomato farmers also selected for shape. To discover those genes, Esther van der Knaap, a Tanksley alumnus now at The Ohio State University, took a gene from one heirloom tomato and inserted it into a wild relative. She observed that, as a result, the tiny fruits became shaped like pears.

E The selection of these traits has, however, affected the heirloom's hardiness. They often suffer from infections that cause the fruit to crack, split and otherwise rot quickly. Wild plants must continuously evolve to fend off such infections, points out Roger Chetelat of the Tomato Genetics Resource Center at the University of California. But in their quest for size, shape and flavor, humans have inadvertently eliminated defensive genes. As a result, most possess only a single disease-resistant gene. Chetelat elaborates that heirlooms' taste may have less to do with their genes than with the productivity of the plant and the growing environment. Any plant that produces only two fruits, as heirlooms sometimes do, is highly likely to produce juicier, sweeter and more flavorful fruit than varieties that produce 100, as commercial types do. In addition, heirlooms are sold ripened on the vine, a certain way to get tastier results than allowing them to mature on the shelf. This means breeders feel confident that getting germ-beating genes back into heirlooms won't harm the desirable aspects of the fruit. Modern breeding has resuscitated grocery store tomatoes with an influx of wild genes; in the past 50 years, as many as 40 disease-resistant genes have been bred back into commercial crops.

F In 1996 a tomato breeder and former Tanksley student named Doug Heath began a favorite project. After 12 years of traditional breeding with the help of molecular markers, he created a new multi-colored tomato less prone to cracking and also endowed with 12 disease-resistant genes. The original heirloom plant, Heath explains, had defective flowers, which is one reason why it produced only two fruits compared with the 30 he gets from his new variety. He claims he is also able to maintain a comparable flavor and sugar profile even on productive plants. The heirloom's defects are, after all, just an accident of a narrow breeding strategy left over from the very beginning of genetic modification.

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Questions 14–17

Reading Passage 2 has six paragraphs, **A–F**. *Which paragraph contains the following information?*

Q	Statement	Answer
14	an explanation of research aimed at restoring the health of the heirloom tomato	
15	a reference to a false belief about the heirloom tomato	
16	a description of the flavour of the heirloom tomato	
17	a reference to a single gene that significantly improves the cultivation of tomatoes	

Questions 18–21

Look at the following statements and the list of researchers below. Match each statement with the correct researcher, **A–D**.

LIST OF RESEARCHERS

A Steven Tanksley
B Esther van der Knaap
C Roger Chetelat
D Doug Heath

Q	Statement	Answer
18	The transplanting of certain genes into tomatoes can change their shape.	
19	The flavour of the heirloom tomato is largely dependent on actual yield and cultivation.	
20	A new type of tomato can be produced that is stronger than the original heirloom tomato yet equally sweet and flavoursome.	
21	The wide variety of heirloom tomatoes is due to only a small number of genes.	

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Questions 22–26

Complete the sentences below. Choose **ONE WORD ONLY** *from the passage for each answer.*

There is little information on the origin of the tomato despite the existence of (22) _____ data on the growing of other New World crops.

Although it is uncertain, the tomato is thought to have first grown in the (23) _____.

In regard to genetic similarities, the (24) _____ tomato is the nearest to the earliest type of tomato.

A genetic (25) _____ which is evident in pomodoro produced larger tomatoes.

(26) _____ are a problem for heirloom tomatoes because they frequently lead to damage and deterioration.

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Reading Passage 3

Crossing the Threshold

The renovated Auckland Art Gallery in New Zealand unites old and new, creating an irresistible urge to step inside

Architects are finding it very difficult in today's cultural landscape. The profession faces a three-way threat: a public that apparently doesn't understand what architects do, developers who couldn't care less what they do, and overbearing councils micromanaging every single aspect of what they do. According to sources within the architectural profession, the situation is much worse when architects work on municipal buildings, as architects FJMT and Archimedia discovered with their Auckland Gallery makeover, where a vast number of external pressures threatened the project, and with so many bureaucratic difficulties it looked doomed to fail.

The major challenge of the gallery renovation project was that it involved two parts. The first was to restore the heritage building, dating back to 1888, which contained a network of small spaces, refurbished so often it contained 17 different floor heights. The second was to deliver a new extension that would not only double floor and exhibition space but also attract new patrons, a total necessity. While the old building's circulation was off-putting, so was something intangible yet just as powerful: its atmosphere. For many, Auckland Art Gallery was just an old building that served a limited range of patrons with highbrow interests, missing its chance to engage with new audiences. A 2003 survey of young people's impressions of the gallery confirmed this opinion, sounding more like references to an abandoned building. For the survey authors, 'threshold fear' - where certain groups are intimidated from entering certain spaces by their off-putting atmosphere - was the institution's undoing, something no architect wants anything to do with. For those young people Auckland Art Gallery was undemocratic, 'dusty' and 'cold' - the epitome of threshold fear. Also, 16% of the sample group had no idea where it even was, despite being interviewed on the pavement right outside it. Clearly, the gallery was fatally out of step at a time when New Zealand's national museum in Wellington was successfully engaging broader audiences with contemporary branding and marketing, interactive displays and temporary events.

The decision to evolve the gallery was actually made in 2000, although it took eight years for building to commence, as the architects fought off heritage committees and conservationists trying to stop them. The architects were not just dealing with a disillusioned public, but also with precious timber and the parkland which surrounds the building. Pushing the design through the Environment Court, the body which approves

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renovations of this scale, alone took three years. During this time the budget blew out by several million dollars, the funding dried up, and the new wing had to be completely redesigned.

Even after the redesign, the use of kauri timber, with its significance to New Zealand's Maori people, stirred up political debate. In the new building the architects have used kauri to produce a canopy with a curving interior roof supported by tapered, steel columns, also clad in kauri. The canopy represents a signature public face, its curvature filtering light to the forecourt to the west and creating a visual echo of the canopy of pohutukawa trees in Albert Park to the east. The park also has cultural significance to Maori as it was the site of early settlements. The connection is reinforced by sculptures from Maori artist Arnold Wilson decorating the columns, while fellow artist Bernard Makoare was a consultant, ensuring the gallery emphasised Maori beliefs. Still that didn't stop the conservationist Stephen King from accusing the architects of 'throwing' kauri at a 'mediocre building' and of misappropriating the 'mana' (spiritual energy) of the precious material (which is almost extinct: harvesting of both petrified and swamp kauri has been likened to a gold rush). However, the kauri that was used here was from the forest floor, and King's misconceptions sum up the prejudice that surrounded the project.

Objections also came from the Auckland Regional Council, worrying about the extension's impact on Albert Park, yet the project's relationship with parkland is one of the most successful outcomes. Impact is not only minimal, but improves the park's social function. The extension's enormous glass atrium opens up the building by directing the gaze from street level to the parkland beyond, while inside, the new art space is fronted along the east by a continuous glass wall incorporating the park into the gallery. The glass becomes a 'screen' for viewing the outside world and makes the art accessible to those in the park, a far cry from both 'white cube' galleries worldwide, the plain boxes where paintings are hung in antiseptic white surroundings, and also the dusty impermeable Auckland Gallery of old.

Another success is the refurbishment of the heritage building, especially the Mackelvie Gallery, in disrepair after its characteristic early twentieth-century Edwardian decoration had been stripped out or walled away in previous renovations. Remarkably the Mackelvie space has been reconstructed from two old photos, although the problem of multiple floor levels was so serious that scaffolding had to be erected at the highest level, with work progressing downwards, the reverse of normal practice. When it was over, a fascinating detail was retained: the lowest level visible under glass embedded in the new floor, the building itself as artwork, while elsewhere columns from the old gallery have been exposed in the walls of the new wing.

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In 2008 the gallery averaged just 190,000 visitors annually. After reopening, it had 300,000 in five months. Cynics will chalk that up to novelty of the new, but the fact is the gallery is now an alluring cultural space which is crawling with young people.

Questions 27–31

Choose the correct letter, **A, B, C** *or* **D**.

27 What is the writer's main point in the first paragraph?

A Criticism of architects by different groups is unfair.
B The architectural profession is generally well respected.
C The most difficult projects for architects are public buildings.
D Failure to deliver buildings is a result of poor communication.

28 The Auckland Gallery project was particularly difficult because

A the existing building was old and parts of it had fallen down.
B there was a high number of floors in the building.
C it needed to satisfy the requirements of the existing patrons.
D it involved renovating the existing building and adding a new one.

29 What disturbing information did the architects find out from the survey of young people?

A They did not visit the gallery because of the way it made them feel.
B They thought that the gallery buildings were not in use.
C The gallery had the reputation of being dirty.
D They did not like the entrance.

30 What point is the writer making when he says that 16% of the sample group did not know where the museum was?

A Young people are not interested in galleries.
B The gallery was not reaching out to involve young people.
C The entrance to the gallery was not well signposted.
D The location of the gallery was difficult to access.

31 Maori artists were used on this project to

A satisfy the concerns of conservationists.
B protect sacred materials in the Albert Park site.
C make sure the gallery respects Maori culture.
D ensure that certain sources of kauri were not used.

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Questions 32–36

Do the following statements agree with the claims of the writer in Reading Passage 3?

YES *if the statement agrees* • **NO** *if it contradicts* • **NOT GIVEN** *if impossible to say*

32 Before the renovation, the Auckland Art Gallery was regarded as an elitist institution.

33 Stephen King's intervention in the project shows his understanding of the architects' use of kauri.

34 The way the building interacts with its surroundings is a triumph.

35 The glass screen along the east wall was one of the most costly items in the rebuild.

36 The design of the extension to the Auckland Art Gallery is similar to the design of 'white cube' galleries in other parts of the world.

Questions 37–40

Complete each sentence with the correct ending, **A–F**, *below.*

ENDINGS

A resulted in work being done in the opposite direction to that usually followed.

B is more than cosmetic and has improved the circulation.

C was the clue to rebuilding the Mackelvie Gallery successfully.

D has resulted in the building itself becoming a work of art.

E means that you should be able to tell whether you are in the old wing or the new one.

F was the result of earlier attempts to modernise the Gallery.

37 The destruction of Edwardian ornamentation ____

38 It is extraordinary that a limited number of photographs ____

39 The problem of having so many floor levels to deal with ____

40 The glass flooring in the Mackelvie Gallery which reveals old features ____

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IELTS Academic Reading — Test 4

Volume 2 · 3 Passages · 40 Questions

Reading Passage 1

Jewels from the Sea

Indigenous necklaces in Tasmania

Necklace-making is the most significant cultural tradition of Tasmanian Aboriginal women. It is also one of the few traditions that have continued without interruption since before the Europeans settled in Tasmania, the island to the south of the mainland of Australia, in 1803. Whereas colonisation disrupted or destroyed so much of Tasmanian Aboriginal life and heritage, necklace-making has not only continued but also evolved in various ways.

The knowledge and skills of shell processing and stringing have been passed down through generations of women, particularly the women of the Furneaux Islands, off Tasmania's north-east coast. Necklace-making is an opportunity for women of all ages to get together and share stories, pass knowledge to younger generations and continue to affirm their culture. Aunty Dulcie Greeno, an elder of the Tasmanian Aboriginal community, has been making necklaces for more than 40 years, but first began practising as a child. 'My grandmother used to do shell necklaces and a couple of my aunties did too,' she said in an interview for Australian Museums and Galleries Online. 'We'd go round with them on the beach and collect shells with them.' Now her sister (Corrie Fullard), daughter (Betty Grace), daughter-in-law (Lola Greeno) and niece (Jeanette James) - all celebrated artists in their own right - make necklaces.

Shell necklaces were originally made as an adornment, as gifts and tokens of honour, and as objects to be exchanged with other groups for tools or for ochre, which could be used in important ceremonies. Archaeologist Rhys Jones found shells that had been pierced for a necklace dating back at least 2,000 years. After European colonisation, necklaces were also sold or exchanged for food, clothing and other essential supplies. Now, the artists are often commissioned to create necklaces for museums, galleries and private collectors.

Early European explorers remarked on the beauty of these treasures, and held them in high esteem. The French naturalist Jacques Labillardière, travelling with the d'Entrecasteaux expedition of 1791-94, observed women wearing 'strings of brilliant pearly blue spiral shells upon their bare heads'. 18th- and 19th-century images show Tasmanian Aboriginal people wearing necklaces, including a photograph taken around 1866 of the leader and spokeswoman Truganini.

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Shell-stringing was, and remains, a painstaking process, requiring knowledge of coastal resources and a great deal of patience. Aunty Dulcie's daughter Patsy Cameron has explained how the women pierced each shell with a tool consisting of a jawbone and sharpened tooth of a kangaroo or wallaby. The shells were then threaded on string made from natural fibres, smoked over a fire, and rubbed in grass to remove their outer coating and reveal the pearly surface. The shells were later treated with penguin or muttonbird oil.

European colonisation introduced new tools and materials, including acidic liquids such as vinegar to clean the shells and steel punches to make holes in them. Needles and cotton or synthetic thread enabled the women to incorporate smaller shells into increasingly intricate designs.

Necklace-making is dependent on the availability of shells, and shell collection has its season. Aunty Dulcie regularly returns to the Furneaux Islands to replenish her supplies. 'We still walk for miles on the beach,' she said. 'We take our lunch and crawl along on our hands and knees to get the shells.'

Men often help women collect the shells, especially the maireeners (rainbow shells), which are found on kelp, a type of seaweed. These shells are best when picked directly from the sea. 'We don't use the ones we pick up on the beach because they are too brittle and they lose their colour,' said Aunty Dulcie.

After colonisation, women started making longer necklaces. In 1835, Benjamin Duterrau sketched Tanleboneyer, 'a native of the district of Oyster Bay', and Bruny Island man Woorraddy, Truganini's husband, with long strands looped around their necks. Wortabowigee, a woman from Port Dalrymple, who is featured in an 1837 portrait by Thomas Bock, wears five loops of what must have been a necklace of astonishing beauty. It is possible that the new European tools adopted by the women enabled them to make longer necklaces, but it also indicates the changing circumstances of Tasmanian Aboriginal people.

Historian Brian Plomley points out that long necklaces would have been impractical for women accustomed to a traditional lifestyle of diving for crabs, crayfish and abalone, digging for root vegetables, hunting seals or climbing trees to catch possums. Men tracking kangaroo, wallaby and emu through the bush would not have risked getting snagged by long necklaces - or risked damaging the valuable necklaces themselves.

The change in necklaces after colonisation also points to the Tasmanians' courageous assertion of their identity, and a continuation of their culture at a time when their world was being taken apart. The connection of shell necklaces with the distinct culture and story of the Tasmanian Aboriginal people and with the Tasmanian natural environment means they have iconic status in the wider Tasmanian community. In 2009, they were listed as a

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Tasmanian Heritage Icon by the National Trust of Australia. The cultural and aesthetic value of the necklaces is also demonstrated by their inclusion in many national and international museum, gallery and private collections.

Questions 1–6

Do the following statements agree with the information given in Reading Passage 1?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

1 After European settlement, Tasmanian Aboriginals stopped making necklaces for a short time.

2 Aboriginal women on the Furneaux Islands made the most beautiful necklaces.

3 An 1866 photograph of the leader Truganini shows her wearing a necklace she had made herself.

4 Men assist in gathering shells growing on sea plants.

5 Tasmanian Aboriginal men wore long necklaces when hunting.

6 Tasmanian Aboriginal necklaces are appreciated outside Australia.

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Questions 7–13

Complete the notes below. Choose **ONE WORD ONLY** *from the passage for each answer.*

The process of shell-stringing

- Shell-stringers must have patience and an understanding of resources available near the sea.

THE TRADITIONAL PROCEDURE

- hole put in shell with an instrument made of animal's bone and (7) _____
- shells strung on natural fibre and smoked
- shells wiped with (8) _____ to achieve a pearly surface
- animal (9) _____ applied

CHANGES AFTER THE EUROPEANS ARRIVED

- cleaning substances like (10) _____ were used
- more complex designs achieved using needles as a tool

FURNEAUX ISLANDS

- shells need to be gathered in the right (11) _____
- shell collectors walk along the beach then (12) _____ in order to pick up shells
- shells from beach not suitable as do not keep (13) _____ and break easily

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Reading Passage 2

The Return of Monkey Life

Rain forest trees growing anew on Central American farmland are helping scientists find ways for monkeys and agriculture to benefit one another

A Hacienda La Pacífica, a remote working cattle ranch in Guanacaste province of northern Costa Rica, has for decades been home to a community of mantled howler monkeys. Other native primates - white-faced capuchin monkeys and spider monkeys were once common in this area, too, but vanished after the Pan-American Highway was built nearby in the 1950s and most of the surrounding land was cleared for cattle raising. At Hacienda La Pacífica, however, an enlightened ranch owner chose to leave some strips of native trees growing. He used these as windbreaks to protect both cattle and their food crops from dry-season winds. In the process, the farmer unwittingly founded a unique laboratory for the study of monkeys.

B Ken Glander, a primatologist from Duke University in the USA, is studying La Pacífica's monkeys in an effort to understand the relationship between howlers and regenerating forests at the edges of grazing lands. Studying such disturbed woodlands is increasingly important, because throughout much of the New World Tropics, these are the only forests left. In the 18th century, tropical dry forests covered most of Central America, but by the 1980s less than two percent remained undisturbed and less than one percent was protected.

C Howlers persist at La Pacífica, Glander explains, because they are leaf-eaters. They eat fruit when it is available but, unlike capuchin and spider monkeys, do not depend on large areas of fruiting trees. Glander is particularly interested in howlers' ability to thrive on leaves loaded with toxins - poisonous substances designed to protect the plants. For leaf-eaters, long-term exposure to a specific plant toxin can increase their ability to neutralize the poisonous substances and absorb the leaf nutrients. Watching generations of howlers at La Pacífica has shown Glander that the monkeys keep their systems primed by sampling a variety of plants and then focusing on a small number of the most nutritious food items. The leaves that grow in regenerating forests, like those at La Pacífica, are actually more howler-friendly than those produced by the centuries-old trees that survive farther south. In younger forests, trees put most of their limited energy into growing wood, leaves and fruit, so they produce much lower levels of toxin than do well-established, old-growth trees.

D The value of maturing forests to primates is also a subject of study at Santa Rosa National Park, about 35 miles northwest of La Pacífica. Large areas of Santa Rosa's forests had at one time been burnt to make space for cattle ranching and coffee farming, thereby

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devastating local monkey habitat. But in 1971 the government protected the area by designating it a National Park, and species of indigenous trees which had been absent for decades began to invade the abandoned pastures. Capuchins were the first to begin using the reborn forests, followed by howlers. Eventually even spider monkeys, fruit-eaters that need large areas of continuous forest, returned. In the first 28 years following protection of the area, the capuchin population doubled, while the number of howlers increased sevenfold.

E Some of the same traits that allow howlers to survive at La Pacífica also explain their population boom in Santa Rosa. Howler reproduction is faster than that of other native monkey species. They give birth for the first time at about 3.5 years of age, compared with seven years for capuchins, and eight or more for spider monkeys. Also, while a female spider monkey will have a baby about once every four years, well-fed howlers can produce an infant every two years. Another factor is diet. Howlers are very adaptable feeders, and only need a comparatively small home range. Spider monkeys, on the other hand, need to occupy a huge home range. Also crucial is the fact that the leaves howlers eat hold plenty of water, so the monkeys can survive away from open streams and water holes. This ability gives them a real advantage over capuchin and spider monkeys, which have suffered during the long, ongoing drought in the area.

F Alejandro Estrada, an ecologist at Estación de Biología Los Tuxtlas in Veracruz, Mexico, has been studying the ecology of a group of howler monkeys that thrive in a habitat totally altered by humans: a cacao plantation in Tabasco state, Mexico. Cacao plants need shade to grow, so 40 years ago the owners of Cholula Cacao Farm planted fig, monkeypod and other tall trees to form a protective canopy over their crop. The howlers moved in about 25 years ago after nearby forests were cut. This strange habitat seems to support about as many monkeys as would a same-sized patch of wild forest. The howlers eat the leaves and fruit of the shade trees, leaving the valuable cacao pods alone.

G Estrada believes the monkeys bring underappreciated benefits to such plantations, dispersing the seeds of fruits such as fig and other shade trees, and fertilizing the soil. Spider monkeys also forage for fruit here, though they need nearby areas of forest to survive in the long term. He hopes that farmers will begin to see wild monkeys, which could include potential ecotourism projects. 'Conservation is usually viewed as a conflict between farming practices and the need to preserve nature,' Estrada says. 'We're moving away from that vision and beginning to consider ways in which commercial activities may become a tool for the conservation of primates in human-modified landscapes.'

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Questions 14–17

Reading Passage 2 has seven paragraphs, **A–G**. *Which paragraph contains the following information?* **NB** *You may use any letter more than once.*

Q	Statement	Answer
14	a reason why newer forests provide howlers with better feeding opportunities than older forests	
15	a reference to a change in farmers' attitudes towards wildlife	
16	a description of the means by which howlers select the best available diet for themselves	
17	figures relating to the reduction of natural wildlife habitat over a period of time	

Questions 18–21

Complete the summary below. Choose **ONE WORD ONLY** *from the passage for each answer.*

Why do howlers have an advantage over other Central American monkeys?

Howler monkeys have a more rapid rate of (18) _____ than either capuchin or spider monkeys. Unlike the other local monkey species, howlers can survive without eating (19) _____, and so can live inside a relatively small habitat area. Their diet is more flexible, and they are able to tolerate leaves with high levels of (20) _____. Howlers can also survive periods of (21) _____ better than the other monkey species can.

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Questions 22–26

Look at the following features and the list of locations below. Match each feature with the correct location, **A–C**. **NB** *You may use any letter more than once.*

LIST OF LOCATIONS

A Hacienda La Pacífica

B Santa Rosa National Park

C Cholula Cacao Farm

Q	Statement	Answer
22	It has seen the return of native tree species.	
23	It supports only one species of native monkey.	
24	Its monkey population helps the agriculture of the area.	
25	It is home to populations of all three local monkey species.	
26	Its landscape was altered by the construction of a transport link.	

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Reading Passage 3

Innovation in Business

Innovation describes the way that we develop new ideas, products and approaches, and it is one of the most vital human endeavours. Over the history of humanity, innovation has made life better in so many different ways. It is part of human nature to recognise the benefits of innovation; however, the majority of us have little comprehension of the processes that actually lead to innovation occurring. It is a frequent topic for discussion in journals and university lecture theatres. However, in company boardrooms across New Zealand, the term is heard far too frequently and more caution should be applied. Certainly, a recent international survey showed that the idea of innovation is so broad as to appear almost meaningless, with different sectors of the business world - or different divisions inside the same business - often defining innovation in their own way.

While there is an obvious attraction to the new, innovation is not always about large scale research and development projects or revolutionary business models; it can be on a small, highly localised scale and involve a step-by-step approach. The famous case study of the drinks company Lucozade shows it can be as simple as adding a screw top to a bottle. This carefully managed change in packaging shifted the public perception of the product from a medicine to a sports drink and Lucozade profited handsomely, clearly showing their approach was successful. So what steps can businesses take to keep ahead in the innovation game?

There seems to be a widespread belief that the world's best ideas are sudden and intuitive. Thomas Edison supposedly invented the light bulb with a sudden moment of clarity, but in fact the book Edison: His Life and Inventions shows that Edison's own inspiration usually resulted from laborious experimentation, rather than a moment of genius. Thus, business managers are at risk of overestimating the pace of development while underestimating the amount of perseverance needed. This is why we must reject the false assumption that innovation is just a moment of brilliance waiting to strike.

No exploration of innovation is complete without mention of Steve Jobs, the late chief executive officer (CEO) of Apple. No-one encapsulated the creative essence of Silicon Valley, the home of many global tech companies, better than Jobs, but he has also been accused of arrogance. Jobs' managerial approach often irritated his co-workers but, in part, it was the fuel for his visionary innovations. He felt it his duty to make society a better place, but his egotistical behaviour could have undesirable consequences, too. Jobs' former employees have suggested this type of demanding behaviour affects sickness rates and group morale, which in turn may squash creativity and innovation.

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In his 2011 TED (Technology, Entertainment, Design) Talk, economist Tim Harford made this statement: 'Show me a successful complex system, and I will show you a system that has evolved through trial and error.' It sounds like common sense, and some commentators have claimed Harford's TED Talk was not persuasive, because it was stating the obvious. But he responds that a trial and error approach is not all that obvious, because society expects all problems to have a quick and simple solution. Harford is right to point out that both consumers and investors expect consistent and clear breakthroughs. But the value of step-by-step innovation is often underrated and failure may in fact be crucial to eventual success.

Former chief solutions officer at Yahoo, Tim Sanders, believes that innovation is only 'little ideas that combine with other little ideas that improve themselves into game-changing ideas'. This snowballing effect has similarities to the phenomenon that author and broadcaster Steven Johnson calls the 'hummingbird effect of innovation'. This points out that progress in a specific area or discipline can end up triggering small but significant developments elsewhere entirely. A good case to cite in support of this universal truth would be Gutenberg's printing press, which was revolutionary in itself for making books accessible to the masses. However, the press also affected an unrelated industry by causing a rise in demand for reading glasses. This highlights the wider idea of connectivity, which is an essential component of innovation.

When Kode Biotech won a recent innovation award, CEO Steve Henry had this to say: 'Most people think innovation is invention, but they don't understand that innovation is the conversion of invention into something useful.' This is particularly true when it comes to Kode's highly specialised use of nano-technology. Henry says that Kode cannot chase all the opportunities it has created. Instead, Kode's approach is to collaborate with similar nano-technology businesses, which take its innovations and develop them into commodities that can be sold in the marketplace.

Innovation can be hectic and disorganised, but that is the way that radical new ideas sometimes come about. But, for established businesses, the disorderly nature of innovation can seem discouraging. Previously Unavailable, a specialist innovation consultancy, says that while 84% of businesses consider innovation critical to their future, only 6% are satisfied with their innovation efforts. Previously Unavailable uses a process called Black Box, which was developed following a study at Harvard Business School. Black Box was formed to overcome the barriers that prevent innovation in larger organisations. Using this unique system, Previously Unavailable will lease a team of innovation experts on a short-term basis to client organisations to assist with conceiving, developing and launching new products and business innovation. It seems probable that this sort of innovation consultancy will become increasingly common, just as businesses already employ specialist consultants for legal, financial or marketing services.

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Questions 27–30

Do the following statements agree with the claims of the writer in Reading Passage 3?

YES *if the statement agrees* • **NO** *if it contradicts* • **NOT GIVEN** *if there is no information*

27 Most people naturally understand how innovation happens.

28 The word 'innovation' is overused in business in New Zealand.

29 There are many interpretations of what innovation means in business, according to one global study.

30 Research and development projects have a higher priority today than in the past.

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Questions 31–35

Choose the correct letter, **A, B, C** *or* **D**.

31 What is the writer doing in the third paragraph?

A criticising the conclusions of one publication
B praising the contribution of a famous inventor
C dismissing one common idea about innovation
D suggesting that innovation changes over time

32 What is the writer's main point in the fourth paragraph?

A Each industry will innovate in its own way.
B There are pros and cons to strong leaders.
C Some individuals are more innovative than others.
D Leadership requires building teams that work together.

33 The writer refers to Gutenberg's printing press in order to

A compare how two different innovations developed.
B illustrate an exception to his point about innovation.
C show why one approach to innovation was controversial.
D give an example of a general rule about innovation.

34 What is the writer's main point in the seventh paragraph?

A Companies should cooperate to turn ideas into products.
B A company tends to make profits from its own innovations.
C Science and business can often have common goals.
D Competition between businesses produces innovation.

35 According to the writer, companies like Previously Unavailable

A have published valuable research about innovation.
B help companies recruit people with skills in innovation.
C run university courses teaching innovation skills.
D can be hired to give businesses advice about innovation.

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Questions 36–40

Complete each sentence with the correct ending, **A–G**, *below.*

ENDINGS

A innovators do not get enough credit for their own ideas.

B innovation may influence a completely different field.

C expressing what everyone already knows is pointless.

D innovation can be enhanced through collaboration with similar companies.

E innovation requires a great deal of hard work.

F copying ideas that already exist is not true innovation.

G certain personality types can limit innovation.

36 A book about the life and work of Thomas Edison suggests that ____

37 People who have worked for CEOs like Steve Jobs claim that ____

38 Critics of Tim Harford's 2011 TED Talk claimed that ____

39 Steven Johnson's 'hummingbird effect' proposes that ____

40 Kode Biotech CEO Steve Henry believes that ____

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IELTS Academic Reading — Test 5

Volume 2 · 3 Passages · 40 Questions

Reading Passage 1

What Lucy Taught Us

A scientific finding in east Africa has changed our understanding of how humans have developed

On a Sunday morning in late November 1974, a team of scientists were digging in an isolated spot in the Afar region of Ethiopia. Surveying the area, palaeoanthropologist Donald Johanson spotted a small piece of bone. Straight away, he recognised it as coming from the elbow of a human ancestor. And there were plenty more. 'As I looked up the slopes to my left, I saw bits of the skull, a chunk of jaw, a couple of vertebrae,' says Johanson.

It was immediately obvious that the skeleton was a significant find, because the sediments at the site were known to be 3.5 million years old. 'I realised this was part of a skeleton that was older than three million years,' says Johanson. It was the most ancient early human ever found. Later it became apparent that it was also the most complete - 40% of the skeleton had been preserved.

At the group's campsite that night, Johanson played a Beatles song called 'Lucy in the Sky with Diamonds', and, as the feeling was that the skeleton was female due to its size, someone suggested calling it Lucy. The name stuck and Johanson says, 'All of a sudden, she became a person.'

But the morning after the discovery, the discussion was dominated by questions. How old was Lucy when she died? Did she have children? And might she be our direct ancestor? Nowadays, we're starting to get the answers to some of these questions.

According to Johanson, Lucy had an incredible combination of primitive and derived features, which had not been seen before. Her skull and jaws were more ape-like than those of other groups of early humans. Her braincase was also very small, no bigger than that of a chimp. She had a hefty jaw, a low forehead and long dangly arms.

For Johanson, it was immediately apparent that Lucy walked upright. That's because the shape and positioning of her pelvis reflected a fully upright gait. Lucy's knee and ankle were also preserved and seemed to reflect bipedal walking. Later studies of feet offer even more evidence. As an upright walker, Lucy strengthened the idea that walking was one of the selective pressures driving human evolution forwards.

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Early humans did not need bigger brains to take defining steps away from apes. Extra brainpower only came over a million years later with the arrival of the species Homo erectus, meaning upright man. Though big brains would clearly be important later, walking remains one of the traits that makes us uniquely human.

She may have walked like a human, but Lucy spent at least some of her time up in the trees, as chimpanzees and orangutans still do today. It may be that upright walking evolved in the trees, as a way to walk along branches that would otherwise be too flexible. It's not clear why Lucy left the safety of the trees.

It is thought that savannahs were gradually opening up, so trees were spaced further apart. But hunting for food may have been the real reason for heading to the ground, says Chris Stringer of the Natural History Museum in London.

In line with this idea, recent evidence suggests that the diet of early humans was changing at that time. Studies of the remains of food trapped on preserved human teeth indicate that several species, including Lucy's, were expanding their diet around 3.5 million years ago. Instead of mostly eating fruit from trees, they began to include grasses and possibly meat. This change in diet may have allowed them to range more widely, and to travel around more efficiently in a changing environment. Fossilised crocodile and turtle eggs were found near her skeleton, suggesting that Lucy died while foraging for them in a nearby lake.

How did early humans process all these new foods? Later species, like Homo erectus, are known to have used simple stone tools, but no tools have ever been found from this far back. However, in 2010 archaeologists uncovered animal bones with scratches that seem to have been made by stone tools. This suggests that Lucy and her relatives used stone tools to eat meat.

There have since been heated debates over whether or not the marks were really made by tools. But if they were, it is not surprising, says Fred Spoor of the Max Planck Institute for Evolutionary Anthropology in Leipzig, Germany.

It also seems that Lucy's childhood was much briefer than ours and that she had to fend for herself from a young age. We know that Lucy was a full-grown adult because she had wisdom teeth and her bones had fused. But unlike modern humans, she seems to have grown to full size very quickly, and time of death was when she was around 12 years old. In line with that, a recent study of a 3-year-old early human suggested that their brains matured much earlier than ours do.

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Questions 1–5

Do the following statements agree with the information given in Reading Passage 1?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

1. Donald Johanson was uncertain about the nature of the elbow bone he found in Afar.
2. Several bones were found by Donald Johanson at the same site in Afar.
3. The experts realised the importance of the discovery at Afar.
4. It was the upper part of the skeleton that had suffered the least damage.
5. The skeleton's measurements helped Johanson's team to decide if it was male or female.

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Questions 6–13

Complete the notes below. Choose **ONE WORD ONLY** *from the passage for each answer.*

Lucy

PHYSICAL FEATURES

- jaws and skull like those of an ape
- braincase similar in size to that of a chimp
- long arms

MOVEMENT

- the positioning and shape of her pelvis made it clear that she walked like a human
- upright movement possibly started among the (6) _____
- probably moved to the (7) _____ in search of food

DIET AND EATING HABITS

- analysis of food in the (8) _____ of the skeletons of early humans shows changes in their diet
- it is likely that meat and grasses were substituted for (9) _____
- (10) _____ that were located close to Lucy suggest these were also part of her diet
- (11) _____ that were found had marks on them, possibly made by tools used for eating

COMPARISONS WITH MODERN-DAY HUMANS

- modern-day humans have a longer (12) _____ than Lucy did
- the (13) _____ of modern-day humans appear to develop later than Lucy's did

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Reading Passage 2

The Gender Gap in New Zealand's High School Examination Results

Results from New Zealand's new national examinations for secondary schools are giving that country some cause for concern

A The issue is the difference in pass rates between the sexes: at each level of the examination and across all school types, the difference is about 10 percentage points. Girls are doing better in every subject, and those in girls-only schools are taking top honours. The results are not a surprise: high school girls have been outperforming boys academically for more than a decade. It is an international phenomenon, and within Australia was the subject of much debate and controversy. Within New Zealand back in the 1980s, there was a concerted campaign, called 'Girls Can Do Anything', which was aimed at lifting girls' participation rates, achievement levels and aspirations. This was so successful that the pendulum has now swung to the other extreme. Views differ on how worried people should be. After all, for much of history, girls were excluded from any form of education, and this new phenomenon could be seen as a temporary over-correction before the balance is righted.

B However the New Zealand State Ministry of Education says it is taking the issue seriously. It is working with a reference group on boys' education which has been set up, and it has commissioned an Australian academic to report on interventions that have been found to work for boys, drawing particularly on Australia's experience. But some, such as former prison manager Celia Lashlie, the author of a book for parents of teenage boys, believe there is still resistance within the Education Ministry towards doing anything about the problem.

C Education Ministry learning policy manager Steve Benson says that the 'National Certificate in Educational Achievement', or NCEA, as New Zealand's high school exams are called, is useful to employers and to universities because it provides a fine-grained picture of pupils' performance in every aspect of a subject, rather than just a pass or fail in an overall area. 'In most parts of the curriculum, for example in maths, there isn't really a gender gap. But literacy is a different matter. Even boys who are good at writing tend not to write so much. There's actually a quantity issue.'

D The discrepancy in reading and writing skills between males and females shows up as early as preschool, and the difference is clear by the time these children enter high school. Not being good at literacy was not such a problem in the old days when many students left school for manual jobs after Year 11. But nowadays many more stay on to higher education, and almost all jobs require literacy skills. Roger Moses, the headmaster of Wellington

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College, says that the written content of NCEA papers is more demanding than the previous system of secondary school qualifications in New Zealand, even in subjects such as statistics and accounting.

E New Zealand 15-year-olds do well in international reading tests, but beneath this average lies a wide variance, with New Zealand European girls most represented at the top and New Zealand Pacific Island boys at the bottom. Yet some European girls drop out, and some Pacific Island boys excel. In other words, the range in performance within each gender group is much greater than the gender differences. Ethnic differences, and differences in socio-economic status, may be more significant than the simple boy/girl explanation.

F This makes the Education Ministry nervous about pushing solutions that emphasise stereotyped gender differences, rather than looking at under-achievement as a whole. Rob Burroughs, principal of Linwood High School in Christchurch, agrees. For three years his school ran separate boys' classes to try to address the disparity in performance, before abandoning them. The research showed that the boys did better in their own class than in the co-educational environment. But when he looked at which teachers they had, and how well those teachers' other classes did, it became clear that the difference was, instead, to do with the quality of instruction.

G At Onslow College, Dr Stuart Martin would do away with the NCEA Level 1 exam if he could. He says that in Year 11, aged 15, boys are simply not mature enough to cope. 'They tend to think that just passing is enough, and that it's not necessary to work hard for a Merit or an Excellence grade. Often they are busy with other activities and part-time jobs. Boys' competitive instinct tends to come out later in their schooling years, especially if there is money attached or other tangible rewards. By 17, boys are catching up academically with the girls, and by the end of Year 13, boys are again winning the top prizes.'

H Boys in single-sex schools do better in NCEA across levels, something economist Brian Easton reported after analysing data from the first year of NCEA's implementation. He said the results were valid, even when socio-economic status was taken into account. Dr Paul Baker, head of Waitaki Boys' High School in Oamaru, agrees. He thinks that although it is possible for all schools to do more to boost boys' performance, it is easier in a boys' school, where activities cannot be 'captured by girls'.

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Questions 14–16

Complete the summary below. Choose **NO MORE THAN TWO WORDS** *from the passage for each answer.*

High school assessment in New Zealand

New Zealanders are worried at the outcomes of their high school assessment system, because the (14) ———— of girls are higher than those of boys by 10%. A gender gap has been apparent for over a (15) ————. This situation is not unique to New Zealand, and has been noticed in (16) ———— also.

Questions 17–20

Reading Passage 2 has eight paragraphs, **A–H**. *Which paragraph contains the following information?*

Q	Statement	Answer
17	an advantage of New Zealand's secondary school tests	
18	a mention of current government initiatives to boost male achievement	
19	when gender difference in literacy skills first becomes evident	
20	findings that relate academic achievement to race	

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Questions 21–26

Look at the following people and the list of statements below. Match each person with the correct statement, **A–H**.

LIST OF STATEMENTS

A Boys gain lower marks on NCEA if they attend an all-boys' school.

B Boys are disadvantaged by girls tending to take over at school.

C Good teaching is more important than whether classrooms are single-sex or mixed.

D Mathematical skills were not so important in the past.

E The difference in achievement between school boys and girls is only evident in some subjects.

F Older boys are more motivated to study than younger boys.

G The NCEA exams have higher literacy standards than past exams did.

H The New Zealand government is reluctant to take action on behalf of boys.

Q	Person	Answer
21	Celia Lashlie	
22	Steve Benson	
23	Roger Moses	
24	Rob Burroughs	
25	Stuart Martin	
26	Paul Baker	

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Reading Passage 3

The Strange World of Sight

Seeing is believing, it is said. But, asks Richard Gregory, could it be the other way round?

Two of the great British men of the 17th century, the philosopher John Locke and the physicist Isaac Newton, were both aware that objects are not coloured, and that against all appearances light is not coloured either. This is still not generally recognised even now, 400 years later, because it seems so implausible. Yet it tells us something very important - that perceptions are not identical with what we perceive, and may be very different.

The most accurate historical account of perception is that of the 19th-century German scientist Hermann von Helmholtz. However, it was ridiculed at the time. Von Helmholtz thought that perceptions are unconscious inferences we make based on a combination of clues provided by the eyes and other senses, and knowledge of the world. This idea of unconscious inference for perception preceded, by several years, the psychoanalyst Freud's notion of the unconscious, which was also initially treated with derision because it undermined the notion of humans as pre-eminently rational beings who could be held responsible for their actions and awarded blame or praise accordingly.

Crucially, perception of the present depends on rich, though of course not always correct or appropriate, knowledge from the past. We interpret sense data (what we hear, touch, taste, see and smell) from the present according to what we already know. This raises the question: if we see the present through memory, why aren't past and present confused?

The pioneering Russian neurologist Alexander Luria described the case of Mr S, who had a remarkable memory. However, he was prone to just such confusions, for example mistaking seeing his clock for remembering it, and so failing to get up in the morning. This suggests that perhaps an important function of perception is to underline the present. Individual perceptions have a vividness that is rare for memories, which might be how we are able to separate them. Try this: look at something for a few seconds, and then shut your eyes and visualise it in memory. You will almost certainly find that the memory is pale by comparison with the perception. Perhaps this is why past and present are not normally confused. Luria's Mr S had exceptionally vivid memories, and rich synaesthesia (experiencing perceptions from another sense as well as the one being stimulated, such as musical notes experienced as colours), which may be why he confused seeing with having seen.

The complexity of processes involved in how we see first impressed itself on me 45 years ago. With my colleague Jean Wallace, I studied the rare case of Sydney Bradford, a man who had been born blind but, through a corneal graft at the age of 52, suddenly found himself able to see. Almost immediately after the operation he was able to 'see' but he

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could only see those things that he already knew about, having experienced them through touch. It was his touch memories that enabled him to perceive them with his eyes. When Bradford was first taken to the zoo, he proved utterly unable to see an elephant as he had no knowledge to make sense of his perceptions.

The more recent case in California of Mike May, who was also born blind, is similar. Since his operation, his sight has gradually improved as he learns to see, for example, by understanding how shadows represent depth and tell us about the shape of things. Some of the consequences of May's new-found vision were less happy. He had been a champion blind skier, but following the operation, he would have to shut his eyes while skiing to block out what he now found was a terrifying sight.

But acceptance of this intimate connection between memory and perception, even though it was first noticed in the 17th century, has been slow in brain science. Despite the fact that state-of-the-art brain imaging shows that perception animates parts of the brain associated with both present information and memory, most research on memory and perception is still undertaken as if these were separate processes. Seeing used to be thought of as taking place only in the eyes, and in quite specialised brain regions; but now it seems that half the brain is occupied with seeing, requiring a lot of energy. Perhaps this is why we shut our eyes for a rest.

It is not just extreme cases like Mike May, but also much more common errors of seeing - illusions - that can reveal the crucial role of memory in governing what we (think we) see. Perception depends on specific knowledge and probabilities. Our brains calculate the likelihood of what is out there, and when too far-fetched, perceptions are rejected.

A dramatic and discomforting example is looking at the two sides of a face-mask. From the front it is a convex shape with the nose sticking out. Then if the mask is rotated, the back of the mask will be seen as convex, though we know that it must be concave. It is almost, if not quite, impossible to sketch the back of a hollow mask to look as it is hollow. Science often learns from what does not happen: people not seeing a hollow face as hollow is the most revealing experiment on perception. The unsettling truth from brain science is that even people with no visual impairment see what, at some level, they expect to see, and often miss things as they really are.

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Questions 27–30

Choose the correct letter, **A, B, C** *or* **D**.

27 Why does the writer refer to Locke and Newton in the first paragraph?

A to indicate that his article will cover several scientific fields
B to stress how much physics has changed in 400 years
C to persuade the reader to take him seriously
D to point out that his notions are not new

28 According to the writer, why was Freud's theory of the unconscious mocked?

A It was too complex for his contemporaries to understand.
B It involved criticism of the way people behaved in society.
C People felt that it devalued the accepted concept of humanity.
D People assumed that it was intended as a joke.

29 The writer describes Mr S failing to get up in order to demonstrate

A how realistic most people's memories are.
B how hard it is to tell dreaming and waking apart.
C how unusual it is to mistake a perception for a memory.
D how valuable knowledge of the past can be.

30 What point is the writer making in the text as a whole?

A Perception involves much more than the data collected by the eyes.
B Learning to see as an adult can be a time-consuming process.
C Science is failing to devote enough attention to sight.
D Human perception is remarkably reliable.

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Questions 31–36

Do the following statements agree with the views of the writer in Reading Passage 3?

YES *if the statement agrees* • **NO** *if it contradicts* • **NOT GIVEN** *if there is no information*

31 Sydney Bradford relied on recollections of objects he had been told about to help him see after his operation.

32 People who only start to see as adults can learn to see as other people do in time.

33 People who have gained their sight as adults find certain activities harder to do than before.

34 It is evident now that sight involves the eyes and one particular area of the brain.

35 The mask experiment is particularly useful in training people who are regaining their sight.

36 People with perfect vision can fail to interpret objects correctly under certain circumstances.

Questions 37–40

Complete the summary using the list of words, **A–J**, *below.*

The mask experiment

A back **B** brain **C** view **D** convex **E** sight **F** nose **G** ground **H** hollow **I** drawing **J** preconception

In this experiment, having looked at the front of a simple face-mask, subjects look at the reverse. However, the subjects are convinced that they are still looking at a mask which is 37 ———— in shape. They believe that the 38 ———— is poking out in the normal manner because that is what they are used to seeing. Attempting to make a 39 ———— of the mask in this orientation leads to the same problem. The subjects fail to see a concave form because of the 40 ———— they have that the features of a face stick out.

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IELTS Academic Reading — Test 6

Volume 2 · 3 Passages · 40 Questions

Reading Passage 1

Why Should We Study History?

In this paper, I shall present a number of people's opinions on why we should study history. Let me point out at the outset that many of these views overlap, with differences that are ones of emphasis, rather than of principle.

According to some of the greatest historians, such as Arnold Toynbee, the essential value of history is its ability to give us self-awareness, self-knowledge. This occurs because historical figures are presented to us in detail. Studying Alexander the Great, for instance, teaches us that a single person is capable of the whole range of behaviour from cruelty to kindness.

Diane Ravich, an education policy police analyst, gives a practical reason for studying history. Like many others before her, she claims that history doesn't tell us the answers to our questions, but it helps to inform us and thus to improve our judgment, so that we make better choices in the future.

With a different focus, though, we can say that we know nothing about the future, and the present is just a moment, so in fact all there is is history. As the philosopher of education John Dewey suggested, the only tool we have for making sense of the reality we experience is what has occurred in the past.

According to historian Mark M Krug, the knowledge of how people acted in the past may not always suggest clever solutions to present crises, but it makes the task of finding them easier by giving us a starting point, a body of past experience which we can and, if necessary, reject.

Peter N Stearns argues that the wisdom available from history has a personal, everyday application. People need to have some sense of the workings of societies, simply so that they can run their own lives without coming into conflict with the community to which they belong.

The philosopher Friedrich Nietzsche said people need models to imitate, and historical models are particularly powerful because they actually existed. Through her contribution to ending English rule over France, Joan of Arc demonstrates the power of individual belief and action. Galileo's opposition to the prevailing belief that the sun and planets

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orbited the Earth symbolises the fight against authority for freedom of thought. When studying history, we judge the actions of historical figures to be admirable or despicable, good or bad, and choose who we want to emulate.

Some philosophers have identified a pattern in historical events. Both Georg Hegel and Oswald Spengler saw history as a dynamic process, one in which nothing is static, in contrast to the many other philosophers who have believed that history consists of unique events. But there the resemblance ends. In Hegel's view, history is a process of conflict between opposite ideas, leading towards greater freedom, which combines the best elements of the two original ideas. Spengler, on the other hand, developed the organic view that historical cultures, like plants and animals follow a process of growth, flowering and deterioration, a rise inevitably leading to a fall. There are very many historical examples of both individuals and empires that have conformed to this pattern.

Many historians argue that a society which has a shared cultural understanding of its own history is more likely to function smoothly than a society that is ignorant of its past. For example, Beverly Southgate claims that societies, like individuals, need to know who they are, and where they belong. Individuals who lack this sense of uniqueness are likely to feel that life has little meaning or purpose, and so too with societies. A shared knowledge of its history provides a nation with meaning, purpose and cohesion.

Southgate would agree with Conal Furay and Michael J Salevouris, in whose view a society that cannot recall its past is like an individual with amnesia, unaware of the roots that have created it. As a consequence, it has no direction, and this impacts on its further evolution. These two historians identify another value of history. We tend to have a narrow view of our own society, assuming that its ways are the only acceptable ones. By learning how our own country used to be, we may realise that our present-day customs are not the only ones which are acceptable. This clearly has implications for our attitude towards other present-day countries and cultures, too.

Many historians and educators believe that the expert knowledge possessed by historians includes not only the learning of factual information, but also the habit of critically examining and evaluating evidence. These are abilities worth developing, as they are very useful, both in other academic pursuits and in almost any career.

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Questions 1–4

Look at the following people and the list of theories of history below. Match each person with the correct theory, **A–F**.

LIST OF THEORIES OF HISTORY

A It helps us to put problems into perspective.

B It helps us to understand human nature.

C It helps us to understand the present.

D It provides examples to copy.

E It shows that there are alternative explanations of events.

F It helps to improve our ability to make decisions.

1 Arnold Toynbee

2 Diane Ravich

3 John Dewey

4 Friedrich Nietzsche

Questions 5–9

Complete the notes using the list of words, **A–J**, *below.*

A analysis **B** decline **C** peace **D** change **E** identity **F** pleasure **G** communication **H** liberty **I** tolerance **J** conflict

Hegel and Spengler: history is a process of 5 _____.

Hegel: historical process ends in an increase in 6 _____.

Spengler: historical process ends in 7 _____.

Southgate: history helps to create a nation's 8 _____.

Furay and Salevouris: study of history encourages 9 _____ of other cultures.

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Questions 10–13

Complete the sentences below. Choose **ONE WORD ONLY** *from the passage for each answer.*

10 Peter N Stearns believes that the study of history helps people to avoid 10 _____ with the society they live in.

11 Southgate compares societies with 11 _____.

12 Furay and Salevouris believe ignorance of history is like suffering from 12 _____.

13 One educational benefit of studying history is the ability to analyse 13 _____ .

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Reading Passage 2

Violins and Very Cold Weather

Lloyd Burckle and Henri D Grissino-Mayer suggest a link between the great Italian violin-makers and a mini ice-age 300 years ago

A There is considerable controversy surrounding reasons why instruments made by the artisans of Cremona in Italy in the late 17th and early 18th centuries sound superior compared to modern instruments. The famous violin-making families of Amati, Stradivari, and Guarneri certainly included many highly accomplished craftsmen, and the popular belief is that the skills of these Cremonese artisans, combined with either a secret ingredient or undocumented process, gave their instruments the rich sound so highly admired by professional musicians. The nature of that secret ingredient has been the subject of conjecture for many years and is still fuelling discussion. Theories have included the development and use of a specially formulated varnish applied to the wood to protect it, drying the wood in ovens, the re-use of old wood taken from castles, and soaking the wood in water to get rid of supposedly harmful chemicals before seasoning.

B However, no basis has been found for these possible explanations. There is no documented evidence of wood being artificially dried by violin-makers, and this process was most likely unknown to these artisans. Nor can seasoning — the practice of leaving wood for the moisture to evaporate naturally — be invoked as a possible explanation, as considerable variability exists in the lengths of seasoning periods, even among individual makers. Instruments made by the most renowned of all violin-makers, Antonio Stradivari (1644-1737), have seasoning periods as short as seven years and as long as 31 years, for example. A secret varnish has yet to be demonstrated, despite considerable scientific analyses that include ultraviolet photography, electron microscopy, ion backscattering and x-ray investigations. Finally, some have suggested that Stradivari used wood taken from castles. This theory has been discounted because analyses on numerous instruments attributed to Stradivari demonstrated ages for the wood that were contemporary with his lifetime.

C Violin-makers have always known that the secret of a quality instrument lies in the selection of the wood. Maple wood (Acer spp.) is preferred for the back, ribs, and neck of the instrument while spruce (Picea spp.) is often used for the top. These woods have superior acoustical and mechanical properties, having the least loss of energy through internal friction compared to other woods. For example, spruce cells are light, physiologically simple, hollow, and rigid — ideal properties for enhanced acoustic quality. Only trees that have grown in thin, poor soils at higher elevations on north-facing slopes

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have the dense wood grain that helps instill a superior tone in violins. Long winters and cool summers produce wood that has the slow, even growth ideal for producing quality sounding boards.

D Stradivari and other eminent Italian violin-makers of the 17th and early 18th centuries had neighbouring workshops in Cremona and would most probably have used the nearby high forest slopes of the southern Italian Alps for their supplies of spruce wood. A probable location is the famous "Forest of the Violins" in the Parco Naturale Paneveggio in the eastern part of Trentino, long familiar to violin-makers for its 'fir trees of resonance'.

E Wood with high density can, however, be found in trees growing in a variety of habitats and environmental settings in numerous higher-elevation locations throughout the world. Hence, an account of the Cremonese craftsmen's superior sound quality based solely on wood properties is insufficient. Instead, could the superior sound quality be explained by a combination of wood properties, environmental characteristics, and macroclimatic conditions? Could the wood used by the Cremonese makers have had particular physical characteristics that were perhaps time-specific? These were questions that required more thorough research.

F The well-documented Maunder Minimum (1645-1715) was a period characterized by a scarcity of sunspots and a reduction in overall solar activity. It coincided with a sharp dip in temperatures and a period of extremely cold weather in western Europe. Analyses of high-elevation forest areas of the European Alps reveal a long period of reduced growth rates between approximately 1625 and 1720, identified by the narrow rings visible in the cross-sections of tree samples from that era.

G It seems likely that the narrow tree rings that identify the Maunder Minimum in Europe played a significant role in the enhanced sound quality of instruments produced by the Cremonese craftsmen. Narrow rings would not only strengthen the violin, but lend the wood a high degree of density — the property so important for acoustic resonance. It is surely significant that Stradivari's working lifetime coincides exactly with this climatic period, and that to fashion the most prized and valued instruments of his 'Golden Period' (1700-1720), he would have used the only wood available to him, i.e. from trees that grew during the Maunder Minimum. The onset of the Maunder Minimum at a time when the skills of the Cremonese violin-makers reached their peak perhaps made that crucial difference in the violin's tone and brilliance. Furthermore, the conjunction of elevation, topography, soil properties and a deterioration in climate was temporally unique — climate conditions with temperatures such as those that occurred during the Maunder Minimum simply cannot and do not occur today in areas where the Cremonese makers obtained their wood.

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Questions 14–20

Reading Passage 2 has seven paragraphs, **A–G**. *Choose the correct heading for each paragraph from the list of headings below.*

LIST OF HEADINGS

i Propositions that needed further investigation
ii The most valuable instrument ever made
iii An unrepeated mix of professional and environmental circumstances
iv A remarkable woodworking technique
v An ongoing debate
vi Choosing the materials for different parts of a violin
vii Disproving some hypotheses
viii A well-known source of suitable wood
ix Evidence of some exceptional environmental conditions

Q	Paragraph	Answer
14	Paragraph A	
15	Paragraph B	
16	Paragraph C	
17	Paragraph D	
18	Paragraph E	
19	Paragraph F	
20	Paragraph G	

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Questions 21–24

Complete the summary below. Choose **ONE WORD ONLY** *from the passage for each answer.*

The Cremonese violins: fine artistry — but what else?

There is no doubt that the superior quality of the late 17th- and early 18th-century Cremonese violins is due in large part to the (21) _____ of the people who crafted them. But it has long been felt that other factors were involved. There has been extensive research into whether the composition of the varnish that was put on the wood could account for the unique sound of the Cremonese violins. Some people have also put forward the idea that Stradivari made his instruments from wood which had already been used in old buildings such as (22) _____. Scientific analysis has proved this not to be the case.

Other theories have focused on the violin-makers' method of using (23) _____ to dry their wood before use, or using water to remove dangerous (24) _____ from the wood. A further suggestion concerns the amount of time taken for the process known as seasoning. However, craftsmen differed in the time they allowed for this.

Questions 25 and 26

Choose **TWO** *letters,* **A–E**. *Which TWO of these factors are mentioned in the passage as possibly having a significant effect on the eventual quality of a violin?*

FACTORS

A the altitude at which a tree is grown
B the stage of a tree's growth at which it is cut down
C the steepness of the mountains where a tree is grown
D the lack of rain during a tree's growing period
E the nutritional composition of the earth in which a tree is grown

(25)

(26)

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Reading Passage 3

What Should Companies Do to Survive?

A In the world of business, where today's fashionable management theory soon becomes a stale cliché or a quaint archaism, it is essential to have a clear idea about the enduring rules of good management. The company that follows these is the company that is in the strongest position amid the uncertainties and violent upheavals of the world as we know it.

B A company is the sum of what its people understand and know how to do well. Value lies increasingly in creative ideas and knowledge. But ideas have value only if people share and develop them in ways that increase revenue; knowledge is useful only if people can find what they need to know. Getting intelligent people to share what is in their heads is vital, and takes more than mere money or clever software. Ideas must flow in every direction through a company — not merely from the top down. And knowledge is worth storing only if senior staff set careful rules to filter and structure it. What goes into a database determines the value of what comes out. So setting central rules and standards is key to good knowledge management.

C Good judgement is a key skill. Managers constantly blitzed with new information need to build in the data that matters and set aside the rest. Big-bang decisions are generally best avoided — or implemented in small incremental moves that leave room for flexibility and for altering course, if circumstances change.

D Customers matter — but some matter more than others. Acquiring new customers often costs more than making extra sales to existing ones. So companies must build loyalty by providing reliability and good service. Given the welter of product information reaching customers, memorable brands will grow more important. Companies need not only to widen their reach by finding new markets, but also to deepen existing relationships. They have more information than ever before about their customers, and must use this to offer their most profitable customers special deals and to make them feel part of an elite club. Some companies even seek to 'fire' unprofitable customers by charging them higher rates than others.

E Like its customers, some of a company's people matter more than others, and not just at the top: managing talent is also about capturing innovative ideas from middle managers and those further down the line. At every level, managers must identify where most value lies. In some cases, a few stars will encapsulate much of a company's value; in others, teams of employees will matter more. Some companies will want to hire the talents of 'free agents' as and when they are required; others, to employ directly their best brains.

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F Companies need to collaborate more, in alliances that allow them to outsource production or to spread risk or to enter new markets. Effective collaboration calls for trust and shared understanding, rather than the top-down, command-and-control approach of hierarchical structures. Successful collaboration also requires excellent communication, and incentives that reward sharing information and working for common goals. As costs of handling information in a company decline, new opportunities open for redefining corporate shape, and companies are becoming less hierarchical, with more ways to arrange and rearrange structure. Managers must think through from scratch which activities should be kept in-house and which outsourced, and normally a company should keep those activities it does better than its competitors.

G Given the pace of change, bosses need more than ever to be able to communicate persuasively through many channels, with their staff and the outside world. They must also listen: the most valuable communications will frequently be bottom-up, and the people nearest to the customer are best placed to explain what they see and hear. Managers must listen to them.

H Ironically, internet technologies, tools of freedom and decentralisation, call for discipline and standard processes. Only by setting standards and insisting that everyone abide by them will companies reap their potential savings. Companies need to insist on common practices in areas such as purchasing and information technology in order to harvest real productivity gains. As a result, some aspects of centralisation will increase: a key task of top managers is to provide structures and standards, and to insist that they are observed.

I Once standards have been set, openness and freedom should reign. Centralisation of standards makes possible decentralisation of decision making. In addition, internet technologies increase the need for a culture of openness, to foster the sharing of knowledge and effective collaboration. Increasingly, companies will allow their suppliers and customers 'inside the machine,' as it were, by giving them extraordinary access to their databases and inner workings in order to integrate their operations and to make collaboration effective.

J The key to success lies much less in technical know-how than in excellent leadership to push through and build upon organisational change. At some points in a company's life, it will need a leader who can rally staff to push through the trauma of disruptive change. At other times, the right style will be the manager-as-coach, a selfless talent scout who specialises in assembling and motivating great teams. Always, the people at the top will set the tone in a firm. Their skills will determine whether it is a good company to work in and do business with.

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Questions 27–34

Do the following statements agree with the views of the writer in Reading Passage 3?

YES *if the statement agrees* • **NO** *if it contradicts* • **NOT GIVEN** *if there is no information*

27 Too few companies understand how to evaluate fashionable management theories.

28 Certain practices should always be followed when coping with unstable situations.

29 It is important that the development of ideas should lead to improvements in the company's financial position.

30 The structuring of a database should be decided by all its users.

31 More training needs to be provided to help managers assess the usefulness of new information.

32 Any major change should be introduced all at once.

33 Companies can benefit from using different strategies with different customers.

34 The activities which are common to all companies are best kept in-house.

Questions 35–40

Reading Passage 3 has ten paragraphs, **A–J**. *Which paragraph contains the following information?*

Q	Statement	Answer
35	examples of qualities required when co-operating with other companies	
36	examples of business activities where all staff should follow the same practices	
37	examples of alternative employment patterns within a company	
38	a basis for deciding whether a company should carry out an activity itself or ask an outside company to do it	
39	an example of how information can be shared by a company and its customers	
40	a reason why customers might continue to buy from the same companies	

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IELTS Academic Reading — Test 7

Volume 2 · 3 Passages · 40 Questions

Reading Passage 1

Book Review: Triumph of the City

Triumph of the City, by Edward Glaeser, is a thrilling and very readable hymn of praise to an invention so vast and so effective that it is generally taken for granted. More than half the global population already live in urban areas and, every month, five million more flood into the cities of the developed and developing worlds. The crowds and poverty of some of these modern cities may horrify us. They shouldn't, says Glaeser — they are signs of growth, energy and aspiration. Cities are our best and brightest hope.

This idea has had more than two hundred years of resistance. Not long after the Industrial Revolution began in Britain, the Romantic poets turned away from the smoke and factories of their cities to celebrate the air and light of untouched nature. In 19th-century America, the writer Henry David Thoreau retreated to the wilderness of Walden Pond to live the solitary, simple life, and convinced generations of Americans that cities were bad and nature was good.

They had, Glaeser admits, a point. The early industrial cities were dirty, since they lacked efficient waste disposal systems, and disease spread rapidly among the population. But more importantly they were profitable, and there were enormous commercial incentives to make them work, as well as political ones. Their transformation could be achieved at a stroke: in the second half of the 19th century, the French Emperor Napoleon III gave Baron Haussmann unrestricted power to turn the slum-infested city of Paris into one of the wonders and delights of the modern world. Or the transformation could be done by trial and error. Glaeser gives a brilliant account of the stop-start progression of New York to its late 20th-century position as the cultural and economic centre of the world. Either way, Paris, New York and other cities developed because they were truly effective markets of ideas and innovation.

For these and many other reasons, we should not be so upset by the spectacle of urban poverty. The poor flock to cities in the hope of becoming richer (which, by and large, they do). They also reinvigorate the economy of the city. It is folly to drive them away by forcing property prices to soar with unreasonable planning regulations. Instead, cities should build more houses and thereby hold property prices in check.

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It can go wrong, of course. In Glaeser's view, this is primarily because municipal authorities fail to understand the principal virtues of their cities. The heart of Paris, as many Parisians say, is turning into a museum because of the desire to preserve Baron Haussmann's 19th-century boulevards. Glaeser defends their preservation, but argues that in the 1950s the French made a mistake in establishing a huge high-rise commercial development — La Défense — on the outskirts of the city. Far better, he says, to have turned the central area of Montparnasse into a new commercial district. This would have revitalised much of the city centre without destroying its fabric. In India, Mumbai could save itself from ever-more inefficient sprawl over the surrounding area simply by relaxing the rules presently imposed on the height of new constructions.

In America, it is the suburbs that have proved to be the real disaster. Glaeser is repentant on this subject himself. He moved to the suburbs when he had children. His entirely legitimate excuse is that the government made him (and millions like him) do it. By under-taxing petrol and imposing tight planning restrictions on inner cities that drove up the cost of property, it made flight to the suburbs more or less inevitable for the middle classes. This is a disaster because nothing is more inefficient than a suburb. Suburbanites mingle less, and lose the face-to-face contact that makes being an urbanite so much more creative. Moreover, houses are costlier to heat and cool than flats, and suburbanites drive thousands more miles per year than city dwellers. Every aspect of life involves more consumption. This leads to the strongest and newest argument in favour of cities — they are good for the environment. To live in the country or the suburbs is to have a vastly larger carbon footprint than any urbanite.

Full of characters and accessible information, this is a tremendous book, not least because, like me, you will find yourself constantly seeking reasons to disagree. Like the poor in the city, this is a sign of success. If you hate the city and get moist-eyed at the thought of the country then, one way or another, Glaeser is the man you will have to take on.

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Questions 1–8

Complete the notes below. Choose **ONE WORD ONLY** *from the passage for each answer.*

Cities

PROBLEMS WITH EARLY CITIES

- dirt
- (1) _____
- but there were commercial and (2) _____ reasons for improving them

URBAN POVERTY IS NOT A MAJOR PROBLEM BECAUSE POOR PEOPLE

- generally get (3) _____
- help to develop the urban (4) _____

CITIES DO HAVE SOME PROBLEMS — E.G.

- the centre of Paris is becoming a (5) _____
- Mumbai is negatively affected by height restrictions on new buildings
- in the US, the middle classes have moved to the suburbs due to: cheap petrol, high (6) _____ prices in inner cities

DISADVANTAGES OF SUBURBS

- less personal (7) _____
- increased (8) _____ of resources such as heating — which damages the environment

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Questions 9–13

Do the following statements agree with the information given in Reading Passage 1?

TRUE *if the statement agrees* • **FALSE** *if the statement contradicts* • **NOT GIVEN** *if there is no information*

9 Glaeser believes that congestion and poverty in some modern cities indicate serious problems.

10 The writer Henry David Thoreau discussed the ideas of the Romantic poets in his work.

11 Emperor Napoleon III was influenced by the complaints of poor people living in Paris.

12 Strict planning regulations may be beneficial for a city's development.

13 Glaeser argues that the location of a commercial development at La Défense was a bad idea.

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Reading Passage 2

Coral Reefs

Coral reefs are underwater structures made from calcium carbonate secreted by corals. Coral reefs are colonies of tiny living animals found in marine waters that contain few nutrients. Most coral reefs are built from stony corals, which in turn consist of polyps that cluster in groups.

A Coral reefs are estimated to cover 284,300 km² — just under 0.1% of the ocean's surface area, about half the area of France. The Indo-Pacific region accounts for 91.9% of this total area. Southeast Asia accounts for 32.3% of that figure, while the Pacific including Australia accounts for 40.8%. Atlantic and Caribbean coral reefs account for 7.6%. Yet often called "rainforests of the sea", coral reefs form some of the most diverse ecosystems on Earth. They provide a home for 25% of all marine species, including fish, mollusks, worms, crustaceans, echinoderms, sponges, tunicates and other cnidarians. Paradoxically, coral reefs flourish even though they are surrounded by ocean waters that provide few nutrients. They are most commonly found at shallow depths in tropical waters, but deep water and cold water corals also exist on smaller scales in other areas. Although corals exist both in temperate and tropical waters, shallow-water reefs form only in a zone extending from 30°N to 30°S of the equator. Deep water coral can exist at greater depths and colder temperatures at much higher latitudes, as far north as Norway. Coral reefs are rare along the American and African west coasts. This is due primarily to upwelling and strong cold coastal currents that reduce water temperatures in these areas (respectively the Peru, Benguela and Canary streams). Corals are seldom found along the coastline of South Asia from the eastern tip of India (Madras) to the Bangladesh and Myanmar borders. They are also rare along the coast around northeastern South America and Bangladesh due to the freshwater released from the Amazon and Ganges Rivers, respectively.

B Coral reefs deliver ecosystem services to tourism, fisheries and coastline protection. The global economic value of coral reefs has been estimated at as much as $US375 billion per year. Coral reefs protect shorelines by absorbing wave energy, and many small islands would not exist without their reef to protect them.

C The value of reefs in biodiverse regions can be even higher. In parts of Indonesia and the Caribbean where tourism is the main use, reefs are estimated to be worth US$1 million per square kilometer, based on the cost of maintaining sandy beaches and the value of attracting snorkelers and scuba divers. Meanwhile, a recent study of the Great Barrier Reef in Australia found that the reef is worth more to the country as an intact ecosystem than an extractive reserve for fishing. Each year more than 1.8 million tourists visit the reef, spending an estimated AU$4.3 billion on reef-related industries from diving to boat rental

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to posh island resort stays. In the Caribbean, says UNEP, the net annual benefits from diver tourism was US$2 billion in 2000 with US$625 million spent directly on diving on reefs. Further, reef tourism is an important source of employment, especially for some of the world's poorest people. UNEP says that of the estimated 30 million small-scale fishers in the developing world, most are dependent to a greater or lesser extent on coral reefs. In the Philippines, for example, more than one million small-scale fishers depend directly on coral reefs for their livelihoods. The report estimates that reef fisheries were worth between $15,000 and $150,000 per square kilometer a year, while fish caught for aquariums were worth $500 a kilogram against $6 for fish caught as food. The aquarium fish export industry supports around 50,000 people and generates some US$5.5 million a year in Sri Lanka alone.

D Unfortunately, coral reefs are dying around the world. In particular, coral mining, agricultural and urban runoff, pollution (organic and inorganic), disease, and the digging of canals and access into islands and bays are localized threats to coral ecosystems. Broader threats are sea temperature rise, sea level rise and pH changes from ocean acidification, all associated with greenhouse gas emissions. Some current fishing practices are destructive and unsustainable. These include cyanide fishing, overfishing and blast fishing. Although cyanide fishing supplies live reef fish for the tropical aquarium market, most fish caught using this method are sold in restaurants, primarily in Asia, where live fish are prized for their freshness. To catch fish with cyanide, fishers dive down to the reef and squirt cyanide in coral crevices and on the fast-moving fish, to stun the fish making them easy to catch. Overfishing is another leading cause for coral reef degradation. Often, too many fish are taken from one reef to sustain a population in that area. Poor fishing practices, such as banging on the reef with sticks (muro-ami), destroy coral formations that normally function as fish habitat. In some instances, people fish with explosives (blast fishing), which blast apart the surrounding coral.

E Tourist resorts that empty their sewage directly into the water surrounding coral reefs contribute to coral reef degradation. Wastes kept in poorly maintained septic tanks can also leak into surrounding ground water, eventually seeping out to the reefs. Careless boating, diving, snorkeling and fishing can also damage coral reefs. Whenever people grab, kick, and walk on, or stir up sediment in the reefs, they contribute to coral reef destruction. Corals are also harmed or killed when people drop anchors on them or when people collect coral.

F To find answers for these problems, scientists and researchers study the various factors that impact reefs. The list includes the ocean's role as a carbon dioxide sink, atmospheric changes, ultraviolet light, ocean acidification, viruses, impacts of dust storms carrying agents to far flung reefs, pollutants, algal blooms and others. Reefs are threatened well beyond coastal areas. General estimates show approximately 10% of the world's coral reefs

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are dead. About 60% of the world's reefs are at risk due to destructive, human-related activities. The threat to the health of reefs is particularly strong in Southeast Asia, where 80% of reefs are endangered.

G In Australia, the Great Barrier Reef is protected by the Great Barrier Reef Marine Park Authority, and is the subject of much legislation, including a biodiversity action plan. Inhabitants of Ahus Island, Manus Province, Papua New Guinea, have followed a generations-old practice of restricting fishing in six areas of their reef lagoon. Their cultural traditions allow line fishing, but not net or spear fishing. The result is both the biomass and individual fish sizes are significantly larger in these areas than in places where fishing is unrestricted.

Questions 14–19

Reading Passage 2 has seven paragraphs, **A–G**. *Which paragraph contains the following information?*

Q	Statement	Answer
14	Geographical location of world's coral reef	
15	How coral reef benefits the local economy	
16	The statistics of coral reefs' economic significance	
17	The listed reasons for the declining number of coral reefs	
18	Physical approach to coral reef by people	
19	Unsustainable fishing methods applied in regions of the world	

Questions 20–25

Do the following statements agree with the information given in Reading Passage 2?

YES *if the statement agrees* • **NO** *if it contradicts* • **NOT GIVEN** *if there is no information*

20 Coral reefs provide habitat to a variety of marine life.

21 Coral reefs distribute around the ocean disproportionally.

22 Coral reefs are increasingly important for scientific purposes.

23 Coral reefs are greatly exchanged among and exported to other countries.

24 Reef tourism is of economic essence generally for some poor people.

25 As with other fishing businesses, coral fishery is not suitable for women and children.

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Question 26

Choose the correct letter, **A, B, C** *or* **D**.

26 What is the main purpose of this passage?

A Demonstrate how coral reef grow in the ocean

B To tell that coral reef is widely used as a scientific project

C Present the general benefits and an alarming situation of coral reef

D To show the vital efforts made to protect coral reef in Australia

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Reading Passage 3

Whale Culture

A Most social scientists stubbornly resist the idea that animals have culture. Even such advanced cetacean mammals as whales and dolphins clearly don't have art, literature, or architecture. But patient observation over many years has begun to reveal behaviours that can only have been learnt from other whales. And that, say whale biologists, constitutes culture.

B So far, humpback and killer whales provide the best evidence of culture in cetaceans, and the song of the male humpback is among the most striking examples. Humpback populations in different oceans sing different songs, but within the same ocean they all stick to the same one. However, during the breeding season the sounds change, as it appears that females are drawn to novel songs. One male might add an extra set of groans; another might drop a series of grunts. Soon all the other males have altered their own rendition to incorporate the changes until they are once again singing the same song. Since this occurs among thousands of whales spread across a vast part of the planet, the change cannot be in response to any factor in the animals' environment. The latest version of the song can be learnt only from other whales — almost certainly by imitation.

C Culture plays an even bigger part in the life of killer whales. Nowhere is this more obvious than along the north-west coast of America, where killer whales are split into two distinct populations — 'residents' and 'transients'. They live in the same stretch of water, but they don't mingle. In effect, they belong to two quite separate cultures. Residents live in stable groups, or 'pods', made up of two or three mothers and their offspring — perhaps 20 whales in all. Calves stay with their mothers throughout adulthood, and in many years of observation no one has ever seen a whale switch pods. Transients travel in smaller, more changeable groups of between three and six.

D One of the most obvious distinctions between the transient and resident societies is the way they impart information. Killer whales detect prey with a range of echo-locating clicks, but converse with a vocabulary of squeaks, whistles and whines. Transients have only a few such calls, and all transient societies share the same ones. Residents have a much more extensive repertoire, and each family group has its own unique and distinctive set of calls. Despite regular interaction between them, each resident pod sticks firmly to its own dialect. Research shows these dialects are maintained for at least 40 years.

E To qualify as part of killer whale culture, dialects must be learnt from other members of the pod. Animals with different dialects share the same waters, so the variation can't be a product of the physical environment. 'And we can throw out the notion that the dialects are inherited,' says Lance Barrett-Lennard of the University of British Columbia. He has spent

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the past seven years analysing DNA from 270 whales. His paternity tests reveal that female killer whales invariably attract mates from outside their own pod — males with a very different dialect. If dialects were programmed by genetics, call patterns from both father and mother would be passed on to the calf. 'A calf uses the calls of its maternal pod very precisely. There's no input from the father,' says Barrett-Lennard.

F The question still remains — is this culture? It is, according to Frans de Waal of Emory University in Atlanta, who argues that culture is just another biological adaptation that has evolved in many creatures. One benefit of viewing culture in this way is that you can start to understand how and why it might have arisen in these creatures. Whales have several biological attributes that give them an advantage in social learning. Apart from their advanced mental abilities, they are adept at recognising sounds: ideal for communicating in the marine environment. Many species spend years rearing their offspring, and live in small, stable, multi-generational societies, a social system that provides ample opportunity for teaching and learning.

G But why have cetaceans evolved the ability to learn from other group members? Experts in whale biology believe that ecological factors and the need to adapt to sudden changes in the environment played a large part in the emergence of culture. Although the ocean is a relatively stable habitat in many ways, it is highly changeable in one crucial respect — the availability of food. One moment there might be a plentiful supply of fish, the next they've disappeared. When that happens, the past experience of the senior members of the group and the ability to share this knowledge is a huge asset. The dialects of killer whales allow members of the group to identify each other, enabling them to share information about food hot spots. Among resident killer whales, it also allows females to avoid inbreeding by picking out a mate with a strange dialect from outside their pod, says Barrett-Lennard.

H The importance of sharing information seems to have led to biological changes in at least some species of whale. Female killer whales, like humans, are very unusual in that they live up to a quarter of a century after they have had their last offspring. This only makes sense if they have something useful to give their descendants. And what whale matriarchs offer is the most important thing of all — cultural knowledge, vital for the group's survival, passed directly from one generation to the next.

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Questions 27–31

Do the following statements agree with the information given in Reading Passage 3?

TRUE *if the statement agrees* • **FALSE** *if the statement contradicts* • **NOT GIVEN** *if there is no information*

27 Resident killer whales appear to remain with their maternal group for life.

28 Resident killer whales have a more restricted range of calls than transients.

29 There is a vocabulary of sounds which is common to all transient killer whales.

30 Resident killer whales share the dialects of other resident communities living in the same waters.

31 The dialects of transient killer whales remain constant over time.

Questions 32–34

Complete the summary below. Choose **NO MORE THAN TWO WORDS** *from the passage for each answer.*

It has been observed that resident killer whales invariably live in fixed family groups, known as 32 ————————. Each of these has its own unique set of calls, despite close contact with other family groups. As the same areas of ocean contain many different groups with widely varying dialects, it is clear that these differences could not have emerged as a result of the whales' 33 ————————.

According to tests conducted by Lance Barrett-Lennard, a calf communicates exclusively with the dialect of the group to which its 34 ———————— belongs. Barrett-Lennard also rejects the idea that the call patterns are inherited.

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Questions 35–37

Choose **THREE** *letters,* **A–F**. *Which THREE of the following features of whales are mentioned in the passage?*

FEATURES

A intelligence

B physical strength

C sensitivity to sound

D prolonged life span

E lengthy period of fertility

F adaptability to a variety of foods

35

36

37

Questions 38–40

Reading Passage 3 has eight paragraphs, **A–H**. *Which paragraph contains the following information?*

Q	Statement	Answer
38	an example of the kind of information passed by whales to each other	
39	a reference to variations in communication styles between different cultures within one species	
40	ways in which the skills of whales are favourable for the development of culture	

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IELTS Academic Reading — Test 8

Volume 2 · 3 Passages · 40 Questions

Reading Passage 1

The Iceman

A It was late spring or early summer. The man hurried through a forest he knew well, wincing from the pain in his injured right hand and pausing occasionally to listen for sounds that he was being pursued. As he fled up the slope, the yellow pollen of the hornbeam blossoms fell like an invisible rain, salting the water and food he consumed when he stopped to rest. Five thousand years later, the Neolithic hunter we call the Iceman would still bear traces of this ancient dusting inside his body — a microscopic record of the time of year it was when he passed through this forest and into the nearby mountains, where fate would finally catch up with him.

B Since hikers discovered his mummified corpse in 1991 in a rocky hollow high in the Otztal Alps on Italy's border with Austria, scientists have used ever more sophisticated tools and intellectual cunning to reconstruct the life and times of the Iceman, the oldest intact member of the human family. We know that he was a small, sinewy, and, for his times, rather elderly man in his mid-40s. Judging from the precious, copper-bladed ax found with him, we suspect that he was a person of considerable social significance. He set off on his journey wearing three layers of garments and sturdy shoes with bearskin soles. He was well equipped with a flint-tipped dagger, a little fire-starting kit, and a birchbark container holding embers wrapped in maple leaves. Yet he also headed into a harsh wilderness curiously under-armed: the arrows in his deerskin quiver were only half finished, as if he had recently fired all his munitions and was in the process of hastily replenishing them. And he was traveling with a long, roughly shaped stalk of yew — an unfinished longbow, yet to be notched and strung. Why?

C When it comes to the Iceman, there has never been a shortage of questions, or theories to answer them. During the 16 years that scientists have poked, prodded, incised, and x-rayed his body, they have dressed him up in speculations that have not worn nearly as well as his rustic garments. At one time or another, he has been mistakenly described as a lost shepherd, a shaman, a victim of ritual sacrifice, and even a vegan. But all these theories fade in the face of the most startling new fact scientists have learned about the Iceman. Although we still don't know exactly what happened up there on that alpine ridge, we now know that he was murdered, and died very quickly, in the rocky hollow where his body was found.

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D "Even five years ago, the story was that he fled up there and walked around in the snow and probably died of exposure," said Klaus Oeggl, an archaeobotanist at the University of Innsbruck. "Now it's all changed. It's more like a paleo crime scene."

E The object of all this intense scientific attention is a freeze-dried slab of human jerky, which since 1998 has resided in a refrigerated, high-tech chamber in the South Tyrol Museum of Archaeology in Bolzano, Italy. The temptation to conduct fresh experiments on the body rises with every new twist of technology, each revealing uncannily precise details about his life. Using a sophisticated analysis of isotopes in one of the Iceman's teeth, for example, scientists led by Wolfgang Muller (now at the Royal Holloway, University of London) have shown that he probably grew up in the Valle Isarco, an extensive north-south valley that includes the modern-day town of Bressanone. Isotope levels in his bones, meanwhile, match those in the soil and water of two alpine valleys farther west, the Val Senales and the Val Venosta. Muller's team has also analyzed microscopic chips of mica recovered from the Iceman's intestines, which were probably ingested accidentally in food made from stone-ground grain; geologic ages of the mica best match a small area limited to the lower Val Venosta. The Iceman probably set off on his final journey from this very area, near where the modern-day Adige and Senales Rivers meet.

F We also know that he was not in good health when he headed up into the mountains. The one surviving fingernail recovered from his remains suggests that he suffered three episodes of significant disease during the last six months of life, the last bout only two months prior to his death. Doctors inspecting the contents of his intestines have found eggs of the whipworm parasite, so he may well have suffered from stomach distress. But he was not too sick to eat. In 2002, Franco Rollo and colleagues at the University of Camerino in Italy analyzed tiny amounts of food residue from the mummy's intestines. A day or two before his death, the Iceman had eaten a piece of wild goat and some plant food.

G Archaeobotanists have used equally clever analyses of pollen and plant fragments to plot the Iceman's last movements. James Dickson of the University of Glasgow has identified no less than 80 distinct species of mosses and liverworts in, on, or near the Iceman's body. The most prominent moss, Neckera complanata, still grows at several sites in the valleys to the south, in some cases quite near known prehistoric sites. According to Dickson, a clot of stems found in the Iceman's possession suggests he was probably using the moss to wrap food, although other ancient peoples used similar mosses as toilet paper.

H Taken together, the evidence strongly indicates that the Iceman's last journey began in the low-altitude deciduous forests to the south, in the springtime when the hop hornbeams were in bloom. But it may not have been a straight hike into the mountains. Oeggl has also found traces of pine pollen in the Iceman's digestive tract, both above and below the hornbeam pollen. This suggests that he may have climbed to a higher altitude where pine

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trees grow in mixed coniferous forests, then descended to the lower altitude of the hop hornbeams, and finally ascended again into the pine forests in his last day or two. Why? No one knows. But perhaps he wanted to avoid the steep, thickly wooded gorge of the lower Val Senales — especially if he was in a hurry.

Questions 1–5

Reading Passage 1 has eight paragraphs, **A–H**. *Which paragraph contains the following information?*

Q	Statement	Answer
1	the last area in which the Iceman might live and stay	
2	a mass of special plant was discovered and used to analyze the Iceman's movements	
3	a scientist analyzes the Iceman's last hike depending on pollen	
4	the time and area the Iceman was found	
5	the Iceman's body had been out of condition for months before his death	

Questions 6–9

Do the following statements agree with the information in Reading Passage 1?

TRUE *if the statement agrees* • **FALSE** *if the statement contradicts* • **NOT GIVEN** *if there is no information*

6 According to the author, there must be another complete human corpse older than the Iceman.

7 The Iceman might be the leader of his society, and he was very rich.

8 Scientists guessed the Iceman's information perfectly, and finally got the real cause of his death.

9 By testing the Iceman's body, we know where he came from.

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Questions 10–13

Complete the sentences below. Choose **NO MORE THAN TWO WORDS** *from the passage for each answer.*

10 The Iceman has been placed in a 10 _____ room since 1998.

11 The Iceman might get 11 _____ for eggs of the whipworm parasite were found in his gut.

12 There are a variety of mosses and liverworts found around the Iceman such as 12 _____.

13 The route of the Iceman's last movement might not be 13 _____.

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Reading Passage 2

Twin Study: Two of a Kind

A THE scientific study of twins goes back to the late 19th century, when Francis Galton, an early geneticist, realized that they came in two varieties: identical twins born from one egg and non-identical twins that had come from two. That insight turned out to be key, although it was not until 1924 that it was used to formulate what is known as the twin rule of pathology, and twin studies really got going.

B The twin rule of pathology states that any heritable disease will be more concordant (that is, more likely to be jointly present or absent) in identical twins than in non-identical twins — and, in turn, will be more concordant in non-identical twins than in non-siblings. Early work, for example, showed that the statistical correlation of skin-mole counts between identical twins was 0.4, while non-identical twins had a correlation of only 0.2. (A score of 1.0 implies perfect correlation, while a score of zero implies no correlation.) This result suggests that moles are heritable, but it also implies that there is an environmental component to the development of moles, otherwise the correlation in identical twins would be close to 1.0.

C Twin research has shown that whether or not someone takes up smoking is determined mainly by environmental factors, but once he does so, how much he smokes is largely down to his genes. And while a person's religion is clearly a cultural attribute, there is a strong genetic component to religious fundamentalism. Twin studies are also unraveling the heritability of various aspects of human personality. Traits from neuroticism and anxiety to thrill- and novelty-seeking all have large genetic components. Parenting matters, but it does not determine personality in the way that some had thought.

D More importantly, perhaps, twin studies are helping the understanding of diseases such as cancer, asthma, osteoporosis, arthritis and immune disorders. And twins can be used, within ethical limits, for medical experiments. A study that administered vitamin C to one twin and a placebo to the other found that it had no effect on the common cold. The lesson from all today's twin studies is that most human traits are at least partially influenced by genes. However, for the most part, the age-old dichotomy between nature and nurture is not very useful. Many genetic programs are open to input from the environment, and genes are frequently switched on or off by environmental signals. It is also possible that genes themselves influence their environment. Some humans have an innate preference for participation in sports. Others are drawn to novelty. Might people also be drawn to certain kinds of friends and types of experience? In this way, a person's genes might shape the environment they act in as much as the environment shapes the actions of the genes.

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E In the past, such research has been controversial. Josef Mengele, a Nazi doctor working at the Auschwitz extermination camp during the second world war, was fascinated by twins. He sought them out among arrivals at the camp and preserved them from the gas-chambers for a series of brutal experiments. After the war, Cyril Burt, a British psychologist who worked on the heredity of intelligence, tainted twin research with results that appear, in retrospect, to have been rather too good. Some of his data on identical twins who had been reared apart were probably faked. In any case, the prevailing ideology in the social sciences after the war was Marxist, and disliked suggestions that differences in human potential might have underlying genetic causes. Twin studies were thus viewed with suspicion.

F The ideological pendulum has swung back, however, as the human genome project and its aftermath have turned genes from abstract concepts to real pieces of DNA. The role of genes in sensitive areas such as intelligence is acknowledged by all but a few die-hards. The interesting questions now concern how nature and nurture interact to produce particular bits of biology, rather than which of the two is more important. Twin studies, which are a good way to ask these questions, are back in fashion, and many twins are enthusiastic participants in this research.

G Research at the Twinsburg festival began in a small way, with a single stand in 1979. Gradually, news spread, and more scientists began turning up. This year, half a dozen groups of researchers were lodged in a specially pitched research tent. In one corner of this tent, Paul Breslin, who works at the Monell Institute in Philadelphia, watched over several tables where twins sat sipping clear liquids from cups and making notes. It was the team's third year at Twinsburg. Dr. Breslin and his colleagues want to find out how genes influence human perception, particularly the senses of smell and taste and those (warmth, cold, pain, tingle, itch and so on) that result from stimulation of the skin. Perception is an example of something that is probably influenced by both genes and experience. Even before birth, people are exposed to flavors such as chocolate, garlic, mint and vanilla that pass intact into the bloodstream, and thus to the fetus. Though it is not yet clear whether such pre-natal exposure shapes taste-perception, there is evidence that it shapes preferences for foods encountered later in life.

H However, there are clearly genetic influences at work, as well — for example in the ability to taste quinine. Some people experience this as intensely bitter, even when it is present at very low levels. Others, whose genetic endowment is different, are less bothered by it. Twin studies make this extremely clear. Within a pair of identical twins, either both, or neither, will find quinine hard to swallow. Non-identical twins will agree less frequently.

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I On the other side of the tent Dennis Drayna, from the National Institute on Deafness and Other Communication Disorders, in Maryland, was studying hearing. He wants to know what happens to sounds after they reach the ear. It is not clear, he says, whether sound is processed into sensation mostly in the ear or in the brain. Dr. Drayna has already been involved in a twin study which revealed that the perception of musical pitch is highly heritable. At Twinsburg, he is playing different words, or parts of words, into the left and right ears of his twinned volunteers. The composite of the two sounds that an individual reports hearing depends on how he processes this diverse information — and that, Dr. Drayna believes, may well be influenced by genetics.

J Elsewhere in the marquee, Peter Miraldi, of Kent State University in Ohio, was trying to find out whether genes affect an individual's motivation to communicate with others. A number of twin studies have shown that personality and sociability are heritable, so he thinks this is fertile ground. And next to Mr. Miraldi was a team of dermatologists from Case Western Reserve University in Cleveland. They are looking at the development of skin diseases and male-pattern baldness. The goal of the latter piece of research is to find the genes responsible for making men's hair fall out.

K The busiest part of the tent, however, was the queue for forensic science research into fingerprints. The origins of this study are shrouded in mystery. For many months, the festival's organizers have been convinced that the Secret Service — the American government agency responsible for, among other things, the safety of the president — is behind it. When The Economist contacted the Secret Service for more information, we were referred to Steve Nash, who is chairman of the International Association for Identification (IAI), and is also a detective in the scientific investigations section of the Marin County Sheriff's Office in California. The IAI, based in Minnesota, is an organization of forensic scientists from around the world. Among other things, it publishes the Journal of Forensic Identification.

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Questions 14–18

Reading Passage 2 has eleven paragraphs, **A–K**. *Which paragraph contains the following information?*

Q	Statement	Answer
14	Mentioned research conducted in Ohio	
15	Medical contribution to the research for twins	
16	Research situation under life-threatening conditions	
17	Data of similarities of identical twins	
18	Reasons that make one study unconvincing	

Questions 19 and 20

Complete the summary below. Choose **NO MORE THAN TWO WORDS** *from the passage for each answer.*

The first one that conducted research on twins is called (19) _____. He separated twins into two categories: non-identical and identical twins. The twin research was used in medical application as early as the year of (20) _____.

Questions 21–23

Choose **THREE** *research fields that had been carried out in Ohio, Maryland and Twinsburg.*

RESEARCH FIELDS

A Sense
B Cancer
C Allergy to Vitamin D
D Mole heredity
E Sound
F Baldness of men

(21)

(22)

(23)

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Questions 24–26

Choose **THREE** *results that had been verified in this passage.*

RESULTS

A Non-identical twins come from different eggs.

B Genetic relation between identical twins is closer than non-identical ones.

C Vitamin C has an evident effect on a cold.

D Genetic influence on smoking is superior to environment's.

E If a pregnant woman eats too much sweet food, it would lead to skin disease.

F Hair loss has been found to be connected with skin problems.

24

25

26

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Reading Passage 3

The Origin of Ancient Writing

A The Sumerians, an ancient people of the Middle East, had a story explaining the invention of writing more than 5,000 years ago. It seems a messenger of the King of Uruk arrived at the court of a distant ruler so exhausted that he was unable to deliver the oracle message. So the king set down the words of his next messages on a clay tablet. A charming story, whose retelling at a recent symposium at the University of Pennsylvania amused scholars. They smiled at the absurdity of a letter which the recipient would not have been able to read.

B They also doubted that the earliest writing was a direct rendering of speech. Writing more likely began as a separate, symbolic system of communication and only later merged with spoken language.

C Yet in the story the Sumerians, who lived in Mesopotamia, in what is now southern Iraq, seemed to understand writing's transforming function. As Dr. Holly Pittman, director of the University's Center for Ancient Studies, observed, writing 'arose out of the need to store and transmit information ... over time and space'.

D In exchanging interpretations and information, the scholars acknowledged that they still had no fully satisfying answers to the questions of how and why writing developed. Many favored an explanation of writing's origins in the visual arts, pictures becoming increasingly abstract and eventually representing spoken words. Their views clashed with a widely held theory among archaeologists that writing developed from the pieces of clay that Sumerian accountants used as tokens to keep track of goods.

E Archaeologists generally concede that they have no definitive answer to the question of whether writing was invented only once, or arose independently in several places, such as Egypt, the Indus Valley, China, Mexico and Central America. The preponderance of archaeological data shows that the urbanizing Sumerians were the first to develop writing, in 3,200 or 3,300 BC. These are the dates for many clay tablets in an early form of cuneiform, a script written by pressing the end of a sharpened stick into wet clay, found at the site of the ancient city of Uruk. The baked clay tablets bore such images as pictorial symbols of the names of people, places and things connected with government and commerce. The Sumerian script gradually evolved from the pictorial to the abstract, but did not at first represent recorded spoken language.

F Dr. Peter Damerow, a specialist in Sumerian cuneiform at the Max Planck Institute for the History of Science in Berlin, said, 'It is likely that there were mutual influences of writing systems around the world. However, their great variety now shows that the development of writing, once initiated, attains a considerable degree of independence and

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flexibility to adapt to specific characteristics of the sounds of the language to be represented.' Not that he accepts the conventional view that writing started as a representation of words by pictures. New studies of early Sumerian writing, he said, challenge this interpretation. The structures of this earliest writing did not, for example, match the structure of spoken language, dealing mainly in lists and categories rather than in sentences and narrative.

G For at least two decades, Dr. Denise Schmandt-Besserat, a University of Texas archaeologist, has argued that the first writing grew directly out of a system practised by Sumerian accountants. They used clay tokens, each one shaped to represent a jar of oil, a container of grain or a particular kind of livestock. These tokens were sealed inside clay spheres, and then the number and type of tokens inside was recorded on the outside using impressions resembling the tokens. Eventually, the token impressions were replaced with inscribed signs, and writing had been invented.

H Though Dr. Schmandt-Besserat has won much support, some linguists question her thesis, and others, like Dr. Pittman, think it too narrow. They emphasize that pictorial representation and writing evolved together. 'There's no question that the token system is a forerunner of writing', Dr. Pittman said, 'but I have an argument with her evidence for a link between tokens and signs, and she doesn't open up the process to include picture making.'

I Dr. Schmandt-Besserat vigorously defended her ideas. 'My colleagues say that pictures were the beginning of writing' she said, 'but show me a single picture that becomes a sign in writing. They say that designs on pottery were the beginning of writing, but show me a single sign of writing you can trace back to a pot — it doesn't exist'. In its first 500 years, she asserted, cuneiform writing was used almost solely for recording economic information, and after that its uses multiplied and broadened.

J Yet other scholars have advanced different ideas. Dr. Piotr Michalowski, Professor of Near East Civilizations at the University of Michigan, said that the proto-writing of Sumerian Uruk was 'so radically different as to be a complete break with the past'. It no doubt served, he said, to store and communicate information, but also became a new instrument of power. Some scholars noted that the origins of writing may not always have been in economics. In Egypt, most early writing is high on monuments or deep in tombs. In this case, said Dr. Pascal Vernus from a university in Paris, early writing was less administrative than sacred. It seems that the only certainty in this field is that many questions remain to be answered.

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Questions 27–30

Choose the correct letter, **A, B, C** *or* **D**.

27 The researchers at the symposium regarded the story of the King of Uruk as ridiculous because

A writing probably developed independently of speech.
B clay tablets had not been invented at that time.
C the distant ruler would have spoken another language.
D evidence of writing has been discovered from an earlier period.

28 According to the writer, the story of the King of Uruk

A is a probable explanation of the origins of writing.
B proves that early writing had a different function to writing today.
C provides an example of symbolic writing.
D shows some awareness amongst Sumerians of the purpose of writing.

29 There was disagreement among the researchers at the symposium about

A the area where writing began.
B the nature of early writing materials.
C the way writing began.
D the meaning of certain abstract images.

30 The opponents of the theory that writing developed from tokens believe that it

A grew out of accountancy.
B evolved from pictures.
C was initially intended as decoration.
D was unlikely to have been connected with commerce.

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Questions 31–36

Look at the following statements and the list of people below. Match each statement with the correct person, **A–E**.

LIST OF PEOPLE

A Dr. Holly Pittman
B Dr. Peter Damerow
C Dr. Denise Schmandt-Besserat
D Dr. Piotr Michalowski
E Dr. Pascal Vernus

31 There is no proof that early writing is connected to decorated household objects.

32 As writing developed, it came to represent speech.

33 Sumerian writing developed into a means of political control.

34 Early writing did not represent the grammatical features of speech.

35 There is no convincing proof that tokens and signs are connected.

36 The uses of cuneiform writing were narrow at first, and later widened.

Questions 37–40

Complete the summary using the list of words, **A–M**, *below.*

The earliest form of writing

A cuneiform **B** pictorial **C** tomb walls **D** urban **E** legible **F** stone blocks **G** simple **H** Mesopotamia **I** abstract **J** papyrus sheets **K** decorative **L** clay tablets **M** Egypt

Most archaeological evidence shows that the people of 37 _____ invented writing in around 3,300 BC. Their script was written on 38 _____ and was called 39 _____. Their script originally showed images related to political power and business, and later developed to become more 40 _____.

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IELTS Academic Reading — Test 9

Volume 2 · 3 Passages · 40 Questions

Reading Passage 1

Becoming an Expert

A Expertise is commitment coupled with creativity. Specifically, it is the commitment of time, energy, and resource to a relatively narrow field of study and the creative energy necessary to generate new knowledge in that field. It takes a considerable amount of time and regular exposure to a large number of cases to become an expert.

B An individual enters a field of study as a novice. The novice needs to learn the guiding principles and rules of a given task in order to perform that task. Concurrently, the novice needs to be exposed to specific cases, or instances, that test the boundaries of such principles. Generally, a novice will find a mentor to guide her through the process of acquiring new knowledge. A fairly simple example would be someone learning to play chess. The novice chess player seeks a mentor to teach her the object of the game, the number of spaces, the names of the pieces, the function of each piece, how each piece is moved, and the necessary conditions for winning or losing the game.

C In time, and with much practice, the novice begins to recognize patterns of behavior within cases and, thus, becomes a journeyman. With more practice and exposure to increasingly complex cases, the journeyman finds patterns not only within cases but also between cases. More importantly, the journeyman learns that these patterns often repeat themselves over time. The journeyman still maintains regular contact with a mentor to solve specific problems and learn more complex strategies. Returning to the example of the chess player, the individual begins to learn patterns of opening moves, offensive and defensive game-playing strategies, and patterns of victory and defeat.

D When a journeyman starts to make and test hypotheses about future behavior based on past experiences, she begins the next transition. Once she creatively generates knowledge, rather than simply matching superficial patterns, she becomes an expert. At this point, she is confident in her knowledge and no longer needs a mentor as a guide — she becomes responsible for her own knowledge. In the chess example, once a journeyman begins competing against experts, makes predictions based on patterns, and tests those predictions against actual behavior, she is generating new knowledge and a deeper understanding of the game. She is creating her own cases rather than relying on the cases of others.

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E The chess example is a rather short description of an apprenticeship model. Apprenticeship may seem like a restrictive 18th century mode of education, but it is still a standard method of training for many complex tasks. Academic doctoral programs are based on an apprenticeship model, as are fields like law, music, engineering, and medicine. Graduate students enter fields of study, find mentors, and begin the long process of becoming independent experts and generating new knowledge in their respective domains.

F Psychologists and cognitive scientists agree that the time it takes to become an expert depends on the complexity of the task and the number of cases, or patterns, to which an individual is exposed. The more complex the task, the longer it takes to build expertise, or, more accurately, the longer it takes to experience and store a large number of cases or patterns.

G An expert perceives meaningful patterns in her domain better than non-experts. Where a novice perceives random or disconnected data points, an expert connects regular patterns within and between cases. This ability to identify patterns is not an innate perceptual skill; rather it reflects the organization of knowledge after exposure to and experience with thousands of cases. Experts have a deeper understanding of their domains than novices do, and utilize higher-order principles to solve problems. A novice, for example, might group objects together by color or size, whereas an expert would group the same objects according to their function or utility. Experts comprehend the meaning of data and weigh variables with different criteria within their domains better than novices. Experts recognized variables that have the largest influence on a particular problem and focus their attention on those variables.

H Experts have better domain-specific short-term and long-term memory than novices do. Moreover, experts perform tasks in their domains faster than novices and commit fewer errors while problem solving. Interestingly, experts go about solving problems differently than novices. Experts spend more time thinking about a problem to fully understand it at the beginning of a task than do novices, who immediately seek to find a solution. Experts use their knowledge of previous cases as context for increasing mental models to solve given problems.

I Better at self-monitoring than novices, experts are more aware of instances where they have committed errors or failed to understand a problem. Experts check their solutions more often than novices and recognize when they are missing information necessary for solving a problem. Experts are aware of the limits of their domain knowledge and apply their domain's heuristics to solve problems that fall outside of their experience base.

J The strengths of expertise can also be weaknesses. Although one would expect experts to be good forecasters, they are not particularly good at making predictions about the future. Since the 1930s, researchers have been testing the ability of experts to make forecasts. The

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performance of experts has been tested against actuarial tables to determine if they are better at making predictions than simple statistical models. Seventy years later, with more than two hundred experiments in different domains, it is clear that the answer is no. If supplied with an equal amount of data about a particular case, an actuarial table is as good, or better, than an expert at making calls about the future. Even if an expert is given more specific case information than is available to the statistical model, the expert does not tend to outperform the actuarial table.

K Theorists and researchers differ when trying to explain why experts are less accurate forecasters than statistical models. Some have argued that experts, like all humans, are inconsistent when using mental models to make predictions. A number of researchers point to human biases to explain unreliable expert predictions. During the last 30 years, researchers have categorized, experimented, and theorized about the cognitive aspects of forecasting. Despite such efforts, the literature shows little consensus regarding the causes or manifestations of human bias.

Questions 1–5

Complete the flow-chart below. Choose **NO MORE THAN THREE WORDS** *from the passage for each answer.*

Novice: needs to study (1) _____ under the guidance of a (2) _____

↓

(3) _____ : starts to identify (4) _____ for cases within or between study; more (5) _____ ways of doing things create new knowledge

↓

Expert: performs task independently

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Questions 6–10

Do the following statements agree with the claims of the writer in Reading Passage 1?

TRUE *if the statement agrees* • **FALSE** *if the statement contradicts* • **NOT GIVEN** *if there is no information*

6 Novices and experts use the same system of knowledge to comprehend and classify objects.

7 The focus of novices' training is necessarily on long term memory.

8 When working out the problems, novices want to solve them straight away.

9 When handling problems, experts are always more efficient than novices in their fields.

10 Experts tend to review more than novices on cases when flaws or limits on understanding took place.

Questions 11–13

Complete the summary below. Choose **NO MORE THAN TWO WORDS** *from the passage for each answer.*

While experts outperform novices and machines in pattern recognition and problem solving, expert predictions of future behavior or events are seldom as accurate as simple actuarial tables. Why? Some have tried to explain that experts differ when using cognitive 11 _____ to forecast. Researchers believe it is due to 12 _____. However attempting endeavour of finding answers did not yet produce 13 _____.

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Reading Passage 2

Revolutions in Mapping

A Today, the mapmaker's vision is no longer confined to what the human eye can see. The perspective of mapmaking has shifted from the crow's nest of the sailing vessel, mountain top and airplane to new orbital heights. Radar, which bounces microwave radio signals off a given surface to create images of its contours and textures, can penetrate jungle foliage and has produced the first maps of the mountains of the planet Venus. And a combination of sonar and radar produces charts of the seafloor, putting much of Earth on the map for the first time. 'Suddenly it's a whole different world for us,' says Joel Morrison, chief of geography at the U.S. Bureau of the Census, 'Our future as mapmakers — even ten years from now — is uncertain.'

B The world's largest collection of maps resides in the basement of the Library of Congress in Washington, D.C. The collection, consisting of up to 4.6 million map sheets and 63,000 atlases, includes magnificent bound collections of elaborate maps — the pride of the golden age of Dutch cartography. In the reading room scholars, wearing thin cotton gloves to protect the fragile sheets, examine ancient maps with magnifying glasses. Across the room people sit at their computer screens, studying the latest maps; with their prodigious memories, computers are able to store data about people, places and environments — the stuff of maps — and almost instantly information is displayed on the screen in the desired geographic context, and at the click of a button, a print-out of the map appears.

C Measuring the spherical Earth ranks as the first major milestone in scientific cartography. This was first achieved by the Greek astronomer Eratosthenes, a scholar at the famous Alexandrian Library in Egypt in the third century BC. He calculated the Earth's circumference as 25,200 miles, which was remarkably accurate. The longitudinal circumference is known to be 24,860 miles.

D Building on the ideas of his predecessors, the astronomer and geographer Ptolemy, working in the second century AD, spelled out a system for organizing maps according to grids of latitude and longitude. Today, parallels of latitude are often spaced at intervals of 10 to 20 degrees and meridians at 15 degrees, and this is the basis for the width of modern time zones. Another legacy of Ptolemy's is his advice to cartographers to create maps to scale. Distance on today's maps is expressed as a fraction or ratio of the real distance. But mapmakers in Ptolemy's time lacked the geographic knowledge to live up to Ptolemy's scientific principles. Even now, when surveyors achieve accuracies down to inches and satellites can plot potential missile targets within feet, maps are not true pictures of reality.

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E However, just as the compass improved navigation and created demand for useful charts, so the invention of the printing press in the 15th century put maps in the hands of more people, and took their production away from monks, who had tended to illustrate theology rather than geography. Ocean-going ships launched an age of discovery, enlarging both what could and needed to be mapped, and awakened an intellectual spirit and desire for knowledge of the world.

F Inspired by the rediscovered Ptolemy, whose writing had been preserved by Arabs after the sacking of the Alexandrian Library in AD 931, mapmakers in the 15th century gradually replaced theology with knowledge of faraway places, as reported by travelling merchants like Marco Polo.

G Gerhardus Mercator, the foremost shipmaker of the 16th century, developed a technique of arranging meridians and parallels in such a way that navigators could draw straight lines between two points and steer a constant compass course between them. This distortion formula, introduced on his world map of 1569, created the 'Greenland problem'. Even on some standard maps to this day, Greenland looks as large as South America — one of the many problems when one tries to portray a round world on a flat sheet of paper. But the Mercator projection was so practical that it is still popular with sailors.

H Scientific mapping of the land came into its own with the achievements of the Cassini family — father, son, grandson and great-grandson. In the late 17th century, the Italian born founder, Jean-Dominique, invented a complex method of determining longitude based on observations of Jupiter's moons. Using this technique, surveyors were able to produce an accurate map of France. The family continued to map the French countryside and his great-grandson finally published their famous Cassini map in 1793 during the French Revolution. While it may have lacked the artistic appeal of earlier maps, it was the model of a social and geographic map showing roads, rivers, canals, towns, abbeys, vineyards, lakes and even windmills. With this achievement, France became the first country to be completely mapped by scientific methods.

I Mapmaking has come a long way since those days. Today's surveyors rarely go into the field without being linked to navigation satellites. Their hand-held receivers are the most familiar of the new mapping technologies, and the satellite system, developed and still operated by the US Defense Department, is increasingly used by surveyors. Even ordinary hikers, sailors and explorers can tap into it for data telling them where they are. Simplified civilian versions of the receivers are available for a few hundred dollars and they are also the heart of electronic map displays available in some cars. Cartography is pressing on to cosmic frontiers, but its objective is, and always has been, to communicate a sense of 'here' in relation to 'there', however far away 'there' may be.

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Questions 14–18

Choose the correct letter, **A, B, C** *or* **D**.

14 According to the first paragraph, mapmakers in the 21st century

A combine techniques to chart unknown territory.
B still rely on being able to see what they map.
C are now able to visit the darkest jungle.
D need input from experts in other fields.

15 The Library of Congress offers an opportunity to

A borrow from their collection of Dutch maps.
B learn how to restore ancient and fragile maps.
C enjoy the atmosphere of the reading room.
D create individual computer maps to order.

16 Ptolemy alerted his contemporaries to the importance of

A measuring the circumference of the world.
B organizing maps to reflect accurate ratios of distance.
C working out the distance between parallels of latitude.
D accuracy and precision in mapping.

17 The invention of the printing press

A revitalized interest in scientific knowledge.
B enabled maps to be produced more cheaply.
C changed the approach to mapmaking.
D ensured that the work of Ptolemy was continued.

18 The writer concludes by stating that

A mapmaking has become too specialized.
B cartographers work in very harsh conditions.
C the fundamental aims of mapmaking remain unchanged.
D the possibilities of satellite mapping are infinite.

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Questions 19–21

Look at the following list of achievements and the list of mapmakers below. Match each achievement with the correct mapmaker, **A–D**.

LIST OF MAPMAKERS

A Mercator
B Ptolemy
C Cassini family
D Eratosthenes

19 came very close to accurately measuring the distance round the Earth

20 produced maps showing man-made landmarks

21 laid the foundation for our modern time zones

Questions 22–26

Complete the summary below. Choose **NO MORE THAN TWO WORDS** *from the passage for each answer.*

Ancient maps allow us to see how we have come to make sense of the world. They also reflect the attitudes and knowledge of the day. The first great step in mapmaking took place in 22 _____ in the 3rd century BC. Work continued in this tradition until the 2nd century AD but was then abandoned for over a thousand years, during which time maps were the responsibility of 23 _____ rather than scientists. Fortunately, however, the writings of 24 _____ had been kept, and interest in scientific mapmaking was revived as scholars sought to produce maps, inspired by the accounts of travellers. These days, 25 _____ are vital to the creation of maps and radar has allowed cartographers to map areas beyond our immediate world. In addition, this high-tech equipment is not only used to map faraway places, but cheaper versions have also been developed for use in 26 _____.

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Reading Passage 3

Sir Francis Ronalds and the Telegraph

A RONALDS, Sir FRANCIS (1788-1873), inventor of the electric telegraph and meteorologist, son of Francis Ronalds, a London merchant, and of his wife, Jane, daughter of William Field, was born in London on 21 Feb. 1788. Ronalds was educated at a private school at Cheshunt by the Rev. E. Cogan. At an early age he displayed a taste for experiment, and he acquired great skill later in practical mechanics and draughtsmanship. Under the influence of Jean Andre de Luc (1727-1817), whose acquaintance he made in 1814, he began to devote himself to practical electricity. In 1814 and 1815 he published several papers on electricity in Tilloch's 'Philosophical Magazine,' one of which records an ingenious use of De Luc's 'electric column' as a motive power for a clock.

B Ronalds's name is chiefly remembered as the inventor of an electric telegraph. Since 1753, when the first proposal for an electric telegraph worked by statical electricity was made by a writer signing 'C. M.' (said to be Charles Morrison) in the 'Scots Magazine', successive advances had been made abroad by Volta, Le Sage, Lomond, Cavallo, Salva, and others; but much was needed to perfect the invention.

C In 1816 Francis Ronalds, then living at Upper Mall, Hammersmith, built in his back garden two frames to accommodate eight miles of wire for his new invention of an electrostatic telegraph. It used clockwork-driven rotating dials, engraved with letters of the alphabet and numbers, synchronized with each other, at both ends of the circuit. For the past three or four years, encouraged by the octogenarian Swiss meteorologist, Jean Andre De Luc, Ronalds had been enthusiastically experimenting with electrostatic clockwork devices. When someone desired to send a message he earthed the wire at his end at the moment when the dial indicated the desired letter. At the receiving end the pith balls would fall together when earthed and the recipient noted the letter showing on his dial at that moment. The system was slow and depended on the two dials staying in step, but Ronalds successfully transmitted and received letters over 150 meters of wire; later he succeeded in sending messages through eight miles of iron wire suspended above his garden in London.

D After sending messages along his wires on the frame, he developed another version in which the wires were enclosed in glass tubes buried in the ground. At each end of the line a clockwork mechanism turned synchronously revolving discs with letters on them. A frictional-electricity machine kept the wire continuously charged, while at each end two pith balls hung from the wire on silk threads, and since they were similarly charged from the wire they stayed apart. Ronalds's instrument was of real practical use, and the brilliant

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idea of using synchronously rotating discs, now employed in the Hughes printing apparatus, was entirely his own. The only defect in his invention was the comparative slowness with which a succession of symbols could be transmitted.

E With communications between London and Portsmouth in mind, he believed his telegraph would work over distances of 800km. In the same year, Ronalds wrote to offer his invention to the Admiralty. In fact, in 1806, Ralph Wedgwood submitted a telegraph based on frictional electricity to the Admiralty, but was told that the semaphore was sufficient for the country. In a pamphlet he suggested the establishment of a telegraph system with public offices in different centers. Francis Ronalds, in 1816, brought a similar telegraph of his invention to the notice of the Admiralty, and was politely informed that 'telegraphs of any kind are now wholly unnecessary.' John Barrow, Secretary to the Admiralty, replied that "Telegraphs of any kind are now wholly unnecessary; and no other than the one now in use will be adopted." (The one in use was a semaphore system. Only a year after the end of the Napoleonic Wars, the Admiralty saw no need for improved communications, even though the semaphore was usable only in daylight and good weather.)

F After this disappointment, Ronalds set off for the continent. He travelled throughout Europe and the Eastern Mediterranean, taking notes, sketching and collecting scientific books between 1816 and 1823. He had begun collecting his large library of works on electricity and kindred subjects. The last activity formed the beginnings of the Ronalds Library, left in trust to the IEE (now the IET) after his death. In a small pamphlet published in 1823, Ronalds described his invention and listed some of its possible uses, "Why should not government govern at Portsmouth almost as promptly as in Downing Street? Why should our defaulters escape by default of our foggy climate? Let us have Electrical Conversazione offices communicating with each other all over the kingdom if we can." In 1825 he invented and patented a perspective tracing instrument, intended to facilitate drawing from nature, which he improved about 1828, and described in a work called 'Mechanical Perspective.' These instruments seem to be the only ones for which he took out patents.

G However, Ronalds never patented his invention in electric telegraph. Ronalds seems to have made few or no practical contributions to science. In the meanwhile, one person did benefit from this work — Charles Wheatstone who saw the telegraph as a boy. When Charles Wheatstone was quite a child, his father had seen the Ronalds telegraph at work. Later, the invention of an electric telegraph had been marvelously developed by Wheatstone, who had seen many of the Hammersmith experiments, in conjunction with Mr. William Fothergill Cooke, and these two men together devised and patented in 1837 the first electric telegraph used publicly and commercially in England. When, in 1855, a controversy arose between Wheatstone and Cooke with regard to their respective shares in

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the invention, Wheatstone at once acknowledged his direct debt to Ronalds, and Cooke, though less fully, acknowledged the priority of Ronalds's work; Until 1855 Ronalds's share in the invention had been forgotten by the public.

H Early in 1843 Ronalds was made honorary director and superintendent of the Meteorological Observatory, which was then established at Kew by the British Association for the Advancement of Science. He began work on a system for registering meteorological data using photography and this time was awarded a grant to continue his work. A similar system was developed independently by Charles Brooke, aided like Ronalds by grants from the Royal Society, had invented independently about this time. But the British Association confirmed Ronalds's priority. This was the beginning of automatic, accurate recording of meteorological data and remained in use for some years after Ronalds's death.

I Ronalds lived long enough to see his prophecies come to fruition and to receive belated official recognition: in 1870, three years before he died, he was knighted by Queen Elizabeth I, for his "early and remarkable labors in telegraphic investigations."

Questions 27–31

Match each correct year to the historical event in the passage.

YEARS

A 1753
B 1806
C 1816
D 1823
E 1825
F 1837
G 1843

27 When did Francis Ronalds achieve a satisfactory result in the electricity experiment conducted for the first time?

28 When was the first proposal of an electric telegraph based on static electricity?

29 When did Ronalds get a patent of his invention firstly?

30 Ronalds first made known and revealed the applicable significance of his telegram to the public.

31 The contribution being done by Ronalds' invention in meteorological data.

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Questions 32–35

Answer the questions below. Choose **NO MORE THAN FOUR WORDS AND/OR A NUMBER** *from the passage for each answer.*

32 What were carved in the experimental dials when doing Ronalds' experiment in his garden? 32 _____

33 What were enclosed with the buried telegram wires when Ronalds did the improved experiment? 33 _____

34 What is the greatest distance Ronalds believed his telegram can send?
34 _____

35 What kind of power supplied the keeping of the wire charged continuously?
35 _____

Questions 36–40

Reading Passage 3 has nine paragraphs, **A–I**. *Which paragraph contains the following information?*

Q	Statement	Answer
36	There is a commercial use of the telegram.	
37	There is a contributory influence on Ronalds from a fellow he got to know.	
38	Ronalds's proposal was rejected as the preceding reference to another application.	
39	There existed a drawback of Ronalds's telegram.	
40	Ronalds's contribution in telegraphic investigations was recognized by authority.	

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IELTS Academic Reading — Test 10

Volume 2 · 3 Passages · 40 Questions

Reading Passage 1

Dirty River But Clean Water

Floods can occur in rivers when the flow rate exceeds the capacity of the river channel, particularly at bends or meanders in the waterway. Floods often cause damage to homes and businesses if they are in the natural flood plains of rivers. While riverine flood damage can be eliminated by moving away from rivers and other bodies of water, people have traditionally lived and worked by rivers because the land is usually flat and fertile and because rivers provide easy travel and access to commerce and industry.

A Fire and flood are two of humanity's worst nightmares. People have, therefore, always sought to control them. Forest fires are snuffed out quickly. The flow of rivers is regulated by weirs and dams. At least, that is how it used to be. But foresters have learned that forests need fires to clear out the brash and even to get seeds to germinate. And a similar revelation is now dawning on hydrologists. Rivers — and the ecosystems they support — need floods. That is why a man-made torrent has been surging down the Grand Canyon. By Thursday March 6th it was running at full throttle, which was expected to be sustained for 60 hours.

B Floods once raged through the canyon every year. Spring Snow from as far away as Wyoming would melt and swell the Colorado river to a flow that averaged around 1,500 cubic metres (50,000 cubic feet) a second. Every eight years or so, that figure rose to almost 3,000 cubic metres. These floods infused the river with sediment, carved its beaches and built its sandbars.

C However, in the four decades since the building of the Glen Canyon dam, just upstream of the Grand Canyon, the only sediment that it has collected has come from tiny, undammed tributaries. Even that has not been much use as those tributaries are not powerful enough to distribute the sediment in an ecologically valuable way.

D This lack of flooding has harmed local wildlife. The humpback chub, for example, thrived in the rust-red waters of the Colorado. Recently, though, its population has crashed. At first sight, it looked as if the reason was that the chub were being eaten by trout introduced for sport fishing in the mid-20th century. But trout and chub co-existed until the Glen Canyon dam was built, so something else is going on. Steve Gloss, of the United States' Geological Survey (USGS), reckons that the chub's decline is the result of their losing their most valuable natural defense, the Colorado's rusty sediment. The chub were

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well adapted to the poor visibility created by the thick, red water which gave the river its name, and depended on it to hide from predators. Without the cloudy water the chub became vulnerable.

E And the chub are not alone. In the years since the Glen Canyon dam was built, several species have vanished altogether. These include the Colorado pike-minnow, the razorback sucker and the round-tail chub. Meanwhile, aliens including fathead minnows, channel catfish and common carp, which would have been hard put to survive in the savage waters of the undammed canyon, have moved in.

F So flooding is the obvious answer. Unfortunately, it is easier said than done. Floods were sent down the Grand Canyon in 1996 and 2004 and the results were mixed. In 1996 the flood was allowed to go on too long. To start with, all seemed well. The floodwaters built up sandbanks and infused the river with sediment. Eventually, however, the continued flow washed most of the sediment out of the canyon. This problem was avoided in 2004, but unfortunately, on that occasion, the volume of sand available behind the dam was too low to rebuild the sandbanks. This time, the USGS is convinced that things will be better. The amount of sediment available is three times greater than it was in 2004. So if a flood is going to do some good, this is the time to unleash one.

G Even so, it may turn out to be an empty gesture. At less than 1,200 cubic metres a second, this flood is smaller than even an average spring flood, let alone one of the mightier deluges of the past. Those glorious inundations moved massive quantities of sediment through the Grand Canyon, wiping the slate dirty, and making a muddy mess of silt and muck that would make modern river rafters cringe.

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Questions 1–7

Do the following statements agree with the information given in Reading Passage 1?

TRUE *if the statement agrees* • **FALSE** *if the statement contradicts* • **NOT GIVEN** *if there is no information*

1 Damage caused by fire is worse than that caused by flood.

2 The flood peaks at almost 1500 cubic metres every eight years.

3 Contribution of sediments delivered by tributaries has little impact.

4 The decreasing number of chubs is always caused by the introduction of trout since the mid 20th century.

5 It seemed that the artificial flood in 1996 had achieved success partly at the very beginning.

6 In fact, the yield of artificial flood water is smaller than an average natural flood at present.

7 Mighty floods drove fast moving flows with clean and high quality water.

Questions 8–13

Complete the summary below. Choose **NO MORE THAN TWO WORDS** *from the passage for each answer.*

The eco-impact of the Canyon Dam

Floods are people's nightmare. In the past, the canyon was raged by flood every year. The snow from far Wyoming would melt in the season of 8 ———— and caused a flood flow peak in the Colorado river. In the four decades after people built the Glen Canyon dam, it only could gather 9 ———— together from tiny, undammed tributaries.

Humpback chub population reduced — why? Then, several species disappeared including Colorado pike-minnow, 10 ———— and the round-tail chub. Meanwhile, some moved in such as fathead minnows, channel catfish and 11 ————. The non-stopped flow led to the washing away of the sediment out of the canyon, which poses great threat to the chubs because it has poor 12 ———— away from predators. In addition, the volume of 13 ———— available behind the dam was too low to rebuild the bars and flooding became more serious.

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Reading Passage 2

We Have Star Performers!

A The difference between companies is people. With capital and technology in plentiful supply, the critical resource for companies in the knowledge era will be human talent. Companies full of achievers will, by definition, outperform organisations of plodders. Ergo, compete ferociously for the best people. Poach and pamper stars; ruthlessly weed out second-raters. This, in essence, has been the recruitment strategy of the ambitious company of the past decade. The 'talent mindset' was given definitive form in two reports by the consultancy McKinsey famously entitled The War for Talent. Although the intensity of the warfare subsequently subsided along with the air in the internet bubble, it has been warming up again as the economy tightens: labour shortages, for example, are the reason the government has laid out the welcome mat for immigrants from the new Europe.

B Yet while the diagnosis — people are important — is evident to the point of platitude, the apparently logical prescription — hire the best — like so much in management is not only not obvious: it is in fact profoundly wrong. The first suspicions dawned with the crash to earth of the dotcom meteors, which showed that dumb is dumb whatever the IQ of those who perpetrate it. The point was illuminated in brilliant relief by Enron, whose leaders, as a New Yorker article called 'The Talent Myth' entertainingly related, were so convinced of their own cleverness that they never twigged that collective intelligence is not the sum of a lot of individual intelligence. In fact, in a profound sense, the two are opposites. Enron believed in stars, noted author Malcolm Gladwell, because they didn't believe in systems. But companies don't just create: 'they execute and compete and coordinate the efforts of many people, and the organizations that are most successful at that task are the ones where the system is the star'. The truth is that you can't win the talent wars by hiring stars — only lose it. New light on why this should be so is thrown by an analysis of star behaviour in this months' Harvard Business Review. In a study of the careers of 1,000 star-stock analysts in the 1990s, the researchers found that when a company recruited a star performer, three things happened.

C First, stardom doesn't easily transfer from one organisation to another. In many cases, performance dropped sharply when high performers switched employers and in some instances never recovered. More of success than commonly supposed is due to the working environment — systems, processes, leadership, accumulated embedded learning that are absent in and can't be transported to the new firm. Moreover, precisely because of their past stellar performance, stars were unwilling to learn new tricks and antagonized those (on whom they now unwittingly depended) who could teach them. So they moved, upping their salary as they did — 36 per cent moved on within three years, fast even for Wall Street. Second, group performance suffered as a result of tensions and resentment by rivals

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within the team. One respondent likened hiring a star to an organ transplant. The new organ can damage others by hogging the blood supply, other organs can start aching or threaten to stop working or the body can reject the transplants altogether, he said. 'You should think about it very carefully before you do a transplant to a healthy body.' Third, investors punished the offender by selling its stock. This is ironic since the motive for importing stars was often a suffering share price in the first place. Shareholders evidently believe that the company is overpaying, the hiree is cashing in on a glorious past rather than preparing for a glowing present, and a spending spree is in the offing.

D The result of mass star hirings as well as individual ones seems to confirm such doubts. Look at County NatWest and Barclays de Zoete Wedd, both of which hired teams of stars with loud fanfare to do great things in investment banking in the 1990s. Both failed dismally. Everyone accepts the cliche that people make the organisation — but much more does the organisation make the people. When researchers studied the performance of fund managers in the 1990s, they discovered that just 30 per cent of the variation in fund performance was due to the individual, compared to 70 per cent to the company-specific setting.

E That will be no surprise to those familiar with systems thinking. W Edwards Deming used to say that there was no point in beating up on people when 90 per cent of performance variation was down to the system within which they worked. Consistent improvement, he said, is a matter not of raising the level of individual intelligence, but of the learning of the organisation as a whole. The star system is glamorous — for the few. But it rarely benefits the company that thinks it is working it. And the knock-on consequences indirectly affect everyone else too. As one internet response to Gladwell's New Yorker article put it: after Enron, 'the rest of corporate America is stuck with overpaid, arrogant, underachieving, and relatively useless talent.'

F Football is another illustration of the star vs systems strategic choice. As with investment banks and stockbrokers, it seems obvious that success should ultimately be down to money. Great players are scarce and expensive. So the club that can afford more of them than anyone else will win. But the performance of Arsenal and Manchester United on one hand and Chelsea and Real Madrid on the other proves that it's not as easy as that. While Chelsea and Real have the funds to be compulsive star collectors — as with Juan Sebastian Veron — they are less successful than Arsenal and United which, like Liverpool before them, have put much more emphasis on developing a setting within which stars-in-the-making can flourish. Significantly, Thierry Henry, Patrick Veira and Robert Pires are much bigger stars than when Arsenal bought them, their value (in all senses) enhanced by the Arsenal system. At Chelsea, by contrast, the only context is the stars themselves — managers with different outlooks come and go every couple of seasons. There is no settled system for the stars to blend into. The Chelsea context has not only not added value, but it

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has also subtracted it. The side is less than the sum of its exorbitantly expensive parts. Even Real Madrid's galacticos, the most extravagantly gifted on the planet, are being outperformed by less talented but better-integrated Spanish sides. In football, too, stars are trumped by systems.

G So if not by hiring stars, how do you compete in the war for talent? You grow your own. This worked for investment analysts, where some companies were not only better at creating stars but also at retaining them. Because they had a much more sophisticated view of the interdependent relationship between star and system, they kept them longer without resorting to the exorbitant salaries that were so destructive to rivals.

Questions 14–17

Reading Passage 2 has seven paragraphs, **A–G**. *Which paragraph contains the following information? (You may use any letter more than once.)*

Q	Statement	Answer
14	One example from non-commerce/business settings that better systems win bigger stars	
15	One failed company that believes in stars rather than the system	
16	One suggestion that the author made about how to acquire employees nowadays	
17	One metaphor to human medical anatomy that illustrates the problems of hiring stars	

Questions 18–21

Do the following statements agree with the information given in Reading Passage 2?

YES *if the statement agrees* • **NO** *if it contradicts* • **NOT GIVEN** *if there is no information*

18 McKinsey, who wrote The War for Talent, had not expected the huge influence made by this book.

19 Economic condition becomes one of the factors which decide whether or not a country would prefer to hire foreign employees.

20 The collapse of Enron is caused totally by an unfortunate incident instead of the company's management mistake.

21 Football clubs that focus on making stars in the setting are better than those simply collecting stars.

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Questions 22–26

Complete the summary below. Choose **NO MORE THAN TWO WORDS** *from the passage for each answer.*

An investigation carried out on 1000 (22) _____ participants of a survey by Harvard Business Review found that when a company hires a (23) _____ it has negative effects. For instance, they behave considerably worse in a new team than in the (24) _____ that they used to be in. They move faster than Wall Street and increase their (25) _____. Secondly, they faced rejection or refusal from those (26) _____ within the team. Lastly, the one who made mistakes had been punished by selling his/her stock share.

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Reading Passage 3

Toddlers Bond With Robot

A Will the robot revolution begin in nursery school? Researchers introduced a state-of-the-art social robot into a classroom of 18- to 24-month-olds for five months as a way of studying human-robot interactions. The children not only came to accept the robot, but treated it as they would a human buddy — hugging it and helping it — a new study says. "The results imply that current robot technology is surprisingly close to achieving autonomous bonding and socialization with human toddlers," said Fumihide Tanaka, a researcher at the University of California, San Diego.

B The development of robots that interact socially with people has been difficult to achieve, experts say, partly because such interactions are hard to study. "To my knowledge, this is the first long-term study of this sort," said Ronald Arkin, a roboticist at the Georgia Institute of Technology, who was not involved with the study. "It is groundbreaking and helps to forward human-robot interaction studies significantly," he said.

C The most successful robots so far have been storytellers, but they have only been able to hold human interest for a limited time. For the new study, researchers introduced a toddler-size humanoid robot into a classroom at a UCSD childhood education center. Initially the researchers wanted to use a 22-inch-tall model, but later they decided to use another robot of the QRIO series, the 23-inch-tall (58-centimeter-tall) machine was originally developed by Sony. Children of toddler age were chosen because they have no preconceived notions of robots, said Tanaka, the lead researcher, who also works for Sony. The researchers sent instructions about every two minutes to the robot to do things like giggle, dance, sit down, or walk in a certain direction. The 45 sessions were videotaped, and interactions between toddlers and the robot were later analyzed.

D The results showed that the quality of those interactions improved steadily over 27 sessions. The tots began to increasingly interact with the robot and treat it more like a peer than an object during the first 11 sessions. The level of social activity increased dramatically when researchers added a new behavior to QRIO's repertoire: If a child touched the humanoid on its head, it would make a giggling noise. The interactions deteriorated quickly over the next 15 sessions, when the robot was reprogrammed to behave in a more limited, predictable manner. Finally, the human-robot relations improved in the last three sessions, after the robot had been reprogrammed to display its full range of behaviors. "Initially the children treated the robot very differently than the way they treated each other," Tanaka said. "But by the end they treated the robot as a peer rather than a toy."

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E Early in the study some children cried when QRIO fell. But a month into the study, the toddlers helped QRIO stand up by pushing its back or pulling its hands. 'The most important aspect of interaction was touch', Tanaka said. 'At first the toddlers would touch the robot on its face, but later on they would touch only on its hands and arms, like they would with other humans'. Another robotlike toy named Robby, which resembled QRIO but did not move, was used as a control toy in the study. While hugging of QRIO increased, hugging of Robby decreased throughout the study. Furthermore, when QRIO laid down on the floor as its batteries ran down, a toddler would put a blanket over his silver-colored "friend" and say "night-night."

F "Our work suggests that touch integrated on the time-scale of a few minutes is a surprisingly effective index of social connectedness," Tanaka says. "Something akin to this index may be used by the human brain to evaluate its own sense of social well-being." He adds that social robots like QRIO could greatly enrich classrooms and assist teachers in early learning programs. Hiroshi Ishiguro — robotics expert at Osaka University in Japan — says, "I think this study has clearly reported the possibilities of small, almost autonomous humanoid robots for toddlers. Nowadays robots can perform a variety of functions that were thought to be incident to people only — in short time we'll have electronic baby-sitters and peer-robots in every kindergarten," said Ishiguro, who was not involved with the study but has collaborated with its authors on other projects.

G Now this study has taken a new direction — the researchers are now developing autonomous robots for the toddler classroom. "I cannot avoid underlining how great potential it could have in educational settings assisting teachers and enriching the classroom environment," Tanaka said. However, some scientists don't share his opinion.

H Arkin, the Georgia Tech roboticist, said he was not surprised by the affection showed by the toddlers toward the robot. "Humans have a tremendous propensity to bond with artifacts with any or all sort, whether it be a car, a doll, or a robot," he said. But he also cautioned that researchers don't yet understand the consequences of increased human-robot interaction. "Just studying how robots and humans work together can give us insight into whether this is a good thing or a bad thing for society," Arkin said. "What are the consequences of introducing a robot artifact into a cadre of children? How will that enhance, or potentially interfere with, their social development? It might make life easier for the teacher, but we really don't understand the long-term impact of having a robot as a childhood friend, do we?"

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Questions 27–33

Reading Passage 3 has eight paragraphs, **A–H**. *Which paragraph contains the following information? (You may use any letter more than once.)*

Q	Statement	Answer
27	Changes in toddler-robot interaction quality	
28	Comparison of two different robots	
29	The fact that previous robots could maintain people's interest only for a short time	
30	The importance of touch	
31	The new direction of the study	
32	Technical parameters of the introduced robot	
33	The significance and novelty of the conducted study	

Questions 34–37

Match each statement below with the correct scientist, **A–C**. *(You may use any letter more than once.)*

LIST OF SCIENTISTS

A Fumihide Tanaka

B Ronald Arkin

C Hiroshi Ishiguro

34 Robots will perform the duties of baby-sitters in the nearest future.

35 By the end of the study children treated the robot as a living creature rather than a toy.

36 The conducted study is the first major study of this sort.

37 Robots can be used in classrooms and assist teachers.

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Questions 38–40

Choose the correct letter, **A, B, C** *or* **D**.

38 In the study, researchers introduced a toddler-size humanoid robot that was

A 58-inch-tall
B 22-inch-tall
C 23-inch-tall
D 45-inch-tall

39 The researchers sent instructions to the robot to perform different actions EXCEPT

A laugh
B dance
C sit down
D crawl

40 The toddlers began to increasingly interact with the robot during

A the first 11 sessions
B the next 15 sessions
C the first 27 sessions
D the last 15 sessions

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PART TWO

Answer Keys

Complete answer keys for Tests 1–10 of Volume 2. Check your answers only after finishing each test.

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Answer Key & Explanations

Reading Test 1 (Volume 2)

Passage 1 — Sweet Trouble

1. B — Mike Berwick

QUESTION KEYWORD	practices close to **environmentally friendly**
PASSAGE EVIDENCE	"there's a formula for sustainable cane production and Mossman has **nearly reached it**"

2. F — Doug Crees

QUESTION KEYWORD	**financial return not the only** factor
PASSAGE EVIDENCE	"Economically, it doesn't make much sense. But **there's more to life than money**"

3. C — Karen Benn

QUESTION KEYWORD	cane may **not harm environment** as much
PASSAGE EVIDENCE	"I'm **less worried** about the effect of cane on the environment than... other agriculture"

4. A — Bill Phillips-Turner

QUESTION KEYWORD	local population would **decline**
PASSAGE EVIDENCE	"most people now employed by the industry would have to **leave the area to find work**"

5. D

QUESTION KEYWORD	town "holding its breath" because
PASSAGE EVIDENCE	"For two years the mill... has been **close to bankruptcy**"

6. A

QUESTION KEYWORD	growers refusing offer expected to
PASSAGE EVIDENCE	"allowing those remaining to **buy the vacated land and improve their economies of scale**"

7. D

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QUESTION KEYWORD	Bill found most difficult
PASSAGE EVIDENCE	"the hardest was **slashing the mill workforce**"

8. B

QUESTION KEYWORD	Doug Crees says he
PASSAGE EVIDENCE	"there's more to life than money... I still **don't want to do that**" (work away from the farm)

9. TRUE

QUESTION KEYWORD	2004 cost **similar to** selling price
PASSAGE EVIDENCE	"about $232 a tonne... that was about **what it cost to grow a tonne** of sugar"

10. NOT GIVEN

QUESTION KEYWORD	farmers who left **moved to the city**
PASSAGE EVIDENCE	No information given on where farmers who took the offer relocated.

11. FALSE

QUESTION KEYWORD	**fuel production** most desirable
PASSAGE EVIDENCE	"the most radical but **preferred alternative** is to create a future for the mill as a **food factory**"

12. NOT GIVEN

QUESTION KEYWORD	reduced use of **pesticides**
PASSAGE EVIDENCE	Tom's changes (tree buffer, rock erosion control, minimal soil disturbance) are described, but pesticide use is never mentioned.

13. FALSE

QUESTION KEYWORD	practices **profitable** for Tom Watters
PASSAGE EVIDENCE	"none of these costly initiatives has **helped him get better cane prices**"

Passage 2 — Global Warming in Europe

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14. TRUE

QUESTION KEYWORD	avg temp **almost 4°C higher**
PASSAGE EVIDENCE	"the average temperature for the summer months was **3.78°C higher** than the long-term norm"

15. TRUE

QUESTION KEYWORD	**human impact** one reason
PASSAGE EVIDENCE	"2003's extreme temperatures may be directly attributed to **global warming caused by human actions**"

16. NOT GIVEN

QUESTION KEYWORD	recorded **twice daily** in major cities
PASSAGE EVIDENCE	No detail about recording frequency or specific cities is given.

17. FALSE

QUESTION KEYWORD	temps within **normal variation**
PASSAGE EVIDENCE	"the final degree of it is likely to be due to **global warming**, caused by human action" — not simply normal variation.

18. TRUE

QUESTION KEYWORD	before 2003, **milder winters** not summers
PASSAGE EVIDENCE	"the warming had been manifesting itself mainly in **winters that were less cold** rather than... summers"

19. NOT GIVEN

QUESTION KEYWORD	new resorts built at **higher altitudes**
PASSAGE EVIDENCE	Only mentions that lower-level ski resorts "will die out" — no mention of new higher-altitude construction.

20. 1976 and 1995

QUESTION KEYWORD	TWO years hotter in Britain
PASSAGE EVIDENCE	"the summer itself... comes behind **1976 and 1995**, when there were longer periods of intense heat"

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21. 2000 floods

QUESTION KEYWORD	other event affecting UK planning
PASSAGE EVIDENCE	"much as the **2000 floods** revolutionised the way the Government is thinking about flooding"

22. 1856

QUESTION KEYWORD	records began in ____
PASSAGE EVIDENCE	"the global temperature record (which goes back to **1856**)"

23. 1990

QUESTION KEYWORD	warmest ten years after ____
PASSAGE EVIDENCE	"The ten hottest years in the record have occurred **since 1990**"

24. 500

QUESTION KEYWORD	highest for at least ____ years
PASSAGE EVIDENCE	"the warmest summer in the last **500 years**"

25. France

QUESTION KEYWORD	thousands died in ____
PASSAGE EVIDENCE	"15,000 extra deaths in **France** during August"

26. D

QUESTION KEYWORD	most suitable title
PASSAGE EVIDENCE	The whole passage centres on the exceptional 2003 European summer — "**The Year of European Heat**".

Passage 3 — Improving Patient Safety

27. C — Richard Mawle

QUESTION KEYWORD	elderly benefit from **no force needed**

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PASSAGE EVIDENCE	"suggested that better designs could rely on **cognitive skills**" instead of strength

28. D — Karel van der Waarde

QUESTION KEYWORD	Braille may **disadvantage sighted**
PASSAGE EVIDENCE	"not known... how much the reading of **visually able patients will be impaired**"

29. B — Child Accident Prevention Trust

QUESTION KEYWORD	containers haven't **eliminated** swallowing
PASSAGE EVIDENCE	"child-resistant lids have helped, they are **not yet fully effective**"

30. A — Thea Swayne

QUESTION KEYWORD	consider how drugs used **at home**
PASSAGE EVIDENCE	"packaging should be designed for moments such as these rather than for the **ideal world of a hospital**"

31. D — Karel van der Waarde

QUESTION KEYWORD	compare containers, not individually
PASSAGE EVIDENCE	"they only see **one box at a time and not several together** as pharmacists would do"

32. C — Richard Mawle

QUESTION KEYWORD	info **not in the right order**
PASSAGE EVIDENCE	"why are **company details towards the beginning**" when more vital info should come first

33. E — marketing team

QUESTION KEYWORD	proposal written by ____
PASSAGE EVIDENCE	"The **marketing team** prepares the initial brief"

34. C — external design team

QUESTION KEYWORD	designs produced by ____

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PASSAGE EVIDENCE	"these are usually commissioned from an **external design team**"

35. A — consumers

QUESTION KEYWORD	shown to ____
PASSAGE EVIDENCE	"Two or three of these are then **tested on a consumer group**"

36. D — in-house design team

QUESTION KEYWORD	____ create the design
PASSAGE EVIDENCE	"most designs for prescription-only products are created **in-house**"

37. B — design engineers

QUESTION KEYWORD	design passed to ____
PASSAGE EVIDENCE	"handing it over to **design engineers**"

38. B

QUESTION KEYWORD	design flaw causing the accident
PASSAGE EVIDENCE	"attributed some blame to the **poor choice of typescript** used on the drug container"

39. C

QUESTION KEYWORD	black-and-white design opinion
PASSAGE EVIDENCE	"people would have no choice but to **read every box carefully**"

40. A

QUESTION KEYWORD	why 'the popper'/'Pluspoint' mentioned
PASSAGE EVIDENCE	"On a **positive note**, a recent innovation exhibition revealed several new designs"

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Answer Key & Explanations

Reading Test 2 (Volume 2)

Passage 1 — Good Nuggets of Australia

1. NOT GIVEN

QUESTION KEYWORD	near a **water source**
PASSAGE EVIDENCE	The passage's definition of placer mining never mentions water sources.

2. NOT GIVEN

QUESTION KEYWORD	villages **still attract** diggers today
PASSAGE EVIDENCE	"Some of these places... are **towns today**" — nothing said about attracting diggers now.

3. FALSE

QUESTION KEYWORD	**small minority** found in Australia
PASSAGE EVIDENCE	"**eight out of the top ten** were found there" — a large majority.

4. FALSE

QUESTION KEYWORD	Welcome Stranger **smaller**
PASSAGE EVIDENCE	"An **even bigger** nugget was found ten years later... the Welcome Stranger"

5. TRUE

QUESTION KEYWORD	finders **sent to trick** them
PASSAGE EVIDENCE	sent to a site "believed to be empty of gold... called **duffer claims**"

6. TRUE

QUESTION KEYWORD	Holtermann **kept a piece**
PASSAGE EVIDENCE	"Only the top of the specimen remains, **cut off by Holtermann as a souvenir**"

7. FALSE

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QUESTION KEYWORD	cut into pieces **above ground**
PASSAGE EVIDENCE	"broken up **below ground** as there seemed no point struggling with it up to the surface"

8. three centimetres

QUESTION KEYWORD	just ____ beneath ground
PASSAGE EVIDENCE	"after digging only **three centimetres**, he was delighted to see the twinkle of gold"

9. stone monument

QUESTION KEYWORD	indicated by a ____
PASSAGE EVIDENCE	"A **stone monument** marks where it was found"

10. gold commissioner

QUESTION KEYWORD	named after a ____
PASSAGE EVIDENCE	"named after a **gold commissioner** called Mr Heron"

11. primary school

QUESTION KEYWORD	back of a ____
PASSAGE EVIDENCE	"found behind a **primary school**"

12. casino

QUESTION KEYWORD	seen at a ____
PASSAGE EVIDENCE	"on display at a **casino** in Las Vegas"

13. quartz

QUESTION KEYWORD	mass also contained ____
PASSAGE EVIDENCE	"The specimen also had **quartz** in it"

Passage 2 — Nature's Useful Designs

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14. E

QUESTION KEYWORD	basic material → incredibly strong
PASSAGE EVIDENCE	Para E: abalone shell from calcium carbonate becomes "armor **3000 times harder** than chalk"

15. I

QUESTION KEYWORD	limited commercial impact
PASSAGE EVIDENCE	Para I: "led to surprisingly **few mass-produced products**... only one household word - Velcro"

16. B

QUESTION KEYWORD	difficulty → accidental discovery
PASSAGE EVIDENCE	Para B: de Mestral "**had trouble removing** a plant's prickle... so he studied it"

17. D

QUESTION KEYWORD	more efficient than household object
PASSAGE EVIDENCE	Para D: glowworm light vs "a normal light bulb **wastes 98 percent** of its energy as heat"

18. C

QUESTION KEYWORD	ancient specimen → innovation
PASSAGE EVIDENCE	Para C: "**45-million-year-old fly** trapped in amber... now being built into solar panels"

19. G

QUESTION KEYWORD	replace human with machine
PASSAGE EVIDENCE	Para G: "deployed in places where people would be too conspicuous, **bored to tears, or killed**"

20. B — Andrew Parker

QUESTION KEYWORD	all living things provide something useful
PASSAGE EVIDENCE	"To Parker, every species... is a **success story**"

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21. D — Ronald Fearing

QUESTION KEYWORD	designs impossible to copy
PASSAGE EVIDENCE	"Some things are just **too mysterious and complex** to be able to replicate"

22. C — Robert Cohen

QUESTION KEYWORD	nothing unless practical application
PASSAGE EVIDENCE	"a success only if it has the potential to make a **useful tool** for people"

23. cosmetics

QUESTION KEYWORD	develop ____
PASSAGE EVIDENCE	"working to make **cosmetics** that mimic the natural sheen of diatoms"

24. birds

QUESTION KEYWORD	some ____ get colour
PASSAGE EVIDENCE	"the metallic sheen and dazzling colors of certain **birds** derive... from microstructures"

25. chemicals

QUESTION KEYWORD	shooting hot ____
PASSAGE EVIDENCE	"heats **chemicals** and fires them at would-be predators"

26. fly

QUESTION KEYWORD	studying the ____ to build a robot
PASSAGE EVIDENCE	"create a miniature robotic **fly**... use in surveillance or search-and-rescue"

Passage 3 — The Art of Deception

27. C

QUESTION KEYWORD	question came from his
PASSAGE EVIDENCE	"addressing a group of young psychiatrists **in training**" — i.e. students.

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28. B

QUESTION KEYWORD	12-minute interviews mentioned to
PASSAGE EVIDENCE	The interview footage is where he **first discovered micro-expressions** — the origin of his theory.

29. A

QUESTION KEYWORD	writer's point, third paragraph
PASSAGE EVIDENCE	"essentially standard, **regardless of language and culture**, from the US to Japan"

30. B

QUESTION KEYWORD	conclusions from his tests
PASSAGE EVIDENCE	"Of the 15,000 Ekman has tested, only **50 people**... have been able to do it"

31. B

QUESTION KEYWORD	point about techniques, fourth paragraph
PASSAGE EVIDENCE	"called in by the **FBI and CIA** (among countless more... agencies)" — great demand.

32. C — false beliefs

QUESTION KEYWORD	encouraging viewers ____
PASSAGE EVIDENCE	"create the quite **inaccurate impression** among audiences that criminals could no longer... lie"

33. E — justice

QUESTION KEYWORD	result in ____ not carried out
PASSAGE EVIDENCE	"concerned about **unfair convictions**... wrongly find someone guilty"

34. H — research

QUESTION KEYWORD	single person's ____
PASSAGE EVIDENCE	"a commercial TV drama has been based on the **work of just one scientist**"

35. D — motives

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QUESTION KEYWORD	____ of the producer
PASSAGE EVIDENCE	"impressed with the producer's **manifestly serious and well-intentioned reasons**"

36. F — accuracy

QUESTION KEYWORD	show's overall ____
PASSAGE EVIDENCE	"believes probably **80-90 per cent** of the show is based on fact"

37. NOT GIVEN

QUESTION KEYWORD	**regrets** childhood lies
PASSAGE EVIDENCE	"professes to have been a terrible liar ever since he was a small boy" — no mention of regret.

38. FALSE

QUESTION KEYWORD	good liars, good at **detecting**
PASSAGE EVIDENCE	"the ability to detect a lie and the ability to lie successfully are **completely unrelated**"

39. FALSE

QUESTION KEYWORD	worked with **poker players** to lie better
PASSAGE EVIDENCE	Poker players are mentioned only as an example of people who **aren't "true liars"** — not as trainees.

40. TRUE

QUESTION KEYWORD	more interested in **serious consequences**
PASSAGE EVIDENCE	"prefers to focus on the kinds of lies where the liar would be in **grave trouble** if found out"

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Answer Key & Explanations

Reading Test 3 (Volume 2)

Passage 1 — Footprints in the Muds of Time

1. NO

QUESTION KEYWORD	still **doubt** about asteroid theory
PASSAGE EVIDENCE	"**Everybody knows** that the dinosaurs became extinct as a result of a large asteroid" — stated as settled fact.

2. NO

QUESTION KEYWORD	books/movies **exaggerated** size
PASSAGE EVIDENCE	"turned into the monsters **realistically depicted** in modern books and movies"

3. NOT GIVEN

QUESTION KEYWORD	others **rejected** Olsen's idea
PASSAGE EVIDENCE	"not the first to suggest... But they are the first to show" — no mention of rejection by others.

4. YES

QUESTION KEYWORD	footprints **more frequent** than skeletons
PASSAGE EVIDENCE	"Dinosaur skeletons are rare. Dinosaur footprints are, however, **surprisingly abundant**"

5. NO

QUESTION KEYWORD	ichnotaxa **exact species** ID
PASSAGE EVIDENCE	"cannot be matched precisely within the species... But they can be matched with a **general sort** of animal"

6. YES

QUESTION KEYWORD	groups **survived** Triassic into Jurassic
PASSAGE EVIDENCE	"four march confidently **across the boundary into the Jurassic**"

7. ecological release

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QUESTION KEYWORD	increase in size due to ____
PASSAGE EVIDENCE	"a phenomenon called **ecological release**"

8. dragons

QUESTION KEYWORD	lizards called ____
PASSAGE EVIDENCE	"often referred to as **dragons**"

9. competitors

QUESTION KEYWORD	no ____
PASSAGE EVIDENCE	"reach islands where they face no **competitors**"

10. overlooked

QUESTION KEYWORD	may have been ____
PASSAGE EVIDENCE	"It may, of course, have been **overlooked**"

11. vanished

QUESTION KEYWORD	could have ____
PASSAGE EVIDENCE	"Alternatively, it may have **vanished**"

12. tectonic processes

QUESTION KEYWORD	____ that produce continental drift
PASSAGE EVIDENCE	"the **tectonic processes** that bring about continental drift"

13. misdated

QUESTION KEYWORD	hole exists but ____
PASSAGE EVIDENCE	"the crater is known, but has been **misdated**"

Passage 2 — The Constant Evolution of the Humble Tomato

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14. E

QUESTION KEYWORD	research restoring heirloom's health
PASSAGE EVIDENCE	Para E: "getting **germ-beating genes back** into heirlooms... 40 disease-resistant genes"

15. A

QUESTION KEYWORD	false belief about heirlooms
PASSAGE EVIDENCE	Para A: myth of "more diverse and superior... genes" vs "no more than **10 mutant genes**"

16. E

QUESTION KEYWORD	description of flavour
PASSAGE EVIDENCE	Para E: "**juicier, sweeter and more flavorful** fruit"

17. D

QUESTION KEYWORD	single gene improves cultivation
PASSAGE EVIDENCE	Para D: gene "increases fruit size by **50 percent**... the most important event in domestication"

18. B — Esther van der Knaap

QUESTION KEYWORD	transplanting genes changes shape
PASSAGE EVIDENCE	"inserted it into a wild relative... the tiny fruits became **shaped like pears**"

19. C — Roger Chetelat

QUESTION KEYWORD	flavour depends on yield/cultivation
PASSAGE EVIDENCE	"less to do with their genes than with the **productivity of the plant**"

20. D — Doug Heath

QUESTION KEYWORD	stronger yet equally flavoursome
PASSAGE EVIDENCE	"less prone to cracking... maintain a **comparable flavor and sugar profile**"

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21. A — Steven Tanksley

QUESTION KEYWORD	variety from few genes
PASSAGE EVIDENCE	"diversity of heirlooms can be accounted for by a **handful of genes**"

22. archaeological

QUESTION KEYWORD	existence of ____ data
PASSAGE EVIDENCE	"scientists have uncovered a wealth of **archaeological** evidence"

23. Andes

QUESTION KEYWORD	first grown in the ____
PASSAGE EVIDENCE	"has its origins in the **Andes** in South America"

24. currant

QUESTION KEYWORD	____ tomato nearest earliest type
PASSAGE EVIDENCE	"The closest relative is the **currant tomato**"

25. mutation

QUESTION KEYWORD	genetic ____ in pomodoro
PASSAGE EVIDENCE	"this **mutation**, which enlarges tomatoes... existed back in the same yellow tomatoes"

26. Infections

QUESTION KEYWORD	____ cause damage
PASSAGE EVIDENCE	"suffer from **infections** that cause the fruit to crack, split and otherwise rot"

Passage 3 — Crossing the Threshold

27. C

QUESTION KEYWORD	writer's main point, first paragraph
PASSAGE EVIDENCE	"the situation is much worse when architects work on **municipal buildings**"

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28. D

QUESTION KEYWORD	particularly difficult because
PASSAGE EVIDENCE	"involved two parts. The first was to **restore**... The second was to **deliver a new extension**"

29. A

QUESTION KEYWORD	disturbing info from survey
PASSAGE EVIDENCE	"'threshold fear'... intimidated from entering certain spaces by their **off-putting atmosphere**"

30. B

QUESTION KEYWORD	16% didn't know location, point
PASSAGE EVIDENCE	gallery was "fatally out of step" while the national museum was "**engaging broader audiences**" — young people simply weren't being reached.

31. C

QUESTION KEYWORD	Maori artists used to
PASSAGE EVIDENCE	"ensuring the gallery **emphasised Maori beliefs**"

32. YES

QUESTION KEYWORD	before renovation, **elitist**
PASSAGE EVIDENCE	"served a **limited range of patrons with highbrow interests**"

33. NO

QUESTION KEYWORD	King's intervention shows **understanding**
PASSAGE EVIDENCE	"King's **misconceptions** sum up the prejudice that surrounded the project" — he misunderstood.

34. YES

QUESTION KEYWORD	interaction with surroundings, triumph
PASSAGE EVIDENCE	"the project's relationship with parkland is **one of the most successful outcomes**"

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35. NOT GIVEN

QUESTION KEYWORD	glass screen **most costly**
PASSAGE EVIDENCE	No cost figures are given for the glass wall specifically.

36. NO

QUESTION KEYWORD	**similar** to 'white cube' galleries
PASSAGE EVIDENCE	"a **far cry from** both 'white cube' galleries worldwide"

37. F

QUESTION KEYWORD	destruction of Edwardian ornamentation
PASSAGE EVIDENCE	decoration "had been stripped out or walled away in **previous renovations**" — earlier modernisation attempts.

38. C

QUESTION KEYWORD	limited photographs, extraordinary
PASSAGE EVIDENCE	"the Mackelvie space has been **reconstructed from two old photos**"

39. A

QUESTION KEYWORD	floor levels problem
PASSAGE EVIDENCE	"scaffolding had to be erected at the highest level... the **reverse of normal practice**"

40. D

QUESTION KEYWORD	glass flooring reveals old features
PASSAGE EVIDENCE	"the **building itself as artwork**"

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Answer Key & Explanations

Reading Test 4 (Volume 2)

Passage 1 — Jewels from the Sea

1. FALSE

QUESTION KEYWORD	**stopped** making necklaces briefly
PASSAGE EVIDENCE	"one of the few traditions that have **continued without interruption**"

2. NOT GIVEN

QUESTION KEYWORD	Furneaux women made **most beautiful**
PASSAGE EVIDENCE	Only says knowledge was passed down there — no comparison of beauty across regions.

3. NOT GIVEN

QUESTION KEYWORD	necklace she **made herself**
PASSAGE EVIDENCE	Only states the photo "shows... wearing necklaces" — not that she made it herself.

4. TRUE

QUESTION KEYWORD	men help gather shells on **sea plants**
PASSAGE EVIDENCE	"maireeners... which are found on **kelp**, a type of seaweed"

5. FALSE

QUESTION KEYWORD	men wore long necklaces **hunting**
PASSAGE EVIDENCE	"would not have risked getting **snagged by long necklaces**"

6. TRUE

QUESTION KEYWORD	appreciated **outside Australia**
PASSAGE EVIDENCE	"inclusion in many national and **international** museum, gallery and private collections"

7. tooth

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QUESTION KEYWORD	bone and ____
PASSAGE EVIDENCE	"a jawbone and sharpened **tooth** of a kangaroo or wallaby"

8. grass

QUESTION KEYWORD	wiped with ____
PASSAGE EVIDENCE	"rubbed in **grass** to remove their outer coating"

9. oil

QUESTION KEYWORD	animal ____ applied
PASSAGE EVIDENCE	"treated with penguin or muttonbird **oil**"

10. vinegar

QUESTION KEYWORD	cleaning substance ____
PASSAGE EVIDENCE	"acidic liquids such as **vinegar** to clean the shells"

11. season

QUESTION KEYWORD	right ____
PASSAGE EVIDENCE	"shell collection has its **season**"

12. crawl

QUESTION KEYWORD	walk then ____
PASSAGE EVIDENCE	"**crawl** along on our hands and knees to get the shells"

13. colour

QUESTION KEYWORD	don't keep ____
PASSAGE EVIDENCE	"too brittle and they lose their **colour**"

Passage 2 — The Return of Monkey Life

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14. C

QUESTION KEYWORD	newer forests, better feeding
PASSAGE EVIDENCE	Para C: younger trees "produce much **lower levels of toxin** than... old-growth trees"

15. G

QUESTION KEYWORD	change in farmers' attitudes
PASSAGE EVIDENCE	Para G: "We're **moving away from that vision**" of conservation vs. farming conflict

16. C

QUESTION KEYWORD	howlers select best diet
PASSAGE EVIDENCE	Para C: "sampling a variety of plants and then **focusing on... the most nutritious**"

17. B

QUESTION KEYWORD	habitat reduction figures
PASSAGE EVIDENCE	Para B: "by the 1980s **less than two percent** remained undisturbed"

18. reproduction

QUESTION KEYWORD	more rapid rate of ____
PASSAGE EVIDENCE	"Howler **reproduction** is faster than that of other native monkey species"

19. fruit

QUESTION KEYWORD	survive without eating ____
PASSAGE EVIDENCE	"unlike capuchin and spider monkeys, do not depend on large areas of **fruiting** trees"

20. toxins

QUESTION KEYWORD	tolerate leaves with high ____
PASSAGE EVIDENCE	"thrive on leaves loaded with **toxins**"

21. drought

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QUESTION KEYWORD	survive periods of ____
PASSAGE EVIDENCE	"suffered during the long, ongoing **drought**"

22. **B — Santa Rosa**

QUESTION KEYWORD	return of native tree species
PASSAGE EVIDENCE	"species of indigenous trees which had been absent for decades **began to invade**"

23. **A — Hacienda La Pacífica**

QUESTION KEYWORD	only one native monkey species
PASSAGE EVIDENCE	capuchins and spider monkeys "**vanished**" leaving only howlers there.

24. **C — Cholula Cacao Farm**

QUESTION KEYWORD	monkeys help agriculture
PASSAGE EVIDENCE	"dispersing the seeds... and **fertilizing the soil**"

25. **B — Santa Rosa**

QUESTION KEYWORD	all three local species
PASSAGE EVIDENCE	"Capuchins were the first... followed by howlers. Eventually even **spider monkeys**... returned"

26. **A — Hacienda La Pacífica**

QUESTION KEYWORD	transport link construction
PASSAGE EVIDENCE	"vanished after the **Pan-American Highway** was built nearby"

Passage 3 — Innovation in Business

27. **NO**

QUESTION KEYWORD	most people **understand** innovation
PASSAGE EVIDENCE	"the majority of us have **little comprehension** of the processes"

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28. YES

QUESTION KEYWORD	word **overused** in NZ business
PASSAGE EVIDENCE	"in company boardrooms across New Zealand, the term is heard **far too frequently**"

29. YES

QUESTION KEYWORD	many interpretations, **global study**
PASSAGE EVIDENCE	"a recent international survey showed that the idea... is so broad as to appear **almost meaningless**"

30. NOT GIVEN

QUESTION KEYWORD	R&D **higher priority** today
PASSAGE EVIDENCE	No comparison of R&D priority between past and present is given.

31. C

QUESTION KEYWORD	writer's action, third paragraph
PASSAGE EVIDENCE	"we must **reject the false assumption** that innovation is just a moment of brilliance"

32. B

QUESTION KEYWORD	main point, fourth paragraph (Jobs)
PASSAGE EVIDENCE	"fuel for his visionary innovations... his egotistical behaviour could have **undesirable consequences**, too"

33. D

QUESTION KEYWORD	Gutenberg press referred to
PASSAGE EVIDENCE	"A good case to cite in support of this **universal truth**" (the hummingbird effect)

34. A

QUESTION KEYWORD	main point, seventh paragraph (Kode)
PASSAGE EVIDENCE	"Kode's approach is to **collaborate with similar nano-technology businesses**"

35. D

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QUESTION KEYWORD	companies like Previously Unavailable
PASSAGE EVIDENCE	"will **lease a team of innovation experts**... to assist with conceiving, developing and launching new products"

36. E

QUESTION KEYWORD	Edison book suggests
PASSAGE EVIDENCE	"Edison's own inspiration usually resulted from **laborious experimentation**"

37. G

QUESTION KEYWORD	Jobs' former employees claim
PASSAGE EVIDENCE	"demanding behaviour affects sickness rates and group morale, which in turn may **squash creativity**"

38. C

QUESTION KEYWORD	critics of Harford's talk claimed
PASSAGE EVIDENCE	"was **not persuasive**, because it was **stating the obvious**"

39. B

QUESTION KEYWORD	hummingbird effect proposes
PASSAGE EVIDENCE	"progress in a specific area... can end up **triggering... developments elsewhere** entirely"

40. D

QUESTION KEYWORD	Steve Henry believes
PASSAGE EVIDENCE	"Kode's approach is to **collaborate with similar** nano-technology businesses"

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Answer Key & Explanations

Reading Test 5 (Volume 2)

Passage 1 — What Lucy Taught Us

1. FALSE

QUESTION KEYWORD	Johanson **uncertain** about the bone
PASSAGE EVIDENCE	"Straight away, he **recognised** it as coming from the elbow"

2. TRUE

QUESTION KEYWORD	several bones, **same site**
PASSAGE EVIDENCE	"bits of the skull, a chunk of jaw, a couple of **vertebrae**"

3. TRUE

QUESTION KEYWORD	experts **realised importance**
PASSAGE EVIDENCE	"It was immediately obvious that the skeleton was a **significant find**"

4. NOT GIVEN

QUESTION KEYWORD	**upper part** least damaged
PASSAGE EVIDENCE	Only "40% of the skeleton had been preserved" overall — no comparison between upper and lower parts.

5. TRUE

QUESTION KEYWORD	measurements decided **sex**
PASSAGE EVIDENCE	"the feeling was that the skeleton was female **due to its size**"

6. trees

QUESTION KEYWORD	upright movement among the ____
PASSAGE EVIDENCE	"upright walking evolved in the **trees**"

7. ground

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QUESTION KEYWORD	moved to the ____
PASSAGE EVIDENCE	"the real reason for heading to the **ground**"

8. teeth

QUESTION KEYWORD	food in the ____
PASSAGE EVIDENCE	"food trapped on preserved human **teeth**"

9. fruit

QUESTION KEYWORD	substituted for ____
PASSAGE EVIDENCE	"Instead of mostly eating **fruit** from trees, they began to include grasses and meat"

10. Eggs

QUESTION KEYWORD	____ close to Lucy
PASSAGE EVIDENCE	"Fossilised crocodile and turtle **eggs** were found near her skeleton"

11. Bones

QUESTION KEYWORD	____ found with marks
PASSAGE EVIDENCE	"archaeologists uncovered animal **bones** with scratches"

12. childhood

QUESTION KEYWORD	longer ____
PASSAGE EVIDENCE	"Lucy's **childhood** was much briefer than ours"

13. brains

QUESTION KEYWORD	____ develop later
PASSAGE EVIDENCE	"their **brains** matured much earlier than ours do" (i.e. ours mature later)

Passage 2 — The Gender Gap in New Zealand's High School Examination Results

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14. pass rates

QUESTION KEYWORD	____ higher by 10%
PASSAGE EVIDENCE	"the difference in **pass rates** between the sexes... about 10 percentage points"

15. decade

QUESTION KEYWORD	apparent for over a ____
PASSAGE EVIDENCE	"outperforming boys academically for **more than a decade**"

16. Australia

QUESTION KEYWORD	noticed in ____ also
PASSAGE EVIDENCE	"within **Australia** was the subject of much debate"

17. C

QUESTION KEYWORD	advantage of NZ tests
PASSAGE EVIDENCE	Para C: "provides a **fine-grained picture** of pupils' performance"

18. B

QUESTION KEYWORD	current government initiatives
PASSAGE EVIDENCE	Para B: "working with a **reference group on boys' education**"

19. D

QUESTION KEYWORD	when literacy gap first evident
PASSAGE EVIDENCE	Para D: "shows up as early as **preschool**"

20. E

QUESTION KEYWORD	achievement relates to race
PASSAGE EVIDENCE	Para E: "**Ethnic differences**... may be more significant than the simple boy/girl explanation"

21. H — Celia Lashlie

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QUESTION KEYWORD	resistance within Ministry
PASSAGE EVIDENCE	"believe there is still **resistance within the Education Ministry**"

22. E — Steve Benson

QUESTION KEYWORD	gap only in some subjects
PASSAGE EVIDENCE	"in maths, there isn't really a gender gap. But **literacy is a different matter**"

23. G — Roger Moses

QUESTION KEYWORD	NCEA higher literacy standards
PASSAGE EVIDENCE	"the written content of NCEA papers is **more demanding** than the previous system"

24. C — Rob Burroughs

QUESTION KEYWORD	good teaching more important
PASSAGE EVIDENCE	"the difference was, instead, to do with the **quality of instruction**"

25. F — Stuart Martin

QUESTION KEYWORD	older boys more motivated
PASSAGE EVIDENCE	"competitive instinct tends to come out **later** in their schooling years"

26. B — Paul Baker

QUESTION KEYWORD	disadvantaged by girls taking over
PASSAGE EVIDENCE	"easier in a boys' school, where activities cannot be '**captured by girls**'"

Passage 3 — The Strange World of Sight

27. D

QUESTION KEYWORD	why refer to Locke and Newton
PASSAGE EVIDENCE	"This is still **not generally recognised even now, 400 years later**" — the idea is old, not new.

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28. C

QUESTION KEYWORD	why Freud's theory mocked
PASSAGE EVIDENCE	"undermined the notion of humans as **pre-eminently rational beings**"

29. C

QUESTION KEYWORD	Mr S failing to get up, demonstrates
PASSAGE EVIDENCE	perceptions are normally far more vivid than memories, so confusing the two is **unusual**.

30. A

QUESTION KEYWORD	writer's point, whole text
PASSAGE EVIDENCE	Perception depends on "knowledge from the past" and inference, not eye data alone — **much more than the eyes**.

31. NO

QUESTION KEYWORD	relied on things he'd been **told about**
PASSAGE EVIDENCE	"It was his **touch memories** that enabled him to perceive them" — his own experience, not being told.

32. NOT GIVEN

QUESTION KEYWORD	learn to see **as others do**
PASSAGE EVIDENCE	Only says sight "gradually improved" — no claim they reach parity with typically-sighted people.

33. YES

QUESTION KEYWORD	activities **harder** than before
PASSAGE EVIDENCE	May "would have to **shut his eyes while skiing**" — a task now harder with sight.

34. NO

QUESTION KEYWORD	eyes + **one** brain area
PASSAGE EVIDENCE	"now it seems that **half the brain** is occupied with seeing" — not just one region.

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35. NOT GIVEN

QUESTION KEYWORD	mask experiment, **training** use
PASSAGE EVIDENCE	Used to illustrate normal perception — no claim it's used for training regaining sight.

36. YES

QUESTION KEYWORD	perfect vision, still **fail**
PASSAGE EVIDENCE	"even people with no visual impairment see what... they expect to see"

37. D — convex

QUESTION KEYWORD	mask still ____ in shape
PASSAGE EVIDENCE	"the back of the mask will be seen as **convex**, though we know it must be concave"

38. F — nose

QUESTION KEYWORD	the ____ poking out
PASSAGE EVIDENCE	"a convex shape with the **nose** sticking out"

39. I — drawing

QUESTION KEYWORD	make a ____ of the mask
PASSAGE EVIDENCE	"impossible to **sketch** the back of a hollow mask to look as it is hollow"

40. J — preconception

QUESTION KEYWORD	the ____ they have
PASSAGE EVIDENCE	"people not seeing a hollow face as hollow" because they **expect** faces to stick out.

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Answer Key & Explanations

Reading Test 6 (Volume 2)

Passage 1 — Why Should We Study History?

1. B — Arnold Toynbee

QUESTION KEYWORD	understand **human nature**
PASSAGE EVIDENCE	"a single person is capable of the whole range of behaviour from **cruelty to kindness**"

2. F — Diane Ravich

QUESTION KEYWORD	improve **decision-making**
PASSAGE EVIDENCE	"helps to inform us and thus to **improve our judgment**, so that we make better choices"

3. C — John Dewey

QUESTION KEYWORD	understand the **present**
PASSAGE EVIDENCE	"the only tool we have for making sense of the reality we experience is **what has occurred in the past**"

4. D — Friedrich Nietzsche

QUESTION KEYWORD	examples to **copy**
PASSAGE EVIDENCE	"people need **models to imitate**"

5. D — change

QUESTION KEYWORD	history is a process of ____
PASSAGE EVIDENCE	"saw history as a **dynamic process**, one in which nothing is static"

6. H — liberty

QUESTION KEYWORD	Hegel: increase in ____
PASSAGE EVIDENCE	"leading towards greater **freedom**"

7. B — decline

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QUESTION KEYWORD	Spengler: ends in ____
PASSAGE EVIDENCE	"growth, flowering and deterioration, a rise inevitably leading to a **fall**"

8. E — identity

QUESTION KEYWORD	Southgate: nation's ____
PASSAGE EVIDENCE	"societies... need to know who they are" — provides a nation with meaning, purpose and cohesion (**identity**)

9. I — tolerance

QUESTION KEYWORD	encourages ____ of other cultures
PASSAGE EVIDENCE	"our present-day customs are **not the only ones which are acceptable**"

10. conflict

QUESTION KEYWORD	avoid ____ with society
PASSAGE EVIDENCE	"run their own lives without coming into **conflict** with the community"

11. individuals

QUESTION KEYWORD	compares societies with ____
PASSAGE EVIDENCE	"societies, like **individuals**, need to know who they are"

12. amnesia

QUESTION KEYWORD	suffering from ____
PASSAGE EVIDENCE	"a society that cannot recall its past is like an individual with **amnesia**"

13. evidence

QUESTION KEYWORD	analyse ____
PASSAGE EVIDENCE	"the habit of critically examining and evaluating **evidence**"

Passage 2 — Violins and Very Cold Weather

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14. v

QUESTION KEYWORD	an ongoing debate
PASSAGE EVIDENCE	Para A: "considerable controversy... is still **fuelling discussion**"

15. vii

QUESTION KEYWORD	disproving hypotheses
PASSAGE EVIDENCE	Para B: "**no basis has been found** for these possible explanations"

16. vi

QUESTION KEYWORD	materials for different parts
PASSAGE EVIDENCE	Para C: "Maple wood... for the back... while **spruce**... for the top"

17. viii

QUESTION KEYWORD	well-known source of wood
PASSAGE EVIDENCE	Para D: the famous "**Forest of the Violins**"

18. i

QUESTION KEYWORD	needed further investigation
PASSAGE EVIDENCE	Para E: "questions that **required more thorough research**"

19. ix

QUESTION KEYWORD	exceptional environmental conditions
PASSAGE EVIDENCE	Para F: the Maunder Minimum, "**narrow rings**" in tree cross-sections

20. iii

QUESTION KEYWORD	unrepeated mix of circumstances
PASSAGE EVIDENCE	Para G: "conjunction of elevation, topography, soil properties and... climate was **temporally unique**"

21. skills

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QUESTION KEYWORD	due to the ____ of the people
PASSAGE EVIDENCE	"the **skills** of these Cremonese artisans"

22. castles

QUESTION KEYWORD	wood used in old buildings
PASSAGE EVIDENCE	"the re-use of old wood taken from **castles**"

23. ovens

QUESTION KEYWORD	used to dry wood
PASSAGE EVIDENCE	"drying the wood in **ovens**"

24. chemicals

QUESTION KEYWORD	remove dangerous ____
PASSAGE EVIDENCE	"get rid of supposedly harmful **chemicals**"

25 & 26. A & E

QUESTION KEYWORD	factors affecting wood quality
PASSAGE EVIDENCE	"grown in thin, **poor soils** (E) at **higher elevations** (A) on north-facing slopes"

Passage 3 — What Should Companies Do to Survive?

27. NOT GIVEN

QUESTION KEYWORD	too few companies **understand** theories
PASSAGE EVIDENCE	Passage says it is essential to know the enduring rules, but never states how many companies actually understand them.

28. YES

QUESTION KEYWORD	practices for **unstable situations**
PASSAGE EVIDENCE	"essential to have a clear idea about the **enduring rules** of good management... strongest position amid the uncertainties"

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29. YES

QUESTION KEYWORD	ideas → **financial** improvement
PASSAGE EVIDENCE	"ideas have value only if people... develop them in ways that **increase revenue**"

30. NO

QUESTION KEYWORD	decided by **all users**
PASSAGE EVIDENCE	"**senior staff** set careful rules to filter and structure it" — not all users

31. NOT GIVEN

QUESTION KEYWORD	more **training** needed
PASSAGE EVIDENCE	Managers must "build in the data that matters" — no mention of training being needed.

32. NO

QUESTION KEYWORD	change introduced **all at once**
PASSAGE EVIDENCE	"Big-bang decisions are generally best **avoided** — or implemented in small incremental moves"

33. YES

QUESTION KEYWORD	different strategies, different customers
PASSAGE EVIDENCE	"offer their most profitable customers **special deals**"; even "'fire' unprofitable customers"

34. NO

QUESTION KEYWORD	activities **common to all** companies
PASSAGE EVIDENCE	"a company should keep those activities it does **better than its competitors**" — not defined by commonality

35. F

QUESTION KEYWORD	qualities for co-operating
PASSAGE EVIDENCE	Para F: "Effective collaboration calls for **trust and shared understanding**... excellent communication"

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36. H

QUESTION KEYWORD	all staff, same practices
PASSAGE EVIDENCE	Para H: "common practices in areas such as **purchasing and information technology**"

37. E

QUESTION KEYWORD	alternative employment patterns
PASSAGE EVIDENCE	Para E: "hire the talents of '**free agents**'... others, to employ directly their best brains"

38. F

QUESTION KEYWORD	in-house vs outsource basis
PASSAGE EVIDENCE	Para F: "a company should keep those activities it **does better than its competitors**"

39. I

QUESTION KEYWORD	info shared with customers
PASSAGE EVIDENCE	Para I: "allow their suppliers and customers '**inside the machine**'... access to their databases"

40. D

QUESTION KEYWORD	why customers keep buying
PASSAGE EVIDENCE	Para D: "companies must build loyalty by providing **reliability and good service**"

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Answer Key & Explanations

Reading Test 7 (Volume 2)

Passage 1 — Book Review: Triumph of the City

1. disease

QUESTION KEYWORD	problems with early cities
PASSAGE EVIDENCE	"lacked efficient waste disposal systems, and **disease** spread rapidly"

2. political

QUESTION KEYWORD	commercial and ____ reasons
PASSAGE EVIDENCE	"enormous commercial incentives... as well as **political** ones"

3. richer

QUESTION KEYWORD	poor people generally get ____
PASSAGE EVIDENCE	"the hope of becoming **richer** (which, by and large, they do)"

4. economy

QUESTION KEYWORD	help develop the urban ____
PASSAGE EVIDENCE	"reinvigorate the **economy** of the city"

5. museum

QUESTION KEYWORD	centre of Paris becoming a ____
PASSAGE EVIDENCE	"The heart of Paris... is turning into a **museum**"

6. property

QUESTION KEYWORD	high ____ prices in inner cities
PASSAGE EVIDENCE	"drove up the cost of **property**"

7. contact

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QUESTION KEYWORD	less personal ____
PASSAGE EVIDENCE	"lose the face-to-face **contact** that makes being an urbanite so much more creative"

8. consumption

QUESTION KEYWORD	increased ____ of resources
PASSAGE EVIDENCE	"Every aspect of life involves more **consumption**"

9. FALSE

QUESTION KEYWORD	congestion/poverty indicate **serious problems**
PASSAGE EVIDENCE	"They shouldn't, says Glaeser — they are **signs of growth, energy and aspiration**"

10. NOT GIVEN

QUESTION KEYWORD	Thoreau **discussed** Romantic poets' ideas
PASSAGE EVIDENCE	Thoreau is mentioned separately; no statement that his work discussed the Romantic poets.

11. NOT GIVEN

QUESTION KEYWORD	Napoleon III influenced by **complaints**
PASSAGE EVIDENCE	Only says he "gave Baron Haussmann unrestricted power" — no mention of complaints from the poor.

12. FALSE

QUESTION KEYWORD	strict planning regulations **beneficial**
PASSAGE EVIDENCE	"It is folly to drive them away by forcing property prices to soar with **unreasonable planning regulations**"

13. TRUE

QUESTION KEYWORD	La Défense location a **bad idea**
PASSAGE EVIDENCE	"argues that in the 1950s the French made **a mistake** in establishing... La Défense"

Passage 2 — Coral Reefs

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14. A

QUESTION KEYWORD	geographical location of reefs
PASSAGE EVIDENCE	Para A: "shallow-water reefs form only in a zone extending from **30°N to 30°S**"

15. C

QUESTION KEYWORD	benefits local economy
PASSAGE EVIDENCE	Para C: reef tourism as "an important source of **employment**" for local people

16. C

QUESTION KEYWORD	economic **statistics**
PASSAGE EVIDENCE	Para C: "AU$4.3 billion", "US$2 billion", "$625 million" figures

17. D

QUESTION KEYWORD	reasons for decline
PASSAGE EVIDENCE	Para D: "coral mining, agricultural and urban runoff, pollution... disease"

18. E

QUESTION KEYWORD	physical approach by people
PASSAGE EVIDENCE	Para E: "people **grab, kick, and walk on**... or drop anchors"

19. D

QUESTION KEYWORD	unsustainable fishing, by region
PASSAGE EVIDENCE	Para D: cyanide fishing "sold in restaurants, primarily in **Asia**"

20. YES

QUESTION KEYWORD	habitat to variety of marine life
PASSAGE EVIDENCE	"provide a home for **25% of all marine species**"

21. YES

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QUESTION KEYWORD	distribute **disproportionally**
PASSAGE EVIDENCE	"The Indo-Pacific region accounts for **91.9%** of this total area" vs. much smaller shares elsewhere

22. NOT GIVEN

QUESTION KEYWORD	**increasingly** important scientifically
PASSAGE EVIDENCE	Para F mentions scientific study of reefs but not that this importance is increasing.

23. NOT GIVEN

QUESTION KEYWORD	exchanged/exported among countries
PASSAGE EVIDENCE	Aquarium fish export is mentioned (Sri Lanka), but no general claim about reefs being exchanged among countries.

24. YES

QUESTION KEYWORD	economic essence for poor people
PASSAGE EVIDENCE	"an important source of employment, especially for some of the world's **poorest people**"

25. NOT GIVEN

QUESTION KEYWORD	not suitable for **women and children**
PASSAGE EVIDENCE	No comparison of coral fishery's suitability for women/children is made anywhere in the passage.

26. C

QUESTION KEYWORD	main purpose of passage
PASSAGE EVIDENCE	Passage covers benefits (B, C) then threats (D–F) — "**general benefits and an alarming situation**"

Passage 3 — Whale Culture

27. TRUE

QUESTION KEYWORD	remain with maternal group **for life**

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PASSAGE EVIDENCE	"Calves stay with their mothers throughout adulthood... **no one has ever seen a whale switch pods**"

28. FALSE

QUESTION KEYWORD	residents more **restricted** calls
PASSAGE EVIDENCE	"Residents have a much **more extensive repertoire**" than transients — the opposite

29. TRUE

QUESTION KEYWORD	vocabulary common to **all** transients
PASSAGE EVIDENCE	"Transients have only a few such calls, and **all transient societies share the same ones**"

30. FALSE

QUESTION KEYWORD	share dialects of other communities
PASSAGE EVIDENCE	"each resident pod sticks **firmly to its own dialect**"

31. NOT GIVEN

QUESTION KEYWORD	transient dialects constant **over time**
PASSAGE EVIDENCE	Only resident dialects are said to be "maintained for at least 40 years" — nothing stated about transients over time.

32. pods

QUESTION KEYWORD	family groups known as ____
PASSAGE EVIDENCE	"Residents live in stable groups, or '**pods**'"

33. genetics

QUESTION KEYWORD	differences not from whales' ____
PASSAGE EVIDENCE	"we can throw out the notion that the dialects are **inherited**"

34. mother

QUESTION KEYWORD	dialect of group its ____ belongs to

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PASSAGE EVIDENCE	"A calf uses the calls of its **maternal pod** very precisely"

35–37. **A, C, D**

QUESTION KEYWORD	features of whales mentioned
PASSAGE EVIDENCE	Para F/H: "**advanced mental abilities**" (A), "adept at **recognising sounds**" (C), females "live up to a quarter of a century after their last offspring" — **prolonged life span** (D)

38. **G**

QUESTION KEYWORD	example of information passed
PASSAGE EVIDENCE	Para G: "share information about **food hot spots**"

39. **D**

QUESTION KEYWORD	variation in communication between cultures
PASSAGE EVIDENCE	Para D: "each family group has its own unique and distinctive set of **calls**"

40. **F**

QUESTION KEYWORD	skills favourable for culture
PASSAGE EVIDENCE	Para F: "several biological attributes that give them an **advantage in social learning**"

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Answer Key & Explanations

Reading Test 8 (Volume 2)

Passage 1 — The Iceman

1. E

QUESTION KEYWORD	last area he might live/stay
PASSAGE EVIDENCE	Para E: "The Iceman probably set off on his **final journey from this very area**, near where the... Rivers meet"

2. G

QUESTION KEYWORD	mass of special plant, used to analyze movements
PASSAGE EVIDENCE	Para G: "the most prominent moss, **Neckera complanata**... a clot of stems found in the Iceman's possession"

3. H

QUESTION KEYWORD	scientist analyzes last hike via pollen
PASSAGE EVIDENCE	Para H: "Oeggl has also found traces of **pine pollen** in the Iceman's digestive tract"

4. B

QUESTION KEYWORD	time and area found
PASSAGE EVIDENCE	Para B: "hikers discovered his mummified corpse in **1991** in a rocky hollow high in the **Otztal Alps**"

5. F

QUESTION KEYWORD	body out of condition for months
PASSAGE EVIDENCE	Para F: "suffered three episodes of significant disease during the **last six months of life**"

6. FALSE

QUESTION KEYWORD	another corpse must be **older**
PASSAGE EVIDENCE	He is described as "the **oldest intact** member of the human family" — implying no older complete corpse exists

7. NOT GIVEN

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QUESTION KEYWORD	leader, **very rich**
PASSAGE EVIDENCE	Only "a person of considerable social significance" from the ax — no mention of wealth or leadership specifically

8. FALSE

QUESTION KEYWORD	scientists guessed **perfectly**
PASSAGE EVIDENCE	Past theories "have not worn nearly as well as his rustic garments" — many were wrong (shepherd, shaman, vegan, etc.)

9. TRUE

QUESTION KEYWORD	testing tells us where he **came from**
PASSAGE EVIDENCE	Isotope analysis "shown that he probably **grew up in the Valle Isarco**"

10. refrigerated, high-tech

QUESTION KEYWORD	placed in a ____ room
PASSAGE EVIDENCE	"resided in a **refrigerated, high-tech chamber**" since 1998

11. stomach distress

QUESTION KEYWORD	get ____ from whipworm eggs
PASSAGE EVIDENCE	"eggs of the whipworm parasite, so he may well have suffered from **stomach distress**"

12. Neckera complanata

QUESTION KEYWORD	mosses/liverworts such as ____
PASSAGE EVIDENCE	"The most prominent moss, **Neckera complanata**"

13. straight

QUESTION KEYWORD	route might not be ____
PASSAGE EVIDENCE	"it may not have been a **straight** hike into the mountains"

Passage 2 — Twin Study: Two of a Kind

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14. J

QUESTION KEYWORD	research in Ohio
PASSAGE EVIDENCE	Para J: "Peter Miraldi, of **Kent State University in Ohio**"

15. D

QUESTION KEYWORD	medical contribution
PASSAGE EVIDENCE	Para D: "twin studies are helping the understanding of **diseases such as cancer, asthma**"

16. E

QUESTION KEYWORD	research under life-threatening conditions
PASSAGE EVIDENCE	Para E: Mengele's experiments at the "**Auschwitz extermination camp**"

17. B

QUESTION KEYWORD	data on identical twins' similarities
PASSAGE EVIDENCE	Para B: "correlation of skin-mole counts between identical twins was **0.4**"

18. E

QUESTION KEYWORD	study made **unconvincing**
PASSAGE EVIDENCE	Para E: Cyril Burt's data "were **probably faked**"

19. Francis Galton

QUESTION KEYWORD	first to conduct research
PASSAGE EVIDENCE	"**Francis Galton**, an early geneticist, realized that they came in two varieties"

20. 1924

QUESTION KEYWORD	medical application as early as
PASSAGE EVIDENCE	"it was not until **1924** that it was used to formulate... the twin rule of pathology"

21–23. A, E, F

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QUESTION KEYWORD	fields researched at Ohio/Maryland/Twinsburg
PASSAGE EVIDENCE	Sense — taste/smell (Breslin, Para G); Sound — hearing (Drayna, Para I); Baldness — "male-pattern baldness" (Para J)

24–26. **A, B, D**

QUESTION KEYWORD	results verified in passage
PASSAGE EVIDENCE	Non-identical twins from different eggs (Para A); identical twins more genetically concordant (Para B); smoking amount "largely down to his genes" (Para C)

Passage 3 — The Origin of Ancient Writing

27. **A**

QUESTION KEYWORD	story regarded as ridiculous because
PASSAGE EVIDENCE	Para A/B: scholars "doubted that the earliest writing was a **direct rendering of speech**" — so the recipient couldn't have simply "read" spoken words

28. **D**

QUESTION KEYWORD	story of King of Uruk shows
PASSAGE EVIDENCE	Para C: writing "arose out of the need to **store and transmit information**" — Sumerians already grasped this

29. **C**

QUESTION KEYWORD	disagreement about
PASSAGE EVIDENCE	Para D: pictorial-origin theory "clashed with" the token-origin theory — disagreement over **the way writing began**

30. **B**

QUESTION KEYWORD	opponents of token theory believe
PASSAGE EVIDENCE	Para D: "explanation of writing's origins in the visual arts, pictures becoming increasingly abstract"

31. **C — Dr. Denise Schmandt-Besserat**

QUESTION KEYWORD	no proof linked to decorated objects

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PASSAGE EVIDENCE	"show me a single sign of writing you can trace back to a **pot** — it doesn't exist"

32. B — Dr. Peter Damerow

QUESTION KEYWORD	came to represent speech
PASSAGE EVIDENCE	"flexibility to adapt to specific characteristics of the **sounds of the language** to be represented"

33. D — Dr. Piotr Michalowski

QUESTION KEYWORD	means of political control
PASSAGE EVIDENCE	"also became a **new instrument of power**"

34. B — Dr. Peter Damerow

QUESTION KEYWORD	did not represent grammar of speech
PASSAGE EVIDENCE	"dealing mainly in **lists and categories** rather than in sentences and narrative"

35. A — Dr. Holly Pittman

QUESTION KEYWORD	no convincing link tokens/signs
PASSAGE EVIDENCE	"I have an argument with her evidence for a **link between tokens and signs**"

36. C — Dr. Denise Schmandt-Besserat

QUESTION KEYWORD	uses narrow at first, then widened
PASSAGE EVIDENCE	"first 500 years... used almost solely for recording economic information, and after that its uses **multiplied and broadened**"

37. H — Mesopotamia

QUESTION KEYWORD	people of ____
PASSAGE EVIDENCE	"Sumerians, who lived in **Mesopotamia**"

38. L — clay tablets

QUESTION KEYWORD	written on ____

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PASSAGE EVIDENCE	"many **clay tablets** in an early form of cuneiform"

39. **A — cuneiform**

QUESTION KEYWORD	called ____
PASSAGE EVIDENCE	"an early form of **cuneiform**, a script written by pressing... into wet clay"

40. **I — abstract**

QUESTION KEYWORD	developed to become more ____
PASSAGE EVIDENCE	"gradually evolved from the pictorial to the **abstract**"

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Answer Key & Explanations

Reading Test 9 (Volume 2)

Passage 1 — Becoming an Expert

1. guiding principles and rules

QUESTION KEYWORD	novice needs to study ____
PASSAGE EVIDENCE	"needs to learn the **guiding principles and rules** of a given task"

2. mentor

QUESTION KEYWORD	guidance of a ____
PASSAGE EVIDENCE	"a novice will find a **mentor** to guide her"

3. journeyman

QUESTION KEYWORD	next stage after novice
PASSAGE EVIDENCE	"becomes a **journeyman**"

4. patterns of behavior

QUESTION KEYWORD	identify ____
PASSAGE EVIDENCE	"begins to recognize **patterns of behavior** within cases"

5. complex

QUESTION KEYWORD	more ____ ways
PASSAGE EVIDENCE	"exposure to increasingly **complex** cases"

6. FALSE

QUESTION KEYWORD	novices/experts use **same system**
PASSAGE EVIDENCE	"A novice... might group objects together by color or size, whereas an **expert would group** the same objects according to their **function or utility**"

7. NOT GIVEN

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QUESTION KEYWORD	novices' training on **long term memory**
PASSAGE EVIDENCE	The passage discusses experts' memory advantages but never states novices' training focuses on long-term memory.

8. TRUE

QUESTION KEYWORD	novices solve **straight away**
PASSAGE EVIDENCE	"novices, who **immediately seek to find a solution**"

9. FALSE

QUESTION KEYWORD	experts **always** more efficient
PASSAGE EVIDENCE	Experts are "not particularly good at making predictions" — an **actuarial table is as good, or better**, so not "always" more efficient

10. TRUE

QUESTION KEYWORD	review more when flaws occur
PASSAGE EVIDENCE	"Experts **check their solutions more often** than novices"

11. models

QUESTION KEYWORD	cognitive ____ to forecast
PASSAGE EVIDENCE	"inconsistent when using mental **models** to make predictions"

12. human biases

QUESTION KEYWORD	believe it is due to ____
PASSAGE EVIDENCE	"researchers point to **human biases** to explain unreliable expert predictions"

13. consensus

QUESTION KEYWORD	did not produce ____
PASSAGE EVIDENCE	"the literature shows little **consensus** regarding the causes"

Passage 2 — Revolutions in Mapping

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14. A

QUESTION KEYWORD	mapmakers in 21st century
PASSAGE EVIDENCE	"a **combination of sonar and radar** produces charts of the seafloor" and radar mapped Venus

15. D

QUESTION KEYWORD	Library of Congress offers
PASSAGE EVIDENCE	"at the click of a button, a **print-out of the map appears**"

16. B

QUESTION KEYWORD	Ptolemy alerted contemporaries to
PASSAGE EVIDENCE	"advice to cartographers to **create maps to scale**... expressed as a fraction or ratio of the real distance"

17. C

QUESTION KEYWORD	invention of the printing press
PASSAGE EVIDENCE	"put maps in the hands of more people, and took their production away from monks" — **changed the approach**

18. C

QUESTION KEYWORD	writer concludes
PASSAGE EVIDENCE	"its objective is, and **always has been**, to communicate a sense of 'here' in relation to 'there'"

19. D — Eratosthenes

QUESTION KEYWORD	close to measuring Earth's distance
PASSAGE EVIDENCE	"calculated the Earth's circumference as **25,200 miles**, which was remarkably accurate"

20. C — Cassini family

QUESTION KEYWORD	maps showing man-made landmarks
PASSAGE EVIDENCE	"showing roads, rivers, canals, towns, abbeys, vineyards, lakes and even **windmills**"

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21. B — Ptolemy

QUESTION KEYWORD	foundation for time zones
PASSAGE EVIDENCE	"this is the basis for the width of **modern time zones**"

22. Egypt

QUESTION KEYWORD	first great step took place in ____
PASSAGE EVIDENCE	Eratosthenes, "a scholar at the famous Alexandrian Library in **Egypt**"

23. monks

QUESTION KEYWORD	maps were responsibility of ____
PASSAGE EVIDENCE	"took their production away from **monks**"

24. Ptolemy

QUESTION KEYWORD	writings of ____ kept
PASSAGE EVIDENCE	"the rediscovered **Ptolemy**, whose writing had been preserved by Arabs"

25. satellites

QUESTION KEYWORD	____ vital to creation of maps
PASSAGE EVIDENCE	"rarely go into the field without being linked to **navigation satellites**"

26. cars

QUESTION KEYWORD	cheaper versions used in ____
PASSAGE EVIDENCE	"the heart of electronic map displays available in **some cars**"

Passage 3 — Sir Francis Ronalds and the Telegraph

27. C — 1816

QUESTION KEYWORD	first satisfactory electricity result

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PASSAGE EVIDENCE	"In **1816**... Ronalds successfully transmitted and received letters over 150 meters of wire"

28. **A — 1753**

QUESTION KEYWORD	first proposal, static electricity
PASSAGE EVIDENCE	"Since **1753**, when the first proposal for an electric telegraph worked by statical electricity was made"

29. **E — 1825**

QUESTION KEYWORD	Ronalds's first patent
PASSAGE EVIDENCE	"In **1825** he invented and patented a perspective tracing instrument"

30. **D — 1823**

QUESTION KEYWORD	revealed significance to public
PASSAGE EVIDENCE	"In a small pamphlet published in **1823**, Ronalds described his invention and listed some of its possible uses"

31. **G — 1843**

QUESTION KEYWORD	contribution to meteorological data
PASSAGE EVIDENCE	"Early in **1843** Ronalds was made honorary director... of the Meteorological Observatory"

32. **letters and numbers**

QUESTION KEYWORD	carved in the dials
PASSAGE EVIDENCE	"rotating dials, engraved with **letters of the alphabet and numbers**"

33. **glass tubes**

QUESTION KEYWORD	enclosed with buried wires
PASSAGE EVIDENCE	"wires were enclosed in **glass tubes** buried in the ground"

34. **800km**

QUESTION KEYWORD	greatest distance believed

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PASSAGE EVIDENCE	"he believed his telegraph would work over distances of **800km**"

35. frictional-electricity machine

QUESTION KEYWORD	power kept wire charged
PASSAGE EVIDENCE	"A **frictional-electricity machine** kept the wire continuously charged"

36. G

QUESTION KEYWORD	commercial use of telegram
PASSAGE EVIDENCE	Para G: "the first electric telegraph used **publicly and commercially** in England"

37. A

QUESTION KEYWORD	influence from a fellow he met
PASSAGE EVIDENCE	Para A: "Under the influence of **Jean Andre de Luc**... whose acquaintance he made in 1814"

38. E

QUESTION KEYWORD	rejected due to preceding reference
PASSAGE EVIDENCE	Para E: Wedgwood's 1806 proposal was told "the **semaphore** was sufficient" — same reasoning applied to Ronalds

39. D

QUESTION KEYWORD	drawback of telegram
PASSAGE EVIDENCE	Para D: "The only defect in his invention was the **comparative slowness**"

40. I

QUESTION KEYWORD	recognized by authority
PASSAGE EVIDENCE	Para I: "he was **knighted**... for his 'early and remarkable labors in telegraphic investigations'"

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Answer Key & Explanations

Reading Test 10 (Volume 2)

Passage 1 — Dirty River But Clean Water

1. NOT GIVEN

QUESTION KEYWORD	fire **worse** than flood
PASSAGE EVIDENCE	Both called "two of humanity's worst nightmares" — no comparison of which is worse

2. FALSE

QUESTION KEYWORD	peaks at **1500** every eight years
PASSAGE EVIDENCE	Average flow was 1,500; "every eight years or so, that figure rose to almost **3,000** cubic metres"

3. TRUE

QUESTION KEYWORD	tributary sediment, **little impact**
PASSAGE EVIDENCE	"not powerful enough to distribute the sediment in an **ecologically valuable way**"

4. FALSE

QUESTION KEYWORD	chub decline **always** caused by trout
PASSAGE EVIDENCE	"trout and chub co-existed until the Glen Canyon dam was built, so **something else is going on**"

5. TRUE

QUESTION KEYWORD	1996 flood success at **beginning**
PASSAGE EVIDENCE	"To start with, all seemed well. The floodwaters **built up sandbanks** and infused the river with sediment"

6. TRUE

QUESTION KEYWORD	artificial flood **smaller** than natural
PASSAGE EVIDENCE	"this flood is **smaller than even an average spring flood**"

7. NOT GIVEN

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QUESTION KEYWORD	mighty floods, **clean** high-quality water
PASSAGE EVIDENCE	Mighty floods "moved massive quantities of sediment... making a **muddy mess**" — the passage does not make the specific "clean, high-quality water" claim in this statement

8. spring

QUESTION KEYWORD	melt in the season of ____
PASSAGE EVIDENCE	"**Spring** Snow from as far away as Wyoming would melt"

9. sediment

QUESTION KEYWORD	gather ____ together
PASSAGE EVIDENCE	"the only **sediment** that it has collected has come from tiny, undammed tributaries"

10. razorback sucker

QUESTION KEYWORD	species disappeared
PASSAGE EVIDENCE	"the Colorado pike-minnow, the **razorback sucker** and the round-tail chub"

11. common carp

QUESTION KEYWORD	species that moved in
PASSAGE EVIDENCE	"fathead minnows, channel catfish and **common carp**"

12. visibility

QUESTION KEYWORD	poor ____ away from predators
PASSAGE EVIDENCE	"poor **visibility** created by the thick, red water"

13. sand

QUESTION KEYWORD	volume of ____ behind dam
PASSAGE EVIDENCE	"the volume of **sand** available behind the dam was too low to rebuild the sandbanks"

Passage 2 — We Have Star Performers!

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14. F

QUESTION KEYWORD	non-business example, systems beat stars
PASSAGE EVIDENCE	Para F: football — "Arsenal and United... put much more emphasis on developing a setting"

15. B

QUESTION KEYWORD	failed company believing in stars
PASSAGE EVIDENCE	Para B: "**Enron** believed in stars... because they didn't believe in systems"

16. G

QUESTION KEYWORD	suggestion for acquiring employees
PASSAGE EVIDENCE	Para G: "how do you compete in the war for talent? **You grow your own**"

17. C

QUESTION KEYWORD	medical anatomy metaphor
PASSAGE EVIDENCE	Para C: "likened hiring a star to an **organ transplant**"

18. NOT GIVEN

QUESTION KEYWORD	McKinsey didn't expect influence
PASSAGE EVIDENCE	No mention of McKinsey's expectations about the book's influence

19. YES

QUESTION KEYWORD	economy affects hiring foreigners
PASSAGE EVIDENCE	"labour shortages... are the reason the government has laid out the welcome mat for immigrants... **as the economy tightens**"

20. NO

QUESTION KEYWORD	Enron collapse, **unfortunate incident**
PASSAGE EVIDENCE	Attributed to their belief in stars over systems — a **management/philosophy failure**, not chance

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21. YES

QUESTION KEYWORD	develop-stars clubs better than collectors
PASSAGE EVIDENCE	Arsenal/United "less successful"... no — "**less successful**" was Chelsea/Real; Arsenal/United outperform them

22. analysts

QUESTION KEYWORD	1000 ____ participants
PASSAGE EVIDENCE	"study of the careers of 1,000 star-stock **analysts**"

23. star performer

QUESTION KEYWORD	company hires a ____
PASSAGE EVIDENCE	"when a company recruited a **star performer**, three things happened"

24. working environment

QUESTION KEYWORD	behave worse than in ____
PASSAGE EVIDENCE	"success... due to the **working environment** — systems, processes, leadership"

25. salary

QUESTION KEYWORD	increase their ____
PASSAGE EVIDENCE	"upping their **salary** as they did"

26. rivals

QUESTION KEYWORD	rejection from those ____
PASSAGE EVIDENCE	"tensions and resentment by **rivals** within the team"

Passage 3 — Toddlers Bond With Robot

27. D

QUESTION KEYWORD	changes in interaction quality

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PASSAGE EVIDENCE	Para D: "the quality of those interactions **improved steadily over 27 sessions**"

28. E

QUESTION KEYWORD	comparison of two robots
PASSAGE EVIDENCE	Para E: QRIO vs. "another robotlike toy named **Robby**... used as a control"

29. C

QUESTION KEYWORD	previous robots, short interest
PASSAGE EVIDENCE	Para C: "only been able to hold human interest for a **limited time**"

30. E

QUESTION KEYWORD	importance of touch
PASSAGE EVIDENCE	Para E: "The most important aspect of interaction was **touch**"

31. G

QUESTION KEYWORD	new direction of study
PASSAGE EVIDENCE	Para G: "this study has taken a **new direction**"

32. C

QUESTION KEYWORD	technical parameters
PASSAGE EVIDENCE	Para C: "the **23-inch-tall (58-centimeter-tall)** machine"

33. B

QUESTION KEYWORD	significance and novelty
PASSAGE EVIDENCE	Para B: "this is the **first long-term study** of this sort... groundbreaking"

34. C — Hiroshi Ishiguro

QUESTION KEYWORD	robots as babysitters

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PASSAGE EVIDENCE	"we'll have **electronic baby-sitters** and peer-robots in every kindergarten"

35. A — Fumihide Tanaka

QUESTION KEYWORD	treated as living creature
PASSAGE EVIDENCE	"by the end they treated the robot **as a peer rather than a toy**"

36. B — Ronald Arkin

QUESTION KEYWORD	first major study
PASSAGE EVIDENCE	"To my knowledge, this is the **first long-term study** of this sort"

37. A — Fumihide Tanaka

QUESTION KEYWORD	robots assist teachers
PASSAGE EVIDENCE	"social robots like QRIO could greatly **enrich classrooms and assist teachers**"

38. C

QUESTION KEYWORD	robot's height
PASSAGE EVIDENCE	"the **23-inch-tall** (58-centimeter-tall) machine"

39. D

QUESTION KEYWORD	instructions EXCEPT
PASSAGE EVIDENCE	Instructions were "to **giggle, dance, sit down, or walk**" — "crawl" is not among them

40. A

QUESTION KEYWORD	increasing interaction during
PASSAGE EVIDENCE	"began to increasingly interact with the robot... during the **first 11 sessions**"

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